NYANDUNGU ECO-PARK
KIGALI, RWANDA

Conservation Area Business Plan

Date: 14 March 2022

This report was prepared by:

Conservation Capital

k.fitzgerald@conservation-capital.com
www.conservation-capital.com

CONTENTS

Acronyms

CABP	Conservation Area Business Plan
CAPEX	Capital expenditure
CC	Conservation Capital
CIFOR	Centre for International Forestry Research
CMP	Collaborative Management Partnership
CSR	Corporate Social Responsibility
DMC	Destination management company
GHG	Greenhouse gas
GoR	Government of Rwanda
GWP	Global warming potential
HR	Human Resources
IPCC	Intergovernmental Panel on Climate Change
M&E	Monitoring & Evaluation
MOU	Memorandum of Understanding
NCA	Natural Capital Accounting
NEP	Nyandungu Eco-Park
NUWEP	Nyandungu Urban Wetland Ecotourism Park
OPEX	Operational expenditure.
PA	Protected Area
PEC	Proposal Evaluation Committee
PES	Payment for ecosystem service
PPP	Public Private Partnership
PR	Public relations
RALC	Rwanda Academy of Language and Culture
REMA	Rwanda Environment Management Authority
RDB	Rwanda Development Board
RMHC	Rwanda Media High Council
ROI	Return on Investment
RRA	Rwanda Revenue Authority
SMART	Specific, Measurable, Achievable, Relevant and Time-based
SWOT	Strength, Weakness, Opportunity, Threat
USPs	Unique Selling Points
VER	Verified Emission Reductions
WAVES	Wealth Accounting and the Valuation of Ecosystem Services

List of Figures

List of Tables

List of Photos

Executive Summary

1. Introduction

The Government of Rwanda's (GoR) development strategy emphasises the importance of environmental protection, natural resource management and climate change preparedness. The GoR has embraced a Green Growth strategy for the country. The Government's vision for Kigali, outlined in the Kigali Master Plan, mirrors this country-level strategy of Green Growth. In addition to developing Kigali and Rwanda in a way that secures the country's natural assets, Rwanda is also committed to creating accessible green and open spaces for its citizens and visitors. For example, Rwanda's Vision 2050 outlines a target for 25% of Kigali City to be turned into recreational spaces, and to create a park in every neighbourhood with 2,000-50,000 inhabitants.

Rwanda's population is close to 13 million, with an annual growth rate of 2.86%. Its urbanisation rate is increasing, and Kigali city is predicted to have the highest population density in the country. The Kigali Development Strategy for 2050 outlines clear green growth targets, including the development of urban recreation areas and green space for its citizens ensuring healthy and environmentally-friendly lifestyles. The country's wetland system is part of this green development strategy.

Approximately 75.21 km^2 of Kigali city is wetland. These wetlands help to mitigate flooding, secure biodiversity, and provide water filtration services. As the city of Kigali continues to expand, the resulting increase in impervious surfaces and its subsequent increase in rainwater run-off, threatens to damage infrastructure and cause loss of human life. Protecting wetlands to mitigate flooding is a priority for the GoR. For example, the 2018 Rwanda Environmental Law states; "Water resources must be protected from any source of pollution. Swamps with permanent water and full of swamp vegetation must be given special protection considering their role and importance."

Recognising the ecological, economic, and social benefits of wetlands, the GoR has committed to protecting and restoring Kigali's wetlands. The Nyandungu Eco-Park (NEP) (referred to as 'the park') is a 121.8-hectare wetland protected area (PA) located in Kicukiro and Gasabo Districts, just north of the Kigali International Airport. NUWEP was a degraded wetland system that is now managed by the Rwanda Environmental Management Authority (REMA). The development of the park followed a 2017 design plan which centres around the wetland system. During its development phase, the park was referred to as Nyandungu Urban Wetland Ecotourism Park (NUWEP). The name changed to NEP in 2022.

The GoR is in the process of developing NEP with five core goals:


Preserve the ecological function of the wetland


Provide flood mitigation


Provide space for education, recreation, and contemplation


Apply state of the art practices for environmental, economic, and social sustainability


Support low impact amenities and activities

This Conservation Area Business Plan (CABP) outlines how the GoR can achieve its conservation objectives and operate and sustain NEP as a thriving eco-park in Kigali.

Overall, NEP is well positioned to support urban biodiversity and its associated eco-system services. As the first restored wetland in Kigali, NEP is a model that can be replicated across the wetland system in the city. Further, the successful ecological restoration should be promoted globally as best practice and in line with the GoR's leadership in conservation and green growth. If well managed, NEP also can create a conservation culture in Kigali through activities and use designed to expose users to its unique natural attributes.

The Strength, Weakness, Opportunity and Threat (SWOT) analysis summarising Conservation Capital's assessment of Nyandungu Eco-Park.

Strengths	Weaknesses
Environmental • Restored and functioning wetland protecting urban biodiversity, mitigation flooding and providing ecosystem services • Substantial piece of land with diversity of habitat types **Commercial** • Central Location in Kigali makes it easily accessible to residents and visitors • The first park of its kind in Kigali, unique attraction • Clear land tenure and vision of use and management by the GoR • Affordable pricing • Existing infrastructure (restaurant, benches, pathways etc.) paid for by donors; therefore, no debt • Ability to support a diversity of low-impact activities • Low marketing need (word of mouth and current enthusiasm to visit NEP) • Ability to cover costs with conservation fees and corporate sponsorship program	**Environmental** • Presence of alien plants • Proximity to industrial zone and pollution • Adjacent busy road creates noise and pollution • Lack of adequate baseline information for monitoring ecological restoration **Commercial** • Insecurity due to lack of fencing and adequate security staff • Short timeline and pressure to open to the public • Reputational risk if the opening is rushed without adequate management and preparation • Lack of adequate funding for on-going management and development • Management transition, lack of adequate time to transition to a new manager before opening • Currently no wetland access via boardwalks • Not in a prime location in Kigali in terms of proximity to tourist accommodation and attractions
Opportunities	**Threats**
Environmental • Continued restoration will result in natural recolonization of bird, mammal, butterfly and other species • Ability to cultivate a conservation culture in Kigali • Ability to provide formal and informal educational opportunities • Development of classes that promote fitness and conservation • A pilot that is replicated across Kigali wetlands • Promote Rwanda as a leader in wetland and ecological restoration **Commercial** • Donor and corporate sponsorship interest • Engagement with private sector to develop new low-impact activities • Ability to attract an innovative management partner that shares the GoR vision • Fencing will control access, establish security and support user fee revenue generation	**Environmental** • External pressure to develop the park in a way that does not support the conservation vision • Environmental effects: pollution, climate change, run-off • Limited control over upstream effects, including water runoff, pollution and alien plants • High market demand and poor control of visitor influx • Land encroachment (agriculture, unsustainable grass cutting) • Potential widening of the road to Akagera **Commercial** • Insufficient funding and capacity to fund the park management, infrastructure, and activities • Lack of a professional management partner • Community dissatisfaction • A similar development in Gikondo would outcompete NEP in some markets

Table 1: Nyandungu Eco-Park SWOT Analysis

The revenue model for NEP is positive. The operating budget (not including capital costs) ranges from USD 300,000 – 357,000 per year (Section 3.2). Considering revenue from conservation fees, conservation-compatible events, and the restaurant, NEP will generate USD 3.7 million over ten years (Section 4.11). The projected revenue will cover NEP's operational and capital expenditure costs and could cover an operational reserve fund for two years, formal education programmes, and potentially support other Kigali wetlands. In addition to user fees, a corporate sponsorship programme is a very viable revenue model for NEP given its proximity to the industrial area and the presence of potential corporate partners in Kigali (Section 5.1). Other revenue models are also profiled such as biodiversity offsets and payment for ecosystem services (Section 5.2). Some of these models have potential if explored at a larger scale within the greater wetland system in Kigali.

2. Methodology

The approach to developing this business plan follows Conservation Capital's CABP methodology. This emphasises a demand-led and need-based approach – with all components being driven by the initial assessment of the conservation and socio-economic context.

The methodology assesses the wetland's conservation priorities and management and capacity needs, along with the corresponding financing requirements, and potential revenue-generating opportunities. The ability to replicate this model in other urban wetlands in Kigali is also considered.

The methodology for NEP is the following:

- ⇒ Establish current project context and development (NUWEP Feasibility Report 2022)
- ⇒ Determine consequential **conservation and management priorities** from relevant stakeholders
- ⇒ Propose a **management structure and guideline**
- ⇒ Identify wetland **management actions, capacity and development needs**
- ⇒ Assess **associated financing requirements**
- ⇒ Set out relevant and feasible **commercial revenue-generating opportunities**
- ⇒ Assess **residual funding gap and potential additional funding (Commercial – Donors)**
- ⇒ Summarise **overall recommendations**

Conservation Capital (CC) drew on the prior NUWEP assessments, the 2022 updated NUWEP Feasibility Report, results from an on-line survey of various stakeholders, results from direct stakeholder consultation, best practice from other urban protected areas in Africa, and the vision and legal framework of the GoR.

3. Business Planning Outcomes

This CABP aims to specifically provide the following:

- A list of possible management and user activities to NEP
- An implementation plan for the activities
- A list of activities deemed unsuitable for NEP at this time
- A provisional financial and revenue model
- Potential management models
- A recommendation for visitor capacity and operating hours
- Indicators for monitoring and evaluation

SECTION 1: CONSERVATION CONTEXT AND MANAGEMENT PRIORITIES

1.1. Conservation Context

1.1.1. Kigali Wetlands

Rwanda's cities, and Kigali in particular, support expansive wetlands systems, which play a key role in the GoR's Green Growth agenda. Approximately 75.1 km^2 of Kigali city is wetland (Kigali 2050 Development Plan). Across Rwanda, wetlands have declined due to development, encroachment, and other key threats. Rwanda's commitment to protecting and restoring its wetlands is supported by the global recognition of the ecological, economic, and social value of wetlands. Wetlands are rich in biodiversity and ecological goods and services and yet they are threatened globally (CBD Wetland & Ecosystem Services Report 2015)

The GoR, through its wetland policies[1] [2], development plans and commitment to various national and international conservation and development targets, aims to protect its existing wetlands and restore degraded wetlands. As Rwanda develops and urbanises in high rainfall and hilly terrain areas such as Kigali, the GoR recognises the risk of increased runoff into the valleys and the need to protect wetlands to help mitigate flooding and support filtration. Once restored, the GoR aims to utilise the wetlands in Kigali city for education about Rwanda's natural assets and to provide space for recreation and contemplation, and, for some wetlands, managed agriculture.

Kigali is surrounded and traversed by several streams, rivers and lakes and is delineated into 25 watershed areas. Lake Muhazi lies along the border of the city northeast of Gasabo District. Nyabarongo River borders Nyarugenge and Kicukiro Districts along the southwestern edge of the city and is the river to which most of the city's other rivers and streams are tributaries. It flows to Lake Rweru and is part of the Nile River Basin via the Akagera River. The Yanze, Kibumba, Rwazangoro and Ruganwa Rivers all drain into Nyabugogo River, which flows into the Nyabarongo River to the west of Kigali (2050 City of Kigali Master Plan).

These waterbodies provide valuable supplies of water, drain stormwater, and contribute to the physical beauty and character of Kigali. However, there are several challenges, including the turbidity of the Nyabarongo River due to high sediment load from soil erosion in the catchments and other polluting content inflowing from sewage and stormwater runoff. Agricultural areas are a source of sediment, pesticides and chemical fertilizers, which further degrade water quality. Urban development and land cover change are increasing the impermeable surfaces in Kigali's catchments, leading to increased stormwater runoff and localized flooding. This compounds the already existent challenge of flash flooding and landslide risks associated with the hilly topography and steep slopes of the city. Flood risks are high in the valley and wetland areas of Kigali to which the catchments drain, including along the Nyabugogo, Gikondo, and Nyabarongo Rivers. Preservation and restoration of the natural drainage channels across the catchments in Kigali is an important strategy for managing flooding, especially in the absence of developed stormwater drainage infrastructure in much of the city (2050 City of Kigali Master Plan; Niyokwiringirwa 2021).

[1] Rwanda Wetlands Governance Profile - CBD

[2] USAID Rwanda Policy Research brief n. 5: balancing wetland sustainable use and protection through policy in Rwanda

As Rwanda develops and urbanizes in high rainfall and hilly terrain areas such as Kigali, the GoR recognizes the risk of increased runoff into the valleys and the need to protect wetlands to help mitigate flooding and support filtration.

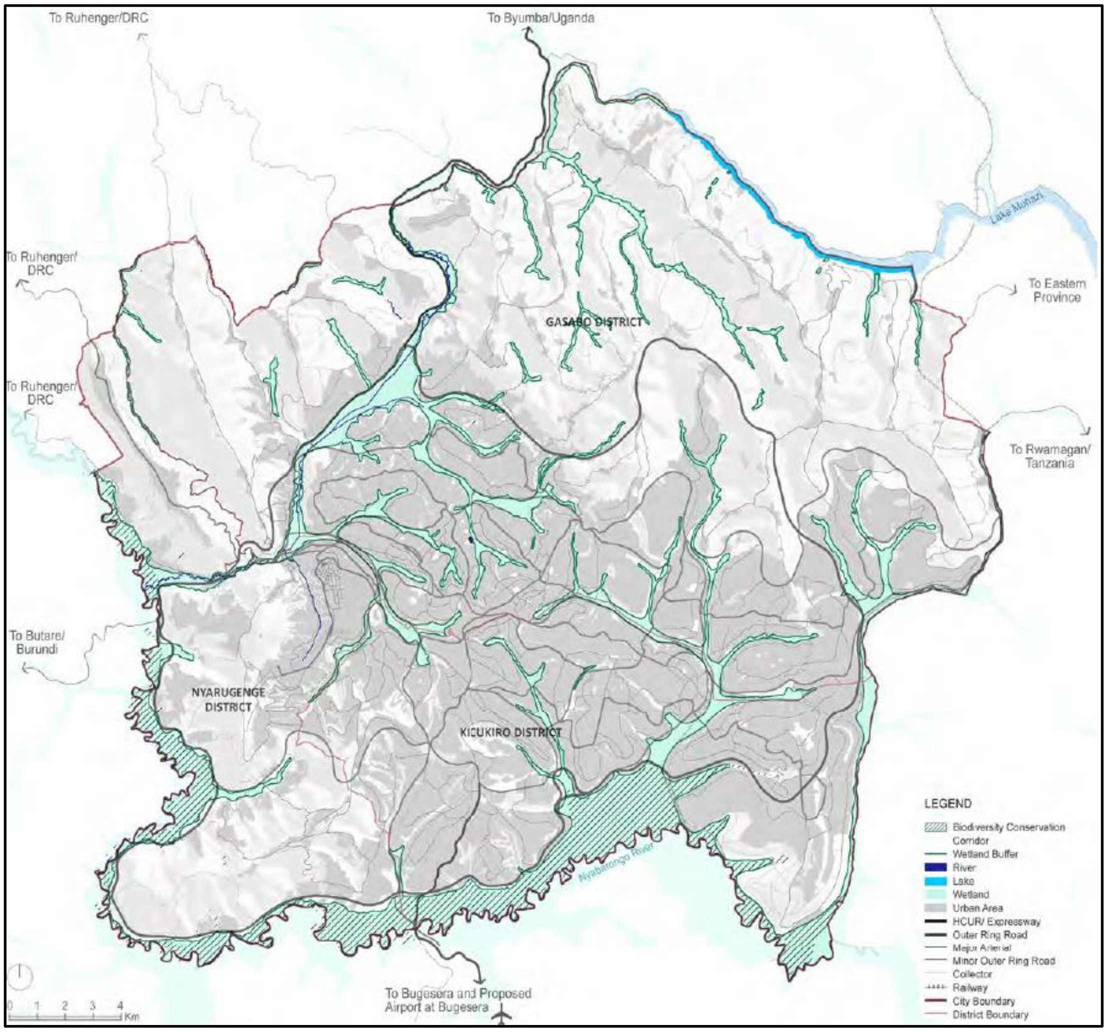


Figure 1: Kigali city wetlands and waterways (2050 City of Kigali Masterplan).

1.1.2. Nyandungu Urban Wetland Ecotourism Park Location

Nyandungu wetland is shared by the administrative sectors of Nyarugunga (Kicukiro District) and Ndera (Gasabo District) in the city of Kigali (Figure 2). The wetland is approximately 121.8 hectares and a total perimeter of approximately 6730.8 metres (Figure 2). Nyandungu wetland is north of the Kigali International Airport. It is bordered to the south by the Kigali-Kayonza Road (RN3) and the road to the Adventist University to the west and the road to Ndera to the east. Kigali's industrial zone abuts Nyandungu to the north. Nyandungu wetland is drained by two streams: Mwanana and Kabagenda. Both flow into the Mulindi stream, a tributary of the Nyabarongo River.

Figure 2: Nyandungu wetland, north of the Kigali International Airport, Kicukiro and Gasabo Districts.

Kigali City has 37 wetlands covering approximately 7,700 hectares. Eight wetlands are highlighted in Figure 3 and Nyandungu wetland is in the centre of the map.

Figure 3: Eight of Kigali's wetlands. Nyandungu wetlands located in the centre of the map. Other Kigali attractions particularly pertinent to international visitors are highlighted in red.

For planning and management purposes NUWEP was delineated into five sectors, that will be referred to throughout this CABP (Figure 4).

Figure 4: Five sections (sectors) that comprise NUWEP, Kigali, Rwanda.

1.1.3. Nyandungu Wetland Land Use

Historically, Nyandungu wetland was highly utilised and cleared for agricultural and pastoral activities. The wetland was polluted with sewage outflow from surrounding neighbours, as well as pollutants from road runoff, the industrial zone and urban development. The poor management of the wetland was leading to flooding of downstream areas. The site is flood-prone due to three factors:

1. It is low-lying land that receives aggressive water run-off from the densely populated and developed surrounding hills.
2. The soil is primarily clay, and the valley receives wastewater from various developments, such as the industrial zones, roads, and nearby households.
3. The Mwanana River is narrow as a result of channelling/canalisation and often flooded during heavy rainfall.

Historically, Nyandungu wetland was managed by the Ministry of Agriculture and Animal Resources (Gabuka 2012). In the 1980s, it was comprised of sugar cane fields and crop nurseries. Thereafter, the wetland was transferred to the Ministry of Defence. Following the Genocide Against the Tutsi in 1994, the land was used for agricultural farming by private individuals, and for sand quarrying. The Ministry of Agriculture and Animal Resources wanted to develop fishponds and a Chinese company proposed to develop a bamboo nursery. In 2018, the wetland was transferred to REMA for management under the Minister of Environment. Subsequently, REMA has actively rehabilitated the wetland, which represents an incredible ecological restoration story, that mirrors Rwanda’s commitment to conservation and restoration.

REMA has actively rehabilitated the wetland, which represents an incredible ecological restoration story, that mirrors Rwanda’s commitment to conservation and restoration.

1.1.4. Nyandungu Eco-Park Land Cover

NEP land cover is delineated as four broad types (Figure 5) and include native and exotic species.

1. Wetland vegetation: the central spine of the park that is closely associated with water flow;
2. Acacia savannah: primarily on the northern sides of Sectors 3 and 4, and with smaller extents in Sectors 2 and 5;
3. Mixed woodlands: present along the fringes of the northern and southern sides of all sectors; and
4. Exotic woodland: primarily along the southern sides of Sectors 3 and 4.

Figure 5: Nyandungu Eco-Park vegetation cover (2021).

The vegetation types are the result of the underlying geology, soil and hydrology; the existing seed bank when restoration started; and the restoration implemented by Afrilandscapes, which largely mirrored this natural base and supplemented it with additional species. The vegetation cover is further broken down by sector (Table 2). Sector 5 includes the largest area of wetland followed by Sector 3.

Sectors	Open Water	Wetland	Exotic woodland	Mixed woodland	Acacia Savanah	Gardens	Total area	Total %
1	0.21	5.06	0	2.86	0.15	0	8.28	7%
2	0	2.33	0	0.82	0.77	0	3.92	3%
3	1.09	8.65	4.75	0.87	13.51	0.1	18.97	23%
4	0.44	6.46	3.31	3.1	16.83	0.12	30.26	24%
5	0.49	31.89	0	13.3	6.84	0	52.52	42%
Total Area	**2.23**	**54.39**	**8.06**	**20.95**	**38.1**	**0.22**	**123.95**[3]	
Total %	**2%**	**44%**	**7%**	**17%**	**31%**	**0%**	**100%**	

Table 2: Breakdown of area of coverage (hectares) of different vegetation types / management zones per sector

[3] The total area is 121.8 hectares. The slight discrepancy is because of the use of Google Maps.

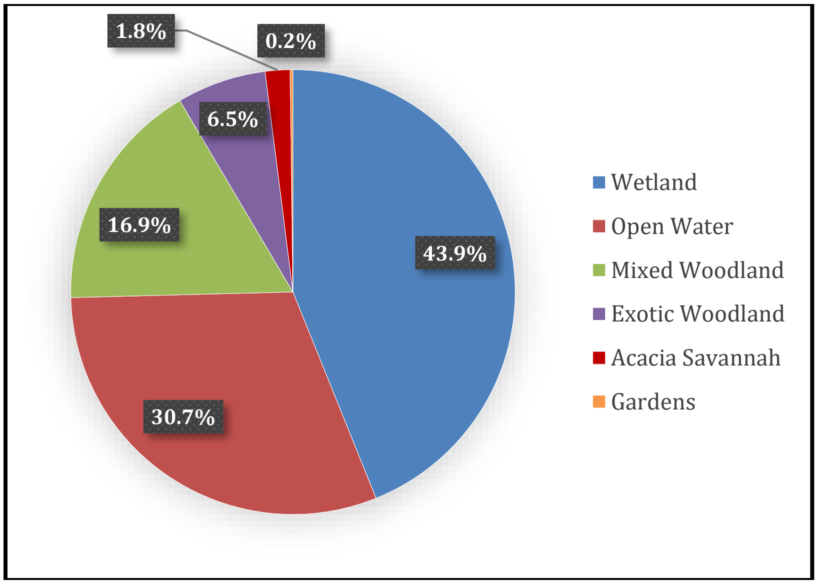


Figure 6: NEP vegetation cover percentage (2021).

1.1.5. Nyandungu Wetland Flora

A survey conducted in 2017 on the species present in Nyandungu wetland identified 96 plant species belonging to 38 families. It found the herbaceous layer as the most dominant with 53 species, followed by shrubs with 21 species. Lianas are less represented in the wetland with only four species. The most represented family was *Asteracea*e (14 species) followed by *Poaceae* (12 species). A far more comprehensive survey conducted in 2019 (Mvukiyumwami 2019) recorded more than 370 species. The full list of plant species is included in the 2022 NUWEP Feasibility Report.

The wetland zone is dominated by reeds in the genus *Typha* with pockets of *Papyrus,* which are adjacent to areas of sedges composed of various *Cyperus* species. Areas of open acacia savannah with quickly maturing specimens of *Acacia hockii* and *Acacia polyacantha are* contiguous to the wetland zone, in most sectors. Two types of woodland are found beyond the savannah areas. To the southern side of the wetland in sectors 3 and 4 there is a lush woodland of primarily exotic tree species including *Casuarina*, *Grevillea*, *Jacaranda,* and others. The dominant feature of this woodland is the stands of exotic *Bambusa* (bamboo) that effectively shade the cycle and walking paths. For the time being, removal of these species is not recommended. The other woodland type is described as mixed woodland and has been specifically planted in the landscaping project to follow the boundaries and paved paths elsewhere in NEP. This vegetation type is currently most developed on the northern side of Sector 5 where stands of *Podocarpus* and *Croton* are already providing significant shade and forming a nascent woodland.

As an urban site that has been significantly disturbed over the years, alien plant species are to be expected. Some alien plants are evident on the fringe of the wetland and in other areas of the wetland and in the park and include most significantly invasive species such as *Lantana camara*, *Ricinus communis* and *Psidium guajava,* which have proven invasive and transformative elsewhere in Africa. *L camara* is also the biggest threat to the savannah areas with some invasion of the areas north of the wetland in Sector 5 evident. Elsewhere on disturbed soils, *Datura* is relatively common. Given the risk of some of these species being dispersed downstream by waterflow and impacting further ecosystems, every effort must be made to remove them. There is also reference in a previous feasibility study to the planting of *Morus alba*, another potentially invasive species - the removal of which is recommended before it is dispersed by fruit-eating birds into areas of the wetland that are difficult to access.

Following the ecological restoration work, it is important to continue to monitor the ecological recovery and the diversity of plans and any vegetation cover change. REMA could partner with local universities to engage their students and train them on ecological assessments and monitoring. This is a great way to collect ecological information, engage students in the park, and use NUWEP as an outdoor classroom.

1.1.6. Nyandungu Eco-Park Fauna

An inventory of mammals has not been completed and is recommended. Several species ranging from rodents to small carnivores, are very likely to occur based on their presence elsewhere in the city of Kigali. Examples are marsh mongoose, blotched genet and the striped weasel. There are reports of vervet monkeys occasionally visiting the site (Afrilandscapes *Per comm, Nov. 2021*). Likewise, it is recommended that baseline surveys of useful indicator species such as amphibians and butterflies are conducted. These baselines will allow for monitoring of changes in species assemblages over time.

At least 100 species of birds have been identified making NEP an interesting and exciting site for urban birdwatching by both Rwandan citizens/residents as well as foreign visitors. The list of birds is provided in the 2022 NUWEP Feasibility Report.

1.1.7. Ecosystem services (ES)

In high rainfall and hilly terrain areas such as Kigali, the GoR recognises the risk of increased runoff into the valleys and the need to protect wetlands to help mitigate flooding and support filtration. In addition to flood mitigation, Nyandungu wetland provides the following regulating services:

- **Air quality:** ecosystems like wetlands influence the local climate and air quality. For example, trees provide shade whilst forests influence rainfall and water availability both locally and regionally. Trees or other plants also play an important role in regulating air quality by removing pollutants from the atmosphere.
- **Carbon sequestration**: ecosystems regulate the global climate by storing greenhouse gases. For example, as trees and plants grow, they remove carbon dioxide from the atmosphere and effectively lock it away in their tissues. Wetlands are highly efficient at capturing atmospheric carbon.
- **Reducing stormwater runoff**: Ecosystems and living organisms, like wetlands, create buffers against natural disasters, such as flooding. They reduce damage from floods, storms, tsunamis, avalanches, landslides and droughts. Nyandungu wetland helps to absorb and filter run off.
- **Preventing soil erosion**: Vegetation cover prevents soil erosion and ensures soil fertility through natural biological processes such as nitrogen fixation. Soil erosion is a key factor in the process of land degradation, loss of soil fertility and desertification, and contributes to decreased productivity of downstream fisheries.
- **Filtering water:** Ecosystems such as wetlands filter effluents, decompose waste through the biological activity of microorganisms, and eliminate harmful pathogens.

- **Pollination:** Wetlands provide habitat for pollinators. Insects pollinate plants and trees which is essential for the development of fruits, vegetables and seeds. Animal pollination provided by insects as well as birds and bats are also supported at Nyandungu. In agro-ecosystems, pollinators are essential for orchard, horticultural and forage production as well as the production of seed for many root and fibre crops. Pollinators such as bees, birds and bats affect 35% of the world's crop production, increasing outputs of around 75% of the leading food crops worldwide.[4]

NEP also provides additional ES such as freshwater (a provisioning service), habitat for species and genetic diversity (supporting services) and recreation, tourism and aesthetic appreciation (cultural services)[5]. By restoring and maintaining Nyandungu wetland, the GoR aims to secure these vital ecosystem services.

1.2. Administrative, political and socio-economic context

1.2.1 Administrative context

The administration in Kigali follows an organisational hierarchy that includes province, district, sector and cell (Table 3).

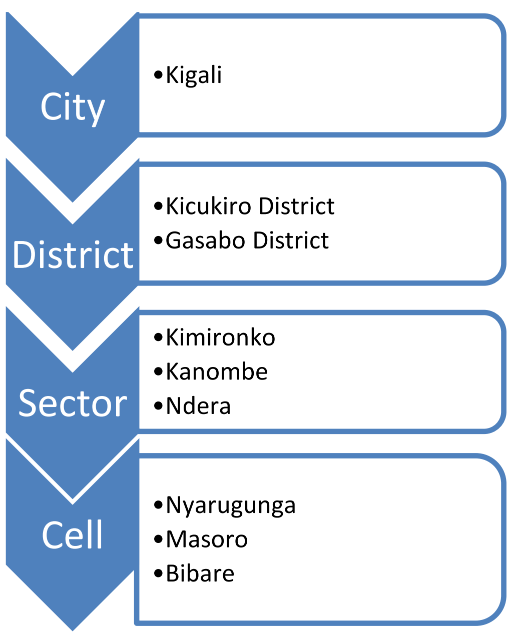


Table 3: Governance of administrative sub-divisions in Kigali (only those relevant to project context are shown).

[4] https://www.fao.org/ecosystem-services-biodiversity/background/regulating-services/en/
[5] https://www.fao.org/ecosystem-services-biodiversity/background/cultural-services/en/

1.3. Nyandungu Eco-Park Threats

CC considered threats to Nyandungu Eco-Park's ecological viability as well as its ability to function as a safe and successful urban recreation park that catalyses support for conservation.

1.3.1 Management

REMA does not intend to manage NEP given it is not part of their mandate. Therefore, a suitable managing entity needs to be identified and engaged with the adequate management rights and resources. The manager of NEP should have a good understanding of wetlands, as well as experience in protected area management and monitoring, urban park management, and business and commercial development.

1.3.2 Security

Security and uncontrolled access around and inside the park are significant threats to the success of NEP. The park is not fenced, providing uncontrolled access to people. For an urban park, NEP is relatively large, with five sectors. The lack of a fence, security personnel, proper lighting and security cameras is a significant risk to the operations of the park.

1.3.3 Pollution

NEP is in a valley surrounded by hills. It is highly prone to runoff and pollutants. For example, the northern boundary of NEP includes one of Kigali's industrial zones and there is a history of pollutants flowing down the hill into the wetland.

1.3.4 Road Expansion

The southern boundary of NEP runs along the Kigali-Kayonza Road (RN3). A walking path just inside the boundary of NEP runs parallel to the road. There is a plan to expand this road by adding more lanes. This could be done by cutting into NEP, which would seriously damage this section of the park. The alternative is for the RN3 Road to be expanded along the southern side of the road, which would not have an impact on NEP.

1.3.5 Community Dissatisfaction

The community living around NEP used to have free access to the wetland before its management and restoration under REMA. Community use historically included grass cutting, livestock grazing and small agriculture activity. Utilisation is now managed and, in some cases, has been halted. For example, cattle grazing is not allowed, but limited grass cutting is permitted. To avoid community dissatisfaction and resentment, it is important to find ways to engage the local community.

1.3.6 Lack of Funding

NUWEP has been supported through donor funding to date, which has resulted in its development and restoration. However, the budget has not been adequate to finish the initial plan. Elements such as a perimeter fence, traffic control, dust bins, solar lighting, and management have not been fully implemented. Sustainable and adequate funding is needed to support the continued development of NEP and its on-going management and maintenance.

1.4. Conservation Priorities

The core conservation priority for NEP is to restore, to the extent that this is possible, and maintain, the ecological function of the wetland. This was clearly articulated by REMA and the Ministry of Environment. This goal is consistent with Rwanda's conservation and wetland laws and its development strategies.

In addition, the GoR wants to:

- Provide accessible space for recreation and relaxation
- Encourage environmental education and research
- Educate Rwandans on the importance of the wetlands and the country's natural resources
- Provide a new attraction for tourists visiting Rwanda

To achieve these objectives, this section highlights conservation management requirements.

Qualified Management

To ensure the sustainable management of the park, the GoR plans to engage, a qualified and experienced park management partner that shares the government's vision for NEP.

Sustainable finance

One of the most important requirements is to ensure that NEP's operations are financially sustainable. The activities and operations to be implemented should be able to cover the operations, development and management costs.

Sustainable development

While the GoR wants to ensure the ecological function of the wetland, it also wants to provide recreation and open space opportunities for citizens and visitors to Rwanda. Striking a balance between conservation and suitable use is going to be very relevant for the sustainable development of NEP and ensuring that certain activities do not compromise the conservation of the wetland system.

Community engagement

NEP is shared by the administrative sectors of Nyarugunga (Kicukiro District) and Ndera (Gasabo District). NEP has been and still is used by the community living around the park. Ensuring that the neighbouring communities are adequately involved in NEP and benefit in some way is important to ensure their support.

Research and monitoring

The effective management of NEP requires on-going monitoring and evaluation of ecological function, user satisfaction, finances and stakeholder engagement. Monitoring and evaluation indicators are outlined in Section 7. Research and monitoring presents NEP management with the opportunity to engage with local universities and research institutions. Telling the restoration story of Nyandungu is important to inform other urban wetland restoration projects and to promote Rwanda's commitment to wetlands and urban green development.

SECTION 2: INSTITUTIONAL ARRANGEMENTS

2.1. Legal basis for wetlands in Rwanda

To ensure the protection of wetlands, various land uses have been delineated in the wetland legal framework of Rwanda. These different land uses are outlined in Table 4 (2050 Kigali Plan).

WETLAND PROTECTION AND USE AS REGULATED BY THE ORGANIC LAW (2005)	
Total Protection wetlands:	Protection as its natural state no agriculture and other activities will be allowed. There are no natural wetlands left within the Kigali City limits except for a few patches of Nyabarongo wetland
Conditional use wetlands:	Allows continuation of existing agricultural activities and conditional use for projects (farming, clay extraction, recreational, etc.) with proper EIA assessment (wetlands mainly along the rural areas). Final authorization of such projects would be granted by REMA
Unconditional use wetlands:	Will not require environmental impact assessment but no development activities will be allowed. These wetlands could be used for recreation, parks, fishing ponds, horticulture etc. (wetlands mainly along the urban areas of Muhima, Kimihurura, Kicukiro etc.)

Table 4: Rwanda (2005) law on wetland protection and land use.

The 2018 Law on the Environment reinforces the prioritisation of wetlands, biodiversity, the natural environment and ecosystems. Article 3 cites the precautionary principle, stating that "activities suspect to have negative impacts on the environment must not be implemented pending results of a scientific assessment ruling out the potentiality of such impacts." This principle will need to be utilised in the development and management of NEP.

The Kigali Development Plan outlines clear wetland protection (net zero loss) and restoration targets. Section 8.2 in the plan specifically mentions ecosystem rehabilitation focusing on urban wetlands (Nyandungu and Kimicanga). The sustainability framework (Tables 3 and 4) highlights wetland and highland conservation targets, and the plan includes specific spatial targets for wetland conservation.

Component	Key issues	Direction	Challenges	Recommendations
Environment Green & Efficient City	Nature areas: ● Urban areas prone to landslides and flooding ● Unplanned developments on steep slopes ● Deforestation ● **Encroachment of wetlands**	● Clearance of development in steep slopes and wetlands and acquire land for relocation ● **Restoration of steep slopes and wetlands** ● Afforestation	● Implementation mechanism and cost of land acquisition & relocation ● Cost for programming and implementation restitution of nature areas	● **Conserve all slopes above 30%** ● **Conserve all wetlands** ● Prepare redevelopment schemes to relocate people from steep slopes and wetlands ● Prepare strategies for **rehabilitation and for management of slopes, forests, and wetlands** ● Explore possibilities for sustainable exploitation of nature areas for economic gain creation of green jobs

Table 5: Updated sustainability framework for city of Kigali. (Bold added for emphasis. Kigali Development Plan).

2.2. Current institutional arrangements

NUWEP is currently managed by REMA under the guidance of the Ministry of the Environment. REMA does not anticipate managing NEP long-term and is therefore seeking a management partner. Given its status as a park and the potential for a public – private – partnership (PPP), the Rwanda Development Board (RDB) will be engaged in the identification of a management partner.

SECTION 3: MANAGEMENT ACTIONS AND COSTS

Based on the clear conservation targets and objectives for NEP (Figure 7), this section presents recommended management activities and operating and capital expenditure (CAPEX) budgets for NEP.

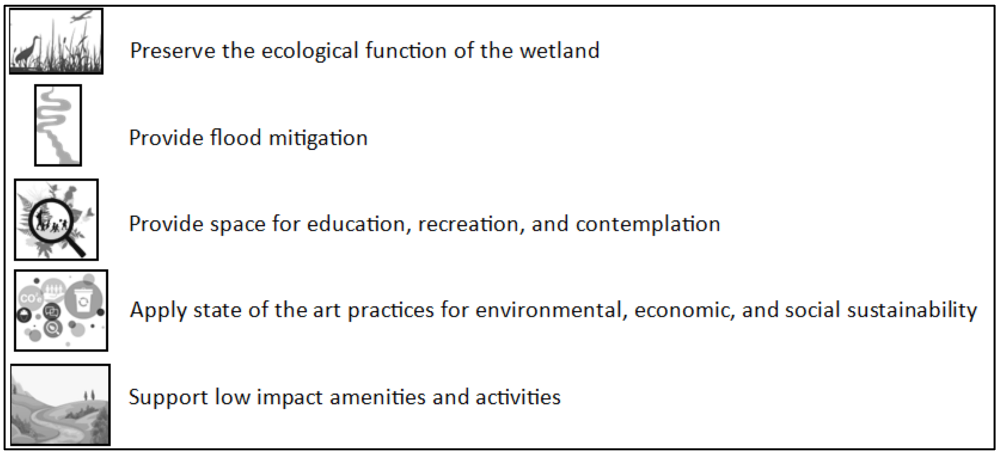


Figure 7: NEP conservation and development goals.

NEP has the potential to be an exceptional example of ecosystem restoration within an African urban environment for the primary purposes of stabilising ecosystem services (for the benefit and cost saving by city management) and providing a nature-based recreational activity area for urban residents (that additionally has potential to be a tourism attraction). While the restoration and rehabilitation work completed to date was donor-funded, ongoing management of the park needs to be self-funded.

The funding mechanism that will support the educational and recreational objectives of NEP is the creation of an urban park where users contribute to the park management costs through a conservation fee in exchange for access and usage. Balancing this need to generate revenue with usage that does not negatively impact the park is important and requires careful consideration of visitor carrying capacity, activities and management. In addition, ensuring that the management partner (Section 6) has the appropriate experience in wetland and park management, tourism and commercial development, and monitoring is important. It is in this context that when considering management, this CABP includes the following critical success factors:

1. successful restoration and maintenance of wetland function
2. creation of an appealing natural environment for Kigali residents and other visitors
3. development of appropriate infrastructure for attracting visitors
4. appointing a qualified management entity
5. effective and targeted marketing
6. ease of access
7. a viable business model
8. increasing understanding of the value of conservation and engagement with nature by visitors

The core components that will result in a successful conservation, visitor experience and management outcome are included in Figure 8.

NEP has the potential to be an exceptional example of ecosystem restoration within an African urban environment for the primary purposes of stabilising ecosystem services and providing a nature-based recreational activity area for urban residents.

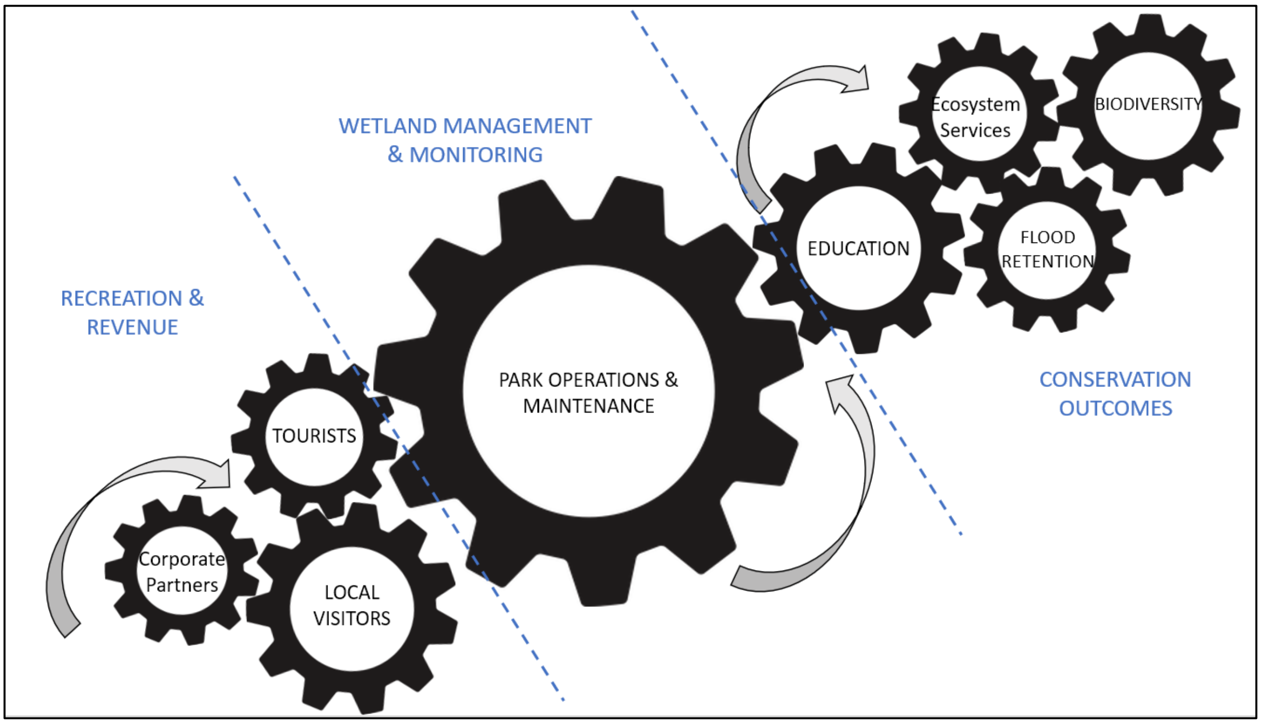
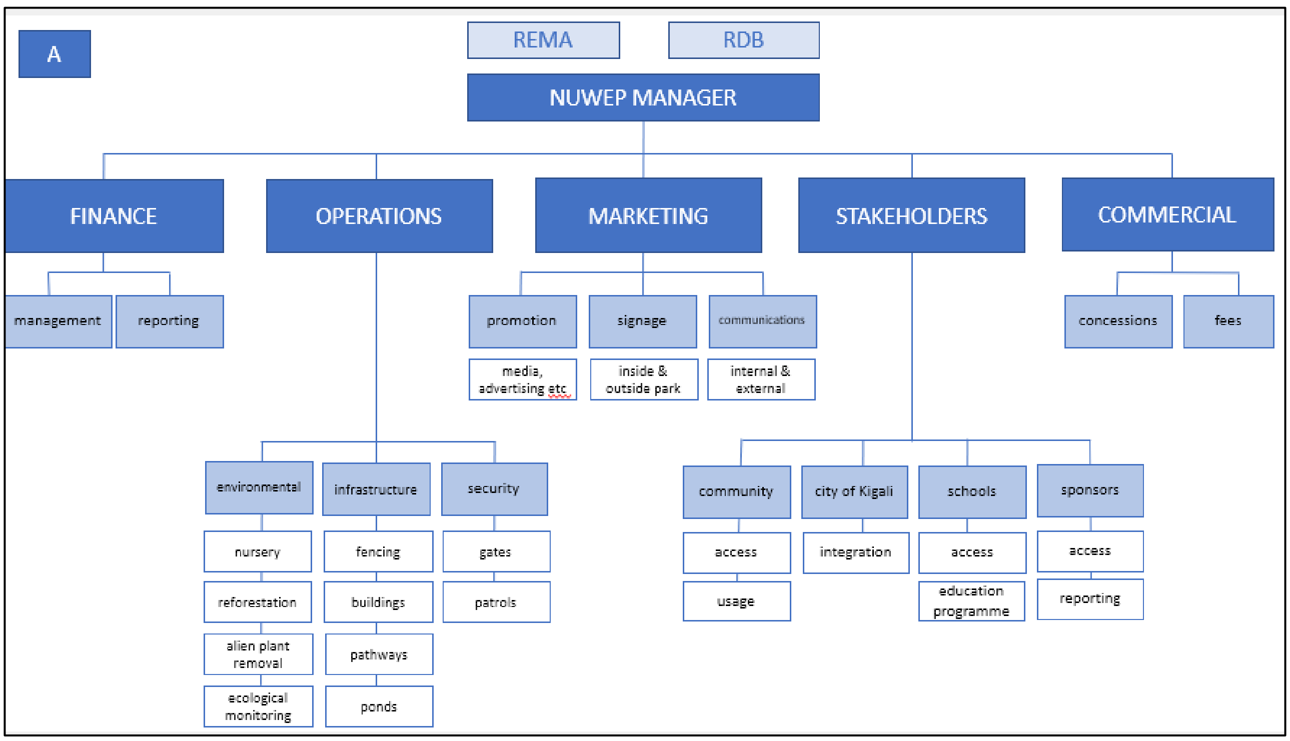


Figure 8: The core components to ensuring a sustainable and successful NEP.

Five main programme areas will be required to ensure the successful development, management and operations of NEP (Figure 9). The organogram can be simplified, as depicted in Figure 9, B.



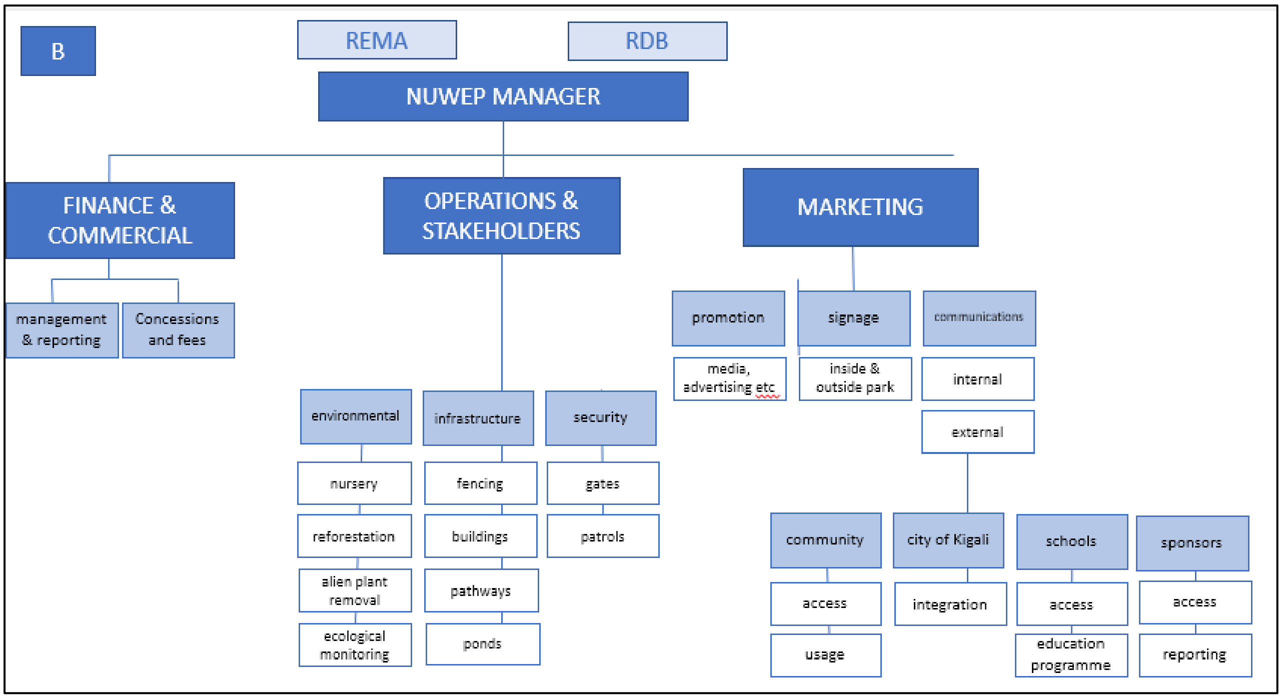


Figure 9: A. NEP management organigram. Various departments might be consolidated as depicted in B. The management partner will want to consider the best alignment of staffing along with staffing requirements.

For each management component, CC considered the desired objectives and the strategy, infrastructure and resources needed.

- **Strategy:** A summary of the optimum strategy for achieving these objectives in the most efficient and effective manner including specifically highlighting how infrastructure, assets and HR resources (and hence costs) can be shared across the different thematic activities.
- **Infrastructure & Assets:** A summary of the infrastructure and assets required to deliver the strategy for NEP.
- **Human Resources (HR):** A quantitative summary of the human resources required to deliver each strategy.

3.1. NEP Management

3.1.1. Overall Management

i) Objectives

- Ensure management coordination: in the field, with key stakeholders, and with REMA
- Maintain staff morale, clarity of roles, clear TORs and clear performance standards
- Generate enough revenue through the park's activities to sustain maintenance and daily operations
- Ensure safety and security for visitors
- Ensure high quality visitor experience
- Ensure effective relations with concession holders in NEP
- Maintain positive relations with stakeholders, local leaders and the local communities

ii) Strategy
- Establish effective management systems
- Establish effective monitoring systems, reporting schedules and adaptive management procedures
- Create a financial system that ensures accurate and transparent financial management and revenue collection
- Create a communication plan that helps promote NEP, keeps stakeholders up to date and effectively engages Rwandans in wetland conservation
- Reduce risk by developing a risk mitigation strategy that is updated annually

iii) Infrastructure & Assets
- Office
- Computers
- Financial software

iv) Human Resources
- NEP manager
- NEP finance officer
- NEP Operations manager

3.1.2. Field Management

i) Objectives
- Manage NUEWP to preserve the ecological function of the wetland system, provide flood mitigation, provide safe recreational opportunities for visitors, develop low impact amenities for visitors and apply state of the art development.
- Restore to the extent possible native plant species and remove exotic species

ii) Strategy
- Assess the roles needed, create hiring profiles and hire adequate and skilled staff
- Ensure management systems are in place to provide effective management of NEP
- Determine the equipment and assets needed for management of NEP

iii) Infrastructure & Assets
- Uniforms
- Trail maintenance equipment
- Wetland management equipment

iv) Human Resources
- Security staff
- Maintenance staff
- Management staff
- Program staff

3.1.3. Community Engagement

v) Objectives
- Effectively involve the community around the park in NEP activities and management to the extent this is possible. Provide limited and managed access, provide alternatives to the natural resources in NEP as appropriate and as feasible, and keep community up to date on activities in the park

vi) Strategy
- Designate community representatives in agreement with the Cell Executive Secretary and the Umudugudu Leaders
- Appoint a community liaison officer to lead community management and coordination

vii) Infrastructure & Assets
- Establish a suitable community meeting space

viii) Human Resources
- Cell
- Umudugudu
- NEP community officer

3.1.4. Research & Monitoring

i) Objectives
- Monitor the ecological recovery in NEP (vegetation, fauna, water quality) to inform management
- Monitor park use to inform management and provision of services and amenities to visitors (number of visitors, preferred sector, favourite activities, amount spent, etc)
- Link ecological restoration results to communication strategy to promote restoration and share lessons learned
- Engage local universities, schools and students where feasible in ecological monitoring

ii) Strategy
- Define monitoring assessment metrics
- Create the monitoring tools
- Regularly analyse the data and produce assessment reports to inform management

iii) Infrastructure & Assets
- Monitoring devices, such as handheld smart phones
- Plant, mammal and bird identification guides

iv) Human Resources.
- Monitoring officer
- NEP Management
- REMA

3.1.5 Stakeholder and Communication Management

i) Objectives
- Maintain positive relationships with key stakeholders
- Ensure stakeholders are well informed and up to date with NEP activities, events and developments

ii) Strategy
- Develop an annual communication strategy and annual communication workplan
- Develop a professional and high-quality website and social media platform and provide regular updates to keep stakeholders informed
- Develop a newsletter that is shared with key stakeholders, sponsors and NEP members
- Provide regular progress reports to key stakeholders
- Engage media in special events and promotional activities

iii) Infrastructure & Assets
- Computer for communication officer
- Office equipment

iv) Human Resources
- Communications officer
- NEP community officer
- REMA
- RDB
- NEP Management

3.1.5 Education Program

i) Objectives
- Ensure that visitors of NEP are well informed about the wetland system, the conservation values and Rwanda's natural resources
- Provide high quality educational opportunities for Rwandans and visitors
- Provide structured educational programmes to students

ii) Strategy
- Develop the information centre to provide high quality educational resources
- Develop outdoor sign boards that provide information about NEP, the wetland system and Rwanda's natural resources
- Develop an educational programme that is offered to Rwandan students

iii) Infrastructure & Assets
- Well-equipped and designed information centre
- Signs
- Educational and outdoor interpretation signs
- Outdoor seating area where students can listen to presentations on NEP

iv) Human Resources
- Education Officer
- NEP Management

3.2. Management Costs

Operational Costs

To effectively manage NEP and develop relevant infrastructure, security and high-quality visitor experiences, CC considered operating and capital expenditure costs over a ten-year period. The detailed budget is included in Annex II. We used an annual inflation of .03%. The largest management cost is staff (54%.) The management partner will likely adjust this management budget; however, this gives REMA a good sense of NEP operational costs and when considering management partners, this will help guide REMA in assessing management proposals.

CC considered the following management needs, which are further broken down in Figure 10:

1. Staffing (i.e., security, management, outreach, communication, education and monitoring) (54% of the budget)
2. Business management (i.e., website, corporate engagement)
3. Civic engagement (i.e., community and district meetings)
4. Field support (i.e., fence maintenance, trainings, uniforms for certain staff)
5. Park and office management (i.e., ecological monitoring, restoration)
6. Fleet management (i.e., vehicle and equipment management)

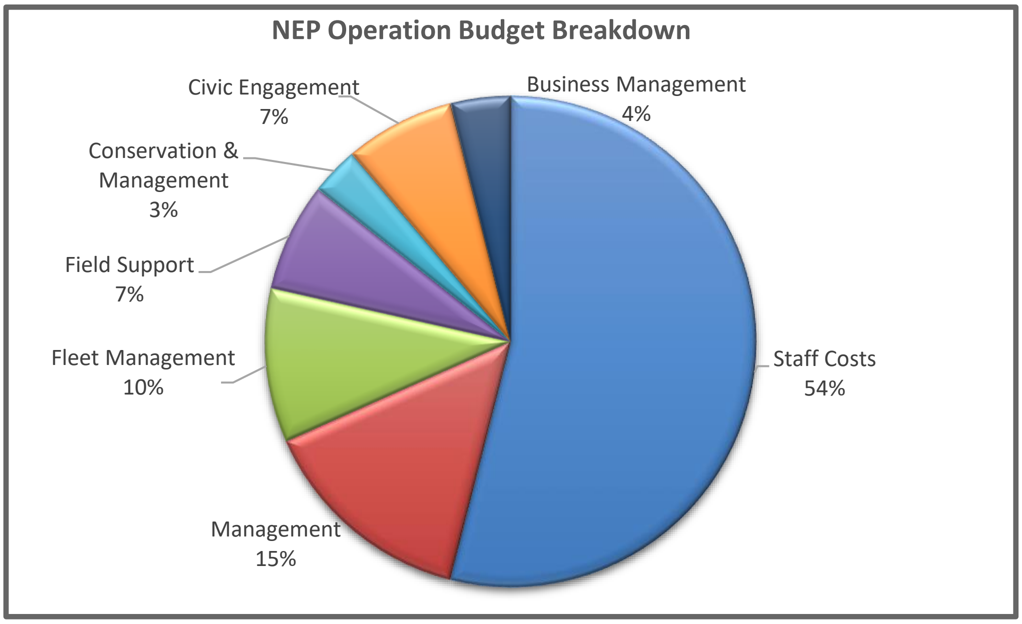


Figure 10: Total NEP operational budget covering seven management areas (with input from Afrilandscapes). Note that conservation and management costs (3%) are also covered under other budget items, such as staffing, management, field support and fleet management.

Figure 11 depicts the operating costs over a ten-year period. Note that in year 4 the budget decreases significantly because in year 3 a perimeter fence is erected around Sectors 3 and 4, which would reduce the security staff budget significantly.

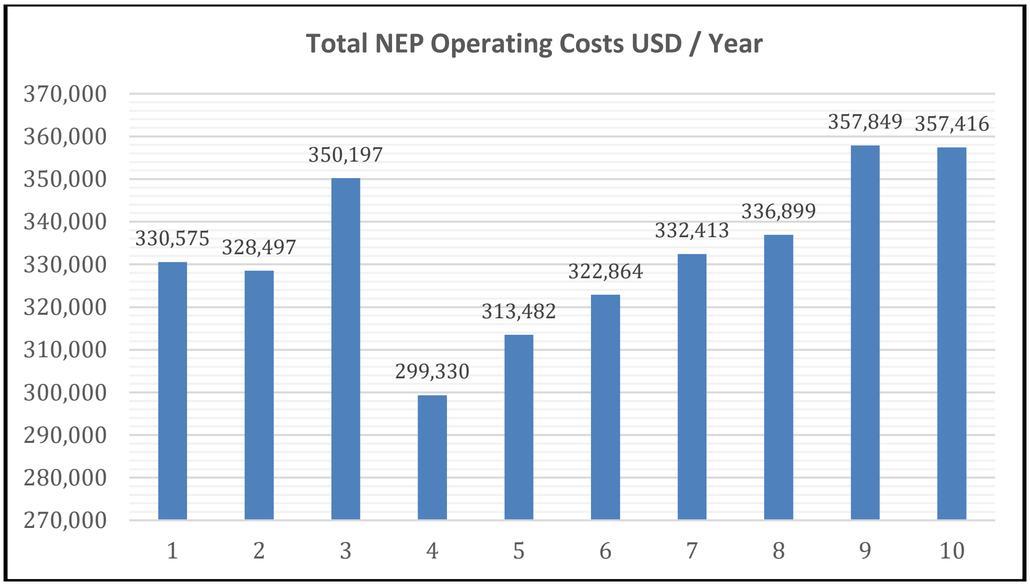


Figure 11: The NEP operating costs over a ten-year period, with an annual inflation rate of .03%.

Capital Expenditure

CC considered the following categories for capital expenditure (CAPEX):

- Management Infrastructure
- Conservation Infrastructure
- Transport
- Communications
- Office and IT Equipment
- Field Equipment
- Roads, bridges, trails and water culverts
- Tourism and Enterprise

Figure 12 depicts the capital costs in year one. The largest percentage of the costs are for conservation infrastructure, followed by management infrastructure and then transport. Figure 13 depicts the capital costs over a ten-year period. This includes replacement cost of relevant infrastructure, such as a vehicle in year 5, and field equipment. Note that while this CABP was being finalised, the GoR was considering a tender for the fence. If this were to be completed prior to engaging a management partner and the costs covered, this would significantly reduce the CABEX budget by 300,000 USD in year 2.

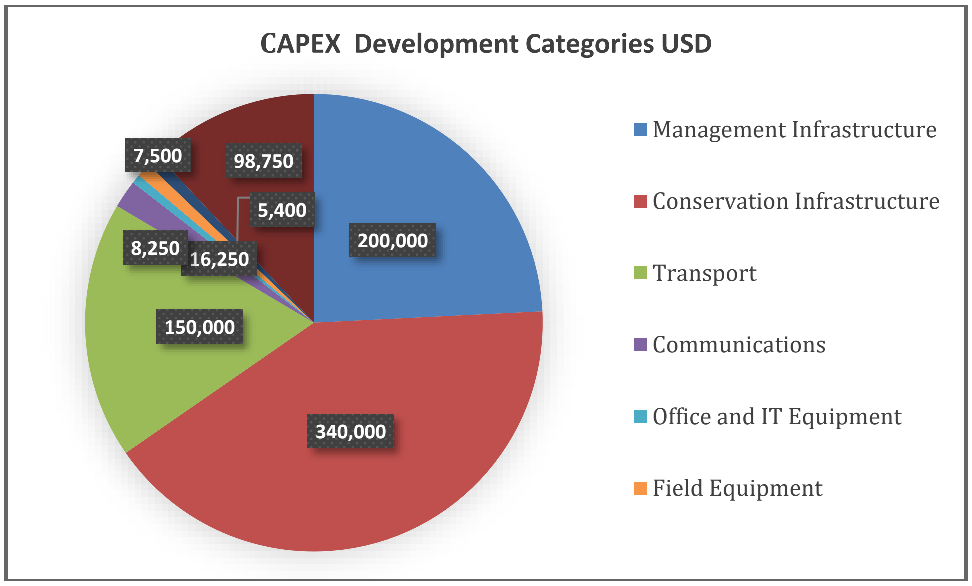


Figure 12: NEP's CAPEX Breakdown (USD).

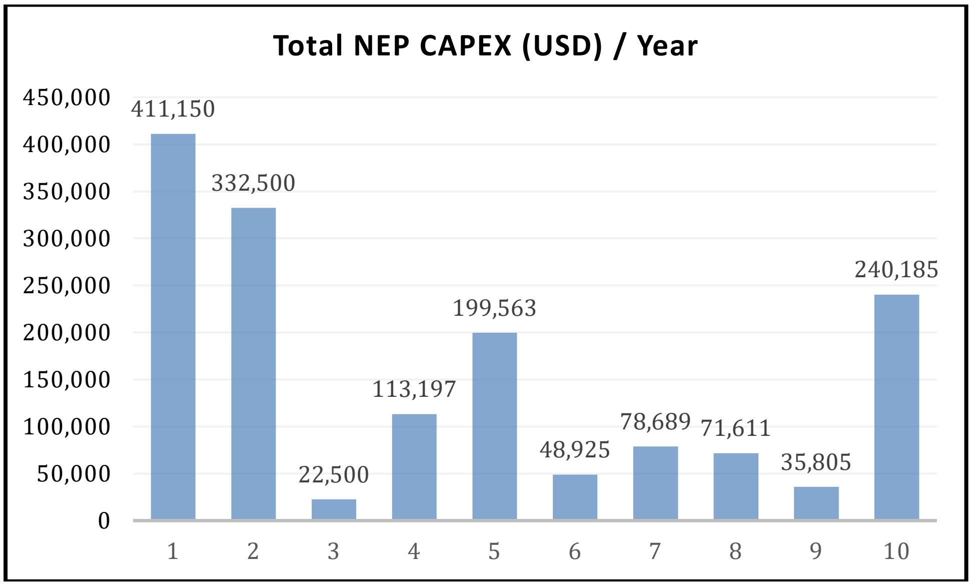


Figure 13: NEP CAPEX costs over a ten-year period (USD).

Annex II includes the combined OPEX and CAPEX budgets over a ten-year period.

SECTION 4: COMMERCIAL REVENUE-GENERATING OPPORTUNITIES

This section provides a broad overview of the potential revenue streams that might be developed to support NEP. Conservation Capital generally examines activities opportunities that meet one or more of the following criteria (Figure 14):

- **Generating finance**: activities that generate a direct monetary return and can support the management of the protected area

- **Promote Environmental Education**: creating incentive to encourage research and education programme in the park

- **Optimising sustainability**: designing or selecting activities that will help maintain the good status of the newly implemented infrastructure

- **Offsetting encroachment:** finding ways to monetize the services provided by the ark as well as offsetting the pollution created by the industrial activities around the park

- **Stimulating engagement:** enabling direct access to, and the enjoyment and appreciation of - nature and wetlands in Kigali in ways which foster an increased interest to Rwanda's natural assets.

- **Encourage safe recreation:** offering a new safe, clean, and eco-friendly environment for Kigali residents and tourists to rest, resource and relax in an urban and fast-developing city

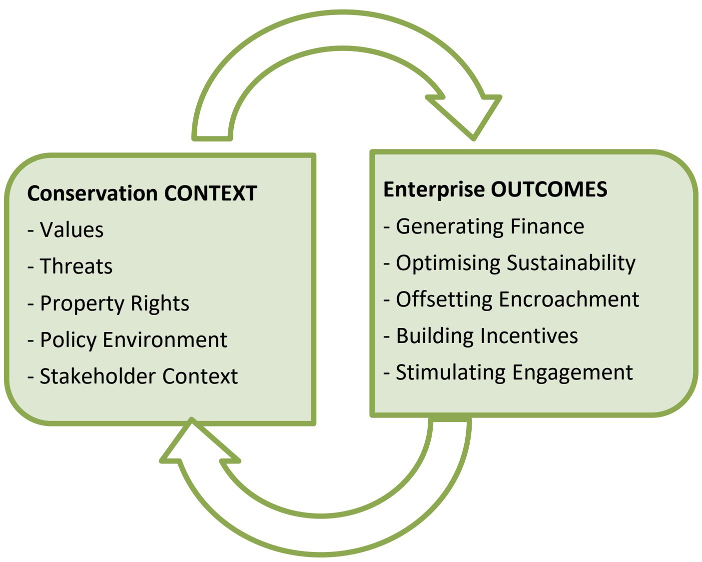


Figure 14: Considerations for NEP commercial opportunity.

Revenue Option 1: Utilisation by Visitors

Development of NEP was initiated in 2017 through donor funding. The development followed a 2017 design plan which centres around the wetland system. Development stopped in 2021 because of lack of funding. The existing infrastructure is high quality and makes NEP ready for use pending the engagement of an experienced management partner, adequate security, and a management budget.

The infrastructure developed includes[6]:

- stone pathways for walking
- stone pathways for biking
- restrooms
- gates
- parking areas
- a restaurant
- benches
- solar lamps

Some of the current challenges NEP faces for visitor use:

- Lack of a perimeter fence exposes all sectors to encroachment
- The proximity to the Free Economic Zone and its industry creates pollutants and noise pollution
- The relatively small size of the park requires adequate visitor management
- The eagerness of Kigali residents to visit the park risks high use and potential negative impacts
- Lack of health and first aid stations in the park
- Lack of proper waste disposal facilities
- Lack of management funding

Striking a balance between protecting the recovering wetland system and providing access to visitors is the greatest challenge in NEP. Generating revenue through user fees is one of the best ways to generate revenue for NEP and to create awareness and support for nature conservation in Rwanda. Table 6 outlines different user fees that might be charged at NEP, the advantages, disadvantages, whether CAPEX is required and the return on investment (ROI).

	Scenarios	Advantages	Disadvantages	Immediate Opportunity	CAPEX Required	ROI
I	**Conservation Fee** A conservation fee suggests that users are paying for conservation, an entrance fee means they are paying to enter a public space	• Revenue generated can be used for management and conservation activities • Pricing can limit numbers of users, which will help to maintain the ecological carrying capacity • A perception of nature's value can be inspired in the public • No perception is created that access is restricted to only a few by 'entrance fees'	• Pricing will impact peoples' choice and frequency of visiting this site • A high conservation fee will raise concerns about equitable access	Immediate	Low (point of sale system; ideally combined with online booking portal; will require employees at relevant gates)	High

[6] The infrastructure is outlined and detailed in the 2022 Updated NUWEP Feasibility Report.

II	Filming and photography fees (for commercial activities like filming of TV advertisements)	• Revenue generated can be used for management and conservation activities • Helps to promote NEP and Rwanda	• If not managed well can disrupt users' visits	Immediate	Low	High
III	Restaurants, coffee bar / lounge	• Concession fee generates revenue for management and conservation activities • Could provide local jobs • Will result in a concentration of many users in one central space; allowing for an effective zonation and separation of different kinds of users and a management plan geared to this	• Requires good waste management • Must be compliant with Rwanda's commitment to no plastic • If expensive, will create animosity between citizens and managers • Attracts people to a certain part of the park which could create a negative impact if not well managed	Immediate	Low	High
IV	Activity Concessions (Bike Rental, for example)	• Concession fee generates revenue for management and conservation activities • Could provide local jobs	• Requires good waste management • Must be compliant with Rwanda's commitment to no plastic • If expensive, will create animosity • Attracts people to a certain part of the park which could create a negative impact	Immediate	Low	High
V	Additional activities such as temporary or permanent exhibitions or events, such as Fun Runs, annual birding count, or a fair for International Wetland Day	• Revenue generated can be used for management and conservation activities • Diversified activities can attract a diversified visitor spectrum • Exhibitions and events can be tailored to amplify the educational and other purposes of NEP	• May require additional capital investment • Large events could result in unnecessary disruption to the environment or other users, and restrict access for the public	Long Term	Potentially High	Potentially High

Table 6: Different utilisation activities for NEP.

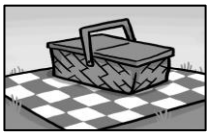
4.1. Activities Summary

Recommended Activities

To attract visitors, several activities will need to be permitted in NEP. Based on the goals of the GoR and the site itself, Conservation Capital assessed different activities that could take place in NEP these are described along with what is needed to support such activities. The activities have been grouped into six categories:

1. Sport and health improvement

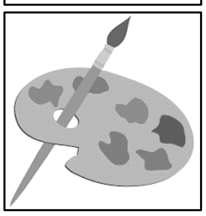
2. Nature & Contemplation



3. Social Recreations

4. Events

5. Learning



6. Other

Figure 15: Six activity categories considered for NEP.

Each category has a set of activities that are described in the following sections of the report and Table 7 provides a description of the activity and what infrastructure might be required at this time to support the activity in NEP. These activities collectively offer a diversity of experiences for visitors, which will help to attract a mix of users.

#	Category	Activity	Infrastructure		Comment	Specific Revenue Generator?
			Already present?	Additional needed?		
1	Sport and health improvement	Walking	Yes - Paved paths (5.4km)	No		Free - Covered by conservation fee
		Running	Yes - Paved paths (5.4km)	No	Restricted to paved pathways	Free - Covered by conservation fee
		Cycling	Yes - Paved paths (5.4km)	No	Restricted to paved pathways	Free - Covered by conservation fee
		Fitness	No	Yes. Kinetic fitness stations (Photo 1)		Free - Covered by conservation fee
		Social sport / Exercise (e.g., frisbee)	Yes – Lawns	No	Zonation required	Free - Covered by conservation fee
		Bike hire	No	Yes	Needs Private Sector Concessionaire	Yes
		Guided walks (Birding/Botany)	No	No	Needs trained community guides	Yes
2	Nature & Contemplation	Bird watching	Yes. Paved & unpaved paths	Yes. Additional unpaved paths, hide, wetland boardwalk		Free - Covered by conservation fee
		Reading	Yes. Benches; lawns	No		Free - Covered by conservation fee
		Meditation	No	No		Free - Covered by conservation fee
		Yoga / Tai Chi	No	No. Although an open-sided pagoda / platform might be used for yoga sessions as well as a corporate meeting space		Free - Covered by conservation fee
		Prayer	Yes. Pope's Garden	No. Although worth consulting with Catholic Church to obtain their views		Free - Covered by conservation fee
3	Social Recreation	Socialising	Yes. Picnic space; restaurant	No		Free - Covered by conservation fee
		Picnic	Yes. Benches	Yes. Tables	Zonation required; will need to prohibit bringing in/consumption of own alcohol	Free - Covered by conservation fee
		Instagram selfies	No	Yes. 'Photo frame' celebrating wetlands, hashtags, etc.		Free - Covered by conservation fee

		Restaurant	Yes– Bourbon (temporarily)	Need to consider satellite kiosks for snacks; or picnic baskets/ takeaways		Yes
		Cocktail bar/Lounge	No	Yes. Attached to restaurant for sundowners / cocktails		Yes
		Kids playing	No	Yes. Nature-themed playground	Zonation required	Depends on development
4	Events	Fun runs / Park runs	Yes - Paved paths	No		Free - Covered by conservation fee
		Corporate meetings / Team building	No	Yes. Open sided pagoda as mentioned above		Yes - Venue hire
		Art installation	No	Yes. Open sided pagoda as mentioned above in Category 2		Free - Covered by conservation fee
5	Learning	Wetland	No	Yes. Wetland boardwalk; interpretation centre & signage	Critical if one of the primary REMA objectives is to be met	Free - Covered by conservation fee
		Ethnobotany	Yes - Medicinal Garden	No		Free - Covered by conservation fee
		Rwanda's National Parks	No	Yes. Interpretive centre with VR experience and expo display	Requires capital investor; potentially RDB does this as part of The Tembera campaign	Depends on development
		Research site	No	No		Free
		Lectures / Enviro functions	No	Yes. Open sided pagoda as mentioned in Categories 2 and 4; or a type of amphitheatre		Depends on needs
6	Other	Nature-themed office space	No	Yes	Requires market research and a capital investor	Potentially
		Art Exhibits	No	No	Clarity required around revenue model	Yes
		Zip Line	No	Yes	Requires market analysis; business plan	Potentially

Table 7: Suitable activities for NEP.

Activities not Recommended

Given NEP'S size and the vision and objectives set forth by the GoR, Conservation Capital vetted the following activities and considers then not suitable at this time.

Activity	Explanation
ATV Riding	Loud, ecologically degrading (especially in the wetland area), polluting, dangerous, and will compromise the experience of other NEP visitors
Kayak / Canoe	The water bodies are too small in NEP
Weddings	The park is too small for large weddings. This will compromise the experience of others. Small weddings might be considered in the future if well managed or users separated spatially. Guidelines will be required and suitable zoning if this is to be considered
Large Private Events	The park is too small for large events. This will compromise the experience of others. Small events might be considered in the future if well managed or if users are separated spatially. Guidelines will be required and suitable zoning if this is to be considered
Fishing	Not recommended as NEP lacks adequate fish resources
Horseback riding	Maintaining horses in NEP is expensive, there is not adequate space for stables or trails for riding
Zoo	Not consistent with the vision of NEP, requires significant and on-going veterinary skills and costs
Large mammal restoration	Requires veterinary team, fencing, and management, which is expensive
Organised sport	Organised sports include soccer, volleyball etc.… Given space limitations at this stage, putting in a football pitch or a volleyball court is not recommended; this does not preclude visitors from playing football, frisbee etc.… although zonation will be required to ensure there is no adverse impact on other users
Hotel/lodge development	Not recommended: too close to industrial zone, has the noise of the airport, and will undoubtedly be uncompetitive with any such development at Gikondo or the other wetlands due for rehabilitation in the more central tourism part of Kigali
Dog Walking	Given the anticipated number of visitors, natural setting and wetland, dog walking is not recommended. In the future, with proper management and guidelines, dog walking may be considered in Sector 5 with leashes, poop-and-scoop policies and on weekdays when there are fewer visitors

Table 8: Activities judged unsuitable for NEP at this time.

4.2. Sport and Health Improvement

4.2.1. Activity Breakdown

Exercise is important to maintain a healthy body and mind. Exercising outdoors is known to promote good health at a higher and faster rate. Visiting a park for outdoors activities or just enjoying nature has been shown to lower stress, blood pressure and heart rate, while encouraging physical activity and buoying mood and mental health. Some research even suggests that green space is associated with a lower risk of developing psychiatric disorders — all findings that doctors are increasingly taking seriously and relaying to their patients. In an article about the factors associated with changes in subjective well-being immediately after urban park visit[7], the authors show that spending just 20 minutes in a park, even if one does not exercise, is enough to improve well-being.

During COVID, the demand to exercise in greener and outdoor spaces[8] exploded around the world and in Rwanda. More people are looking for parks for their regular sport activities or are interested in trying new ones. Another study[9] shows that parks attract and encourage low-income and minority communities to practise some activities freely because of the sense of belonging attached to a "public park."

The GoR recognises the value of urban parks for its citizens and has outlined a clear strategy to increase the number of parks in Kigali to reach the following target:

- One neighbourhood park per 2,000-20,000 people (1 hectare)
- One sub-city park per planning area (6 hectare)
- One sports field per planning area (1.5 hectare)

NEP has the potential to provide outdoor space for citizens and visitors to recreate and spend time in nature.

4.2.2. Infrastructure Available

Different types of exercise are feasible in NEP with the existence of the paved paths for walking, running, roller blading and cycling. Table 9 indicates the number of paths in the park. The park also has several lawns that could be used for group or social activities (ZUMBA, yoga, warm up, frisbee, etc.) Most of the paved paths are in Sectors 3, 4 and 5. Sectors 3, 4 and 5 also have some unpaved pedestrian pathways that could be used for walking and running.

The envisioned partnership between REMA and a local cycling start-up such as Kigali Bikes would give the visitors the opportunity to rent bikes and fully experience the park. Additional infrastructure might include fitness stations (Photo 1.)

Photo 1: Fitness stations a public park in Johannesburg, South Africa.

[7] Hon K. Yuen & Gavin R. Jenkins (2020) Factors associated with changes in subjective well-being immediately after urban park visit, International Journal of Environmental Health Research, 30:2, 134-145, DOI: 10.1080/09603123.2019.1577368
[8] https://fortune.com/2021/12/19/green-exercise-hiking-cycling-outdoor-workouts/
[9] https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1805017/

NEP has the potential to provide outdoor space for citizens and visitors to recreate and spend time in nature.

Trails and paths developed in NEP (excluding the paved roads)					
Sectors 3 & 4	**Existing (m)**				
Cyclist (paved)	3,601	686	168	0	0
Pedestrian (paved)	3,144	691	161	0	0
Pedestrian (unpaved)	88	298	136	131	102

Table 9: Trails and paths in NEP Sectors 3 and 4.

4.2.3. Assessment of opportunity in local context

In Rwanda, physical activities are part of the daily life of residents. Several programmes and events are already held to promote outdoor activities such as:

- Car-Free Day (Photo 2): A national initiative launched by the GoR to encourage non-motorised transportation and fight air pollution, and to promote healthy living through exercises such as running, walking, hiking, soccer, roller skating, street music and dancing
- Marathon: Kigali hosts the full and half International Peace Marathon
- Umuganda[10]: Even though Umuganda is more of a social initiative, it also encourages citizens to be outside and to move around while performing community work

Photo 2: Kigali Car Free day.

[10] A national holiday in Rwanda that takes place on the last Saturday of every month for community work from 8:00 - 11:00 am. Participation is required by law. The program, re-established in 2009, has resulted in improvement in the cleanliness of Rwanda.

Appraisal criteria	Score (1 =low / 5=high) For risk categories: (1 = high risk / 5 = low)	Description
Citizen Interest	5	• Since COVID, citizens are looking for more venues to practise sport outside • Existing exercise activities in Kigali demonstrate a very high level of interest • Strong demand for outdoor, accessible exercise venues
Risk[11] to wetland conservation	3	• There is a risk that visitors go beyond the designated paths and areas in NEP and cause damage to the natural environment • Many visitors for sport could disturb the ecosystem
Financial / revenue potential	3	• Significant opportunity to generate revenue from user / conservation fees • Potential to have additional revenue from athletic events, such as fun runs, and other amenities such as restaurants
Market demand	4	• Rwanda's health and sports culture, the participation level in similar activities and venue, the demand between medium and high • Tourists and business travellers visiting Kigali on weekends may be interested in visiting NEP for a run, walk or birding if promoted by the hotels and travel industry
Risks and external Threats	4	• The designated pathways for exercises minimise the risk to the wetland • Off path activities could have a negative impact • Heavy rains could affect the designated paths and areas • Year one will experience high use. The challenge is maintaining the interest and use in NEP by citizens
Overall assessment	**3.8**	***Exercise is a viable activity for NEP, and the infrastructure is in place to launch it. The designated pathways for exercises minimise the risk to the wetland but there is a need to control the traffic inside the park.***

Table 10: Activity assessment – Sport and Health Improvement.

4.3. Nature & Contemplation

4.3.1. Activities Breakdown

NEP offers a range of activities that would allow visitors to connect with nature and/or relax. Since the restoration of the park, at least 100 bird species have been seen at NEP - making it an excellent and very accessible location for bird watchers. The park itself is a great place for nature lovers to enjoy the trees, ponds and quiet space. As Kigali continues to develop, citizens are seeking open and green spaces to relax, read, meditate or pray.

[11] The risks are weighted contrary to their score as the lower the risk the better. E.g., if score is 1, the value for the overall calculation is 5, 2 is 4, 3 is 3. 4 is 2 and 5 is 1.

According to birding experts in Kigali:

1. NEP is good for a general introduction to Rwandan birds.
2. There are no unusual birds per se (but olive backs likely occur, and as the wetland matures and the papyrus coverage increases, papyrus gonoleks might colonise, and the wetland could be managed to achieve this.)
3. In a similar way to how Umusambi Village is currently used in terms of tourism, NEP could be used as a 'journey breaker' and/or a bird introduction en route to Akagera National Park; and, for short term visitors to Kigali looking for a morning birding outing (day trips to Akagera last too long for birders; Nyungwe is not possible to visit in a day.)
4. Ruhondo Pond (Sector 4) is the best location for a photographic bird hide.
5. There is an opportunity to train local bird guides and create jobs (this is common practice in South Africa and other parts of Africa where there is a big push on birder friendly accommodation and local guides.)

4.3.2. Infrastructure Available

For enjoying nature, NEP has different types of venues that are suitable:

- Lawns
- Ponds
- Woodlands
- Benches
- Pope's Garden
- Medicinal Garden
- Additional unpaved pathways could be added to enhance the experience for those looking to enjoy nature
- A boardwalk crossing an appropriate part of the wetland should be added (with interpretation signs) as an essential part of achieving NEP's educational goals around the importance of wetlands; such a boardwalk also offers another dimension to a multitude of different users including birders

4.3.3. Assessment of opportunity in local context

Kigali is a densely populated place and citizens are seeking open green space. Rwandan citizens are aware of nature and want to interact with it, and while Rwanda supports several spectacular parks, many people are unable to afford travel to these destinations. NEP is accessible and local. Kigali has an active community of bird watchers and is visited by tourists who would be keen to bird in the park.

Christianity is the main religion in Rwanda. The Pope's Garden, located in Sector 3, has historical significance and may attract people for prayer or just to visit.

Another activity that is becoming popular in Kigali is yoga. There is also growing interest in mindfulness, reducing stress and finding work/life balance. NEP offers a good setting for such activities. A very simple platform could be erected for yoga, meditation, or Thai Chi classes.

Appraisal criteria	Score (1 =low / 5=high) For risk categories: (1 = high risk / 5 = low)	Description
Citizen Interest	3	• High interest based on citizens' visits to other green spaces like Sunday Park (which would receive more visitors if the entry price was lower) • High interest given population density in Kigali and demand for green space for family fun, contemplation
Risk to wetland conservation	4	• Relatively quiet activities that will not disrupt the daily activities of NEP so long as people stick to the paths and designated walkways
Financial / revenue potential	2	• Significant opportunity to generate revenue from user / conservation fees • Potential for additional revenue from other amenities such as restaurants • Potential for revenue from specific activities such as company retreats, yoga classes
Market demand	3	• Demand evaluated at 'medium' based on current participation rate to similar activities and consultation with stakeholders
Risks and external Threats	4	• The designated pathways for exercises minimise the risk to the wetland • Off path activities could have a minor (and manageable) negative impact • Heavy rains could affect the designated paths and areas • Year one will experience high use. The challenge is maintaining the interest and use in NEP by citizens
Overall assessment	**3.2**	***Nature experience and contemplation is a viable activity for NEP with most of the infrastructure ready to launch it. Use of trails, platforms and other designated areas will limit any impact. One additional infrastructure development recommended is a timber walkway and interpretation signs crossing the wetland as a means of exposing visitors to a different habitat and to enhance education around wetlands.***

Table 11: Activity assessment – Nature Experience and Contemplation

4.4. Social Recreation

4.4.1. Activities Breakdown

NEP is a perfect venue for social recreation, as it is centrally located and has a diversity of meeting options. From walking, running, group sports, picnics and visiting the restaurant, to sitting on a bench or relaxing on a blanket on the lawn - NEP has a variety of places and activities that could foster communication, engagement, and interaction. There are some simple and fun infrastructure improvements that could be made to attract social interaction, such as a photo frame (Photo 3). Designed well, infrastructure can also inspire appreciation for nature and create awareness of NEP. For example, the photo frame could include NEP's website and promote wetlands.

NEP is a perfect venue for social recreation, as it is centrally located and has a diversity of meeting options.

Photo 3: Left, sample of a Nyandungu selfie-frame; Right: selfie-frame at Zoo Lake, Johannesburg, South Africa.

4.4.2. Infrastructure Available

- Restaurant
- Lawns
- Pathways
- Ponds
- Benches

4.4.3. Assessment of opportunity in local context

Kigali citizens are seeking open spaces where they can meet friends and family members to catch up, walk and talk. This space is currently limited in Kigali, making NEP unique and potentially in high demand.

Appraisal criteria	Score (1 =low / 5=high) For risk categories: (1 = high risk / 5 = low)	Description
Citizen interest	4	• High interest based on citizen visits to other green spaces like Fazenda Sengha, Sunday Park or Umusambi Village for picnics and other activities • High interest given the lack of green spaces in Kigali for social recreation and meeting friends and family
Risk to Wetland conservation	2	• Relatively quiet activities that will not disrupt the daily activities of NEP • Risk of littering • Risk of veering off paths is relatively low
Financial / Revenue potential	3	• Significant opportunity to generate revenue from user / conservation fees • Potential for additional revenue from athletic events, such as fun runs, and other amenities such as restaurants
Market demand	3	• Demand evaluated at 'medium' based on current participation rate to similar activities
Risks and External Threats	1	• Heavy rains could affect the designated paths and areas
Overall assessment	**3.6**	***Social gathering and recreation are viable activities for NEP. Most of the infrastructure is in place to launch it. This activity set has limited risks to the conservation of NEP.***

Table 12: Activity assessment - Social Recreation

4.5. Events

4.5.1. Activities Breakdown

While some large events such as concerts or large weddings in NEP might compromise the experiences of others and potentially negatively impact the ecological value of the park, smaller and more public events have potential. The park is built for walkers and runners, hence hosting some group runs, part of a marathon run, sport activities or specific corporate events and exhibitions are viable models for the park. Nature and environment themed small events should also be encouraged.

For example, in Nairobi's Karura Forest, a 3km run was held and led by the First Lady of Kenya. These events require good management, proper planning and advertising. They can generate revenue, enhance awareness about a location and result in enthusiasm for NEP.

Photo 4: Three-kilometre run led by First Lady Margaret Kenyatta (centre in grey) in Karura Forest, Nairobi.

Other events might include small corporate retreats. These could be physical activities such as a group walk, or a picnic linked to the restaurant. Given the proximity to the industrial area, there is high potential for these kinds of events. In addition, environmental activities and events, such as a fair to celebrate International Wetland Day (2 February) could host food trucks, educational booths, outreach events and educational activities to attract people to NEP and create awareness of wetlands in Rwanda and globally. Partner NGOs based in Kigali might be interested in collaborating on these types of activities.

4.5.2. Infrastructure Available

- Restaurant
- Information Centre
- Pathways
- Lawns

4.5.3. *Assessment of opportunity in local context*

Appraisal criteria	Score (1 =low / 5=high) For risk categories: (1 = high risk / 5 = low)	Description
Citizen interest	4	• High interest based on participation rate in events like 'Kigali Night Run'
Risk to Wetland conservation	2	• Large group of visitors • Risk of littering • Veering off designated paths and areas • Noise pollution
Financial / Revenue potential	4	• Revenue generated through user / conservation fees • Revenue from tickets or event fees for specific activities such as company retreat or Fun Run
Market demand	3	• Demand evaluated as high based on current participation rate to similar activities and frequency of similar events.
Risks and External Threats	2	• The designated pathways for exercises minimise the risk to the wetland • Heavy rains could affect the designated paths and areas • A large group of people entails requires enhanced security
Overall assessment	**3**	***Events in NEP are feasible with most of the infrastructure in place. This could represent a big risk to the wetlands due to the number of people that should be involved as well as the limited capacity to control movement and avoid littering or infrastructure damages. However, this could be planned for and managed.***

Table 13: Activity assessment - Events

4.6. Learning

4.6.1. *Activities Breakdown*

NEP has a strong potential to encourage learning about wetlands and Rwanda's environment. There are multiple levels of educational opportunities in NEP.

- Informal education through exposure to nature, outside interpretive signs (Photo 5), and the visitor centre
- Formal engagement with universities for research and monitoring
- Primary and secondary schools visit as part of their curriculum in biology, research, writing and/or recreation
- Primary and secondary schools visit the park with teachers as an extracurricular activity, which could lead to the creation of 'nature clubs'

The cost of transporting students and paying for conservation fees might be a barrier for conservation education programmes. However, a corporate sponsorship programme (Section 5) is a viable way to finance educational trips for Rwandan students if needed.

Tertiary students and professional researchers could also access the park to conduct academic research. NEP's visitor centre could provide information on Rwanda's other conservation areas to increase awareness about Rwanda's parks. Many citizens do not have the opportunity to visit Rwanda's parks and learning about them at NEP might create support for these conservation areas. Outdoor interpretive signs need to be placed in appropriate locations, made of durable materials, attractively designed and written clearly (Photos 5).

Photo 5: Left: Sample outdoor interpretation sign located in a suitable location in front of a pond; and right: information sign on birds from Umusambi Village, Kigali.

4.6.2. Infrastructure Available

- The environment: wetland ecosystem, trees, birds
- Medicinal Garden
- Information centre
- Restaurant
- Lawns

4.6.3. Assessment of opportunity in local context

Appraisal criteria	Score (1 =low / 5=high) For risk categories: (1 = high risk / 5 = low)	Description
Citizen interest	4	• High interest based on visits and participation in educational activities organised by nature-related institutions, such as Umusambi Village
Risk to Wetland conservation	2	• Risk of littering • Damage to vegetation by veering off trails and pathways • Invasive species introduction through visitation
Financial / Revenue potential	4	• Revenue would be generated through conservation / user fees • Support for programmes could be received through corporate sponsorship programmes and grants to provide Rwandan students with a chance to visit NEP
Market demand	4	• Demand evaluated as 'high' based on current participation rate to similar activities and frequency of organisation of similar events.

Appraisal criteria	Score (1 =low / 5=high) For risk categories: (1 = high risk / 5 = low)	Description
Risks and External Threats	2	• The designated walking areas might not be respected • Heavy rains could affect the designated paths and areas • Weather and seasons might affect the presence of some species and ability to visit in groups
Overall assessment	**3.2**	***Education is a viable activity for NEP with most of the infrastructure needed already in place. A research partnership/s could benefit the park as it would inform management and create awareness through publications. It could also stimulate support for conservation and lead to conservation careers. It is essential to make sure guidelines are followed by the group so as not to disrupt the ecosystem recovery.***

Table 14: Activity assessment - Learning

4.7. Other

4.7.1. Activities Breakdown

Other activities in NEP might include, for example, hosting of art exhibits, development of green office space for lease to NGOs in Kigali. For each of these the opportunity will need to be assessed against the conservation targets and management needs. Consideration of the impact on other activities should also be considered.

A zip line has been suggested for NEP to attract visitors. There are zip lines at Fazenda Sengha, which are very popular with visitors. Conservation Capital did not carry out a feasibility study of a zip line. Should one be considered, a proper feasibility study is needed to assess demand, safety, cost, maintenance, management and suitable location within NEP.

4.8. Carrying Capacity of the Park

Tourism carrying capacity can be a subjective measure and depends on (i) how thresholds of potential concern are viewed, and (ii) the objectives of the area being visited. At the outset of a carrying capacity assessment in the NEP case, it must be noted that historically, the environment suffered significant impact and cannot now, even after years of rehabilitation and restoration, be regarded as a pristine wetland. It should also be understood that carrying capacity can be variable depending on degrees of management. Notwithstanding any of the aforementioned themes, CC attempted to calculate a carrying capacity for NEP that takes its socio-environmental context into consideration as well as its size, infrastructure, likely user profiles and its soil and ecosystem/vegetation composition.

Given the concentration of facilities in Sectors 3, 4 and 5, and the intention to leave Sectors 1 and 2 relatively undeveloped, Conservation Capital considered Sectors 3, 4 and 5 alone in the carrying capacity calculation. Calculating carrying capacity based uniquely on area size is inadequate because certain areas might not be accessible due to water for example. Instead, we chose to calculate carrying capacity based on maximum optimal usage of the pathway infrastructure already in place together with additional pathways suggested. Thus, carrying capacity is calculated per metre of available pathway rather than square metres or hectares. Upper thresholds for people per metre of such infrastructure were based on the average speed of movement and purpose of the user in using that infrastructure. Thus, the lower limit of metres per person of paved cycleway (80m per person) is greater than half of that for paved walkways (30m per person), and this figure varies for unpaved pathways (50m per person) used for exposure to nature rather than for exercise, and the higher density usage of the suggested boardwalk crossing the wetland (20m per person).

Pathway Types	Existing (m)		Suggested Additional (m)		Total (m)	Suggested metres per person	Carrying Capacity		
	Sectors 3 & 4	Sector 5	Sectors 3 & 4	Sector 5			Sectors 3 & 4	Sector 5	Total
Cyclist (paved)	4,455	1,138	-	-	5,593	80	56	14	70
Pedestrian (paved)	4,239	1,130	-	-	5,369	30	141	38	179
Pedestrian (unpaved)	755	750	1,014	-	2,519	50	35	15	50
Pedestrian (boardwalk)	-	-	192	-	192	20	0	10	10
						Total People	**232**	**76**	**309**
						People / ha	**3.9**	**1.5**	**2.8**

Table 15: Calculation of visitor carrying capacity

The calculated carrying capacity as a result of this equation is just over 300 people for Sectors 3, 4 and 5 combined. This can be rounded down to 300 for ease of expression and should be understood as a carrying capacity for any one moment in time, and not as the limit of visitors per day (which could be far greater.) It should also be understood as needing to be applied partially to Sectors 3 & 4 and partially to Sector 5. Thus, at any one time, the maximum number of people permitted in Sectors 3 and 4 combined is suggested as 232 people (just under 4 people per hectare), while the maximum number permitted in Sector 5 at any one time is suggested as 76 people (or 1.5 people per hectare).

Managing these numbers ideally requires that Sectors 3, 4 and 5 are fenced (the cost of which is built into the CAPEX budget), and access controlled at limited numbers of entry points. The number of people within those sectors at any one time would need to be monitored; where new entrants are only permitted when sufficient numbers of people had left. Paying the conservation fee would enable access to both Sectors 3 and 4 as well as Sector 5 for all users within the parameters explained above.

Applying such limits during the week is unlikely to be necessary because we do not anticipate high volumes but doing so on the weekend is anticipated as being essential. Theorising how many visitors per day this would translate into requires an understanding of the user segmentation and the average time spent within NEP. At this juncture this is a hypothetical exercise that could be adapted as visitor patterns are better understood. In building a revenue projection model we have, however, attempted to do just this. Table 16 outlines the different time slots that might be considered for visitor management and planning.

Time slot	User type	Primary activities	Comments
05h30 – 08h00	Exercisers: cyclists, runners, walkers Nature enthusiasts: birders	Exercise on paved paths Exercise at fitness stations Birding on paved and unpaved paths, the wetland boardwalk and bird hide	We envisage this occurring both during the week and at weekends
08h00 – 11h30	School groups Families Friends Couples	Usage of paved paths Concentrations around the restaurant and picnic areas, as well as on the wetland boardwalk and bird hide	Aside from school groups during the week, we envisage this time slot being utilised primarily at weekends
11h30 – 14h30	School groups Families Friends Couples	Usage of paved paths Concentrations around the restaurant and picnic areas, as well as on the wetland boardwalk and bird hide	Aside from school groups during the week, we envisage this time slot being utilised primarily at weekends
14h30 – 17h00	Families Friends Couples	Usage of paved paths Concentrations around the restaurant and picnic areas, as well as on the wetland boardwalk and bird hide	We envisage this time slot being used primarily at weekends
17h00 – closing (19h00)	Friends Couples Exercisers: cyclists, runners, walkers	Usage of paved paths Concentrations around the restaurant and picnic areas, as well as on the wetland boardwalk and bird hide	We envisage this time slot being utilised more heavily at weekends, but also during the week at lower intensity

Table 16: NEP visitor segmentation by time slot and activities for planning and management. *Note that this is not suggested for entry fee; these time slots are used for management purposes only.*

4.9. Operating Hours

NEP will attract a diversity of users across both weekdays and weekends/holidays, ranging from early morning birders and pre-work joggers to midday walkers to evening meetups. The suggestion is to open the park at 05h30 and close in the evening at perhaps 19h00. In terms of capacity, 300 is suggested as the cap to the number of people at any one time. This is based on the total size of the park, the total area that is physically accessible to visitors, the total developed pathway infrastructure, and the intended experience. Ticketing could be done on-line with a platform such as TiCQet.

Security for visitors is very important. Many of the activities proposed require proper fencing, presence of park 'rangers' and good lighting if the park is to be open in the evening. This will require significant investment, which is incorporated into the CAPEX budget in this business plan.

4.10. Soliciting Feedback

Given this is Rwanda's first public recreational urban national park, soliciting feedback from users is important for adaptive management and improvement. We recommend users are provided with a very easy way to provide feedback, via an on-line survey. This can also be provided when they exit the park on a tablet or even at suitable locations in the park, such as a mounted tablet in the restaurant, or through smart phones. A sample survey is provided in Annex I. This information will also be very useful as REMA restores and develops other wetlands in Kigali.

4.11 Revenue Model

A. Conservation Fee

REMA has been clear that ideally access to the site is paid for by the public to instil a sense of value in the park. In addition, they have indicated that pricing must be affordable and in no way a barrier to access - but rather a mechanism to change public perspectives on the value of nature (as well as generating sufficient funds for ongoing operation and maintenance.) Paying for something, even if it is a small fee, creates value. To achieve this, we recommend a **'**conservation fee**'** rather than an 'entrance fee.'

As a public asset, charging user fees to citizens can be controversial, raising issues of equity, accessibility and ability to-pay (Boo 1990). Fees and pricing will affect peoples' choice and frequency of visiting sites (Schwartz and Lin 2006). At a lower price, NEP is more attractive to the public, and can still attract visitors while instilling an awareness about its value and the need to maintain the asset. Studies have shown that when visitors know where their money is going, for example to maintain and protect NUWEP, they are willing to pay. Therefore, clearly communicating how fees are collected and used is critical for NEP. This is an opportunity for the GoR to promote conservation using language such as: 'your conservation fee supports wetland conservation.'

In time, there is a possibility that NEP could be financed through corporate partnerships, building conservation contributions into the price of products in the cafes and into small events, and through undeveloped models such as a natural capital subsidy from the GoR. However, given the immediate interest in opening the park, providing citizens access, and creating a conservation culture - conservation fees are recommended.

Parks in Rwanda and throughout Africa charge different levels of fees for foreigners, residents and citizens and we recommend this for NEP[12]. In addition, family and year passes, which other urban conservation areas and other protected areas offer, is recommended. Tables 17 and 18 provide conservation fee examples from Karura Forest in Nairobi, Kenya. Karura and other urban protected areas are profiled in the 2022 NUWEP Updated Feasibility Report.

Category	Conservation fee		Parking fee	
	Adult	Child	Daily	Annual
Citizen	KES 100	KES 50		
Resident	KES 200	KES 100		
Non-resident	KES 600	KES 300		
Car: 5 doors			KES 100	
Minivan (12-seater)			KES 200	
Minibus (13–32-seater)			KES 300	
Bus (>32-seater)			KES 500	

Table 17: Karura Forest, Nairobi Kenya, Entrance and Parking Fees.

[12] Case studies of urban parks in Africa are included in the Conservation Capital 2022 NUWEP Updated Feasibility Study.

Studies show that when visitors know where their money is going, for example to maintain and protect NEP, they are willing to pay.

ANNUAL PASS TYPES & FEES	INDIVIDUAL	FAMILY
FKF MEMBERS	5,000	10,000
JUNIOR	N/A	3,000
FKF LIFE MEMBERS	4,000	7,000
PARKING	5,000	5,000
NON-MEMBERS	N/A	N/A

Table 18: Karura Forest, Nairobi Kenya: sample entry, parking and annual membership fees for Friends of Karura Forest (FKF) members and non-members (KES).

After stakeholder consultation and analysis of other fees raised in Rwanda and Africa, Conservation Capital recommends the following conservation and parking fees for NEP:

		One Time Entry			**Annual Pass**		
		Adult	**Child (>5 years)**	**Child (<5 years)**	**Adult**	**Child (>5 years)**	**Child (<5 years)**
Conservation Fee	Citizen	RWF 1,000	RWF 500	RWF 0	RWF 10,000	RWF 5,000	RWF 0
	Resident Expatriate	RWF 5,000	RWF 2,500	RWF 0	RWF 50,000	RWF 25,000	RWF 0
	EAC Tourist	RWF 5,000	RWF 2,500	RWF 0	RWF 50,000	RWF 25,000	RWF 0
	Rest of Africa Tourist	RWF 5,000	RWF 2,500	RWF 0	RWF 50,000	RWF 25,000	RWF 0
	Non-African Tourist	RWF 10,000	RWF 5,000	RWF 0	RWF 100,000	RWF 50,000	RWF 0
Car Park Fees (0-3 hours)	Sedan Car	RWF 5,000			RWF 50,000		
	Minibus / safari vehicle	RWF 10,000			RWF 100,000		
	Bus / coach	RWF 25,000			RWF 250,000		

Table 19: NEP suggested conservation and parking fees.

Implementation Steps

To implement the conservation fees, REMA (or the management partner) will need to:

1. Agree on the conservation fee amount.
2. Set up the registration and fee collection system.
3. Establish a good monitoring system to assess use and monitor satisfaction.

4. Market the conservation fee as a mechanism used to generate financial support for the wetland and instil a sense of pride with users — e.g., 'your contribution helps to protect Rwanda wetlands.'
5. Ensure staffing is adequate in NEP to check conservation fees at the entrance.

B. Filming and Photography Fees

Special permits could be sold for filming and photography for professional purposes. RDB has guidelines for filming and photography in Rwanda national parks, which should be reviewed for consistency. A filming permit in Rwanda is managed by the Rwanda Academy of Language and Culture (RALC) and costs USD 30 for 15 days and USD 50 for 30 days. This rate serves as an indication for REMA as it considers filming and photography fees at NEP. In addition to the filming permit, all film crew members must have press accreditation from the Rwanda Media High Council (Figure 16).

A foreign Journalist requesting an accreditation card shall pay as follows to Rwanda Revenue Authority (RRA):

- – USD30 press Accreditation card valid 01 – 15 days.
- – USD50 press Accreditation card valid 15 Days – 01 months/30 Days.
- – USD100 press Accreditation card valid 01 – 03 months.
- – USD200 press Accreditation card valid 03 – 06 months.
- – USD300 press Accreditation card valid 06 Months – 01year/12 Months.

*The rates mentioned above should only be treated as a guide – as they can change from time to time.

Figure 16: Filming and photography costs in Rwanda provided by RRA, RALC and RMHC

C. Restaurant

A restaurant was constructed in NEP using donor funding, which will help attract guests to the park and in turn generate entry fees for the park. The restaurant should be professionally managed by a private sector partner. Additional revenue will be generated by the operating lease that is granted to an operator. Given the CAPEX costs have already been financed through grant funding, a renewable 5-year management lease could be issued to an operator with a track record and experience.

Generally, concessions like restaurants are paid through a fixed fee (lease) and a variable fee (percentage of revenue). In general, variable fees should always be based on top-line revenue (gross) net of sales taxes. When CAPEX is required by the operator, the average fee is approximately 10%. Given CAPEX has been covered in the case of NEP, this fee might be marginally higher, but would need to be decided upon according to the operator's business projections and target markets. The auditing process should be based on quarterly management accounts submitted by the operator to the landowner, audited by a pre-agreed audit firm, which is approved by NEP's manager. The cost of such an audit would be borne by the operator.

The process to engage a restaurant operator is a public tendering process. This will largely follow the steps outlined in Section 6 used to tender and engage a private sector manager.

D. Shop

The information centre has space designed for a shop. Rwanda has exceptional products: art, crafts, t-shirts, and beauty products. In addition, NEP products could be sold, such as a bird guide, NEP logo materials, maps, binoculars and other useful products. The shop will generate revenue for NEP, depending on the nature of the store. As a result, the projected revenue is not included in the revenue model in this CABP.

E. Activity Concession

To attract visitors to NEP certain activities might be developed, such as bike rentals, guided walks, or birding visits. In some cases, revenue is generated by attracting people to the park and collecting conservation fees. However, in some cases a lease fee or concession fee might be charged to provide additional revenue for NEP management. Like the concession model described in C above, a percentage of gross revenue may be considered as a fee. Training local communities as guides is a good way to provide employment and provide a quality experience for visitors.

F. Revenue projections

Conservation Fees

In attempting to estimate potential revenue generation through conservation fees we estimated occupancy usage of the park at different times of the day and by different price categories of user. As indicated prior, for management purposes, we split the day into five brackets ranging in duration from 2.5 – 3.5 hours and hypothesised as being utilised by different user segments. We then applied an occupancy percentage to each of those time slots based on likely popularity (the maximum number of visitors per slot set at the carrying capacity of 300; thus, 1,500 potential visitors per day) and used this to calculate the potential amount of conservation fees paid. These occupancies per time slot are variable depending on weekdays versus weekends. For example, weekdays were projected to operate at 14% capacity while weekends were projected to operate at 57% of capacity. See Table 20.

Time slot	TTL # Tickets	TTL # Tickets	Weekday occupancy	Weekend occupancy	Weekday tickets	Weekend tickets
Early Morning	05.30h00 - 08h00	300	20%	35%	60	105
Morning	08h00 - 11h30	300	10%	50%	30	150
Midday	11h30 -14h30	300	10%	80%	30	240
Afternoon	14h30 - 17h00	300	10%	60%	30	180
Evening	17h00 - 1900	300	20%	60%	60	180
Total		**1,500**	**14%**	**57%**	**210**	**855**

Table 20: Hypothetical visitor numbers and capacity utilisation at NEP.

With these estimated occupancy figures we then applied ratios of citizen, resident and foreign visitors to them so that we could calculate conservation fee totals. We estimated 80% of users to be Rwandan citizens, 18% to be expatriate residents, and 2% to be foreign tourists and applied the suggested conservation fee rates to each group, i.e., RWF 1,000 for Rwandan citizens, RWF 5,000 for expatriate residents and RWF 10,000 for foreign tourists (Table 21).

On this basis we estimated that NEP would generate RWF 272,688,000 per annum, with 42% of this revenue being generated by Rwandan citizens, 47% by expatriate residents (despite being estimated as being only 18% of users) and 11% by foreign tourists (Figure 17).

	Total projected visitors / day	Proportion citizen visitors (80%)	Proportion resident visitors (18%)	Proportion tourist visitors (2%)
Weekday	210	168	38	4
Saturday	855	684	154	17
Sunday	855	684	154	17

	Total RWF projected per day	Total RWF projected per day	Total RWF projected per day	Total RWF projected per day
Weekday	399,000	168,000	189,000	42,000
Saturday	1,624,500	684,000	769,500	171,000
Sunday	1,624,500	684,000	769,500	171,000

	Total RWF projected per year	Total RWF projected per year	Total RWF projected per year	Total RWF projected per year
Weekday	103,740,000	43,680,000	49,140,000	10,920,000
Saturday	84,474,000	35,568,000	40,014,000	8,892,000
Sunday	84,474,000	35,568,000	40,014,000	8,892,000
	Total Conservation Fees	Rwanda citizens conservation fee	Expatriate resident conservation fee	Tourist conservation fee
TOTAL	**RWF 272,688,000**	**RWF 114,816,000**	**RWF 129,168,000**	**RWF 28,704,000**
TOTAL	**USD 272,688**	**USD 114,816**	**USD 129,168**	**USD 28,704**

Table 21: Hypothesised revenue generation through conservation fees of adult visitors in first year.

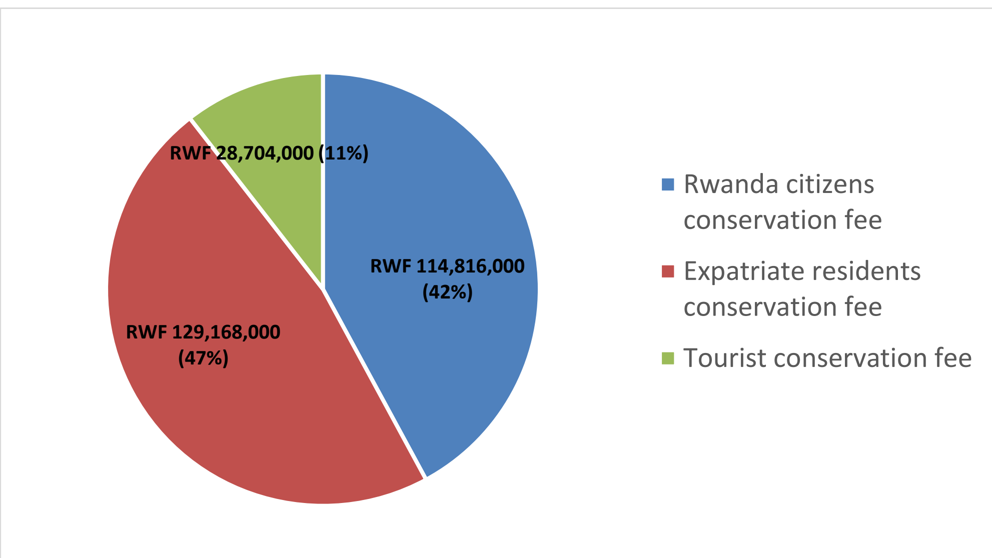


Figure 17: Proportional contribution by different user groups from conservation fees.

This hypothesised revenue generation does not consider parking fees or family passes, both of which are recommended. In addition, we have also not included revenue from certain activities such as birding clinics or yoga classes. These would add to the overall revenue model, but not significantly.

Assuming the same level of visitation and increasing the conservation fees every three years (Table 22), over a ten-year period, a total of USD 3.3 million would be generated for NEP from conservation fees. Figure 18 shows the revenue generated from conservation fees over a ten-year period.

Conservation Fees Conservative Rate Increase Every Three Years (RWF)										
	Year 1	**Year 2**	**Year 3**	**Year 4**	**Year 5**	**Year 6**	**Year 7**	**Year 8**	**Year 9**	**Year 10**
Citizen	1,000	1,000	1,000	2,000	2,000	2,000	3,000	3,000	3,000	3,500
Resident	5,000	5,000	5,000	8,000	8,000	8,000	10,000	10,000	10,000	13,000
Tourist	10,000	10,000	10,000	15,000	15,000	15,000	20,000	20,000	20,000	25,000

Table 22: Conservation fees projected to increase every three years

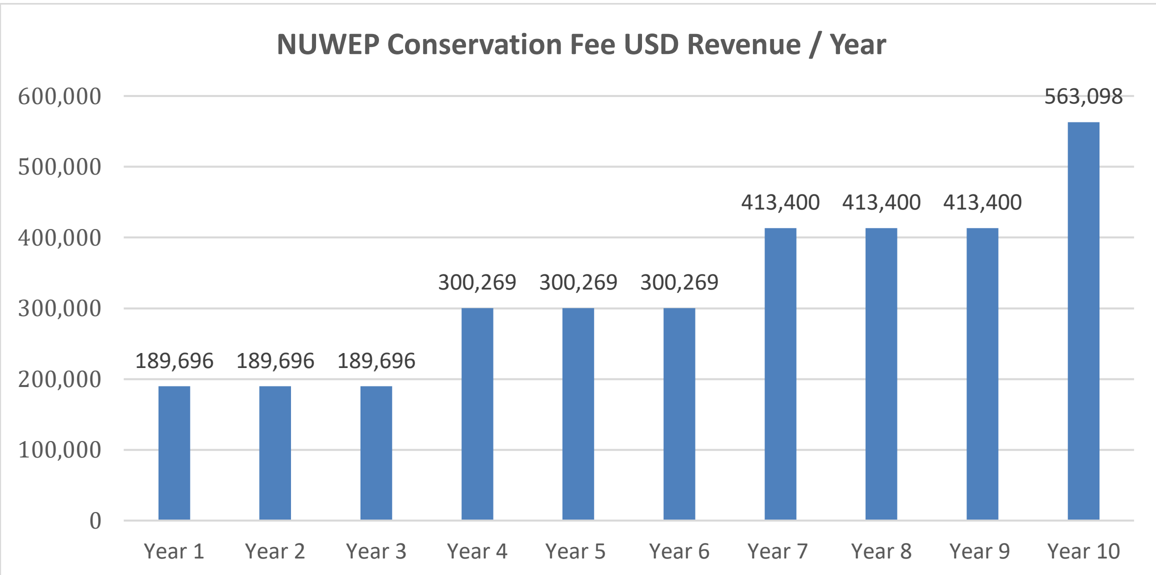


Figure 18: Conservation fee revenue generation over ten years.

Restaurant Revenue

Table 23 shows a projection of the revenue that might be generated by the restaurant in NEP. We used a 10% concession fee on gross revenue (we strongly advise against any kind of percentage based on net revenue.) The modelling of the restaurant's revenue anticipates the expansion of facilities over time to accommodate greater demand and includes the assumption that the café-restaurant will provide catering for different events, while serving a mix of indoor and outdoor seating and take-away.

Restaurant Annual Revenue for NEP										
Year	1	2	3	4	5	6	7	8	9	10
Concession fee to NEP (10%) USD	26,832	31,581	40,550	52,067	66,854	85,840	110,219	141,521	181,712	233,319

Table 23: Annual projection of restaurant concession fee (10% of gross revenue, USD).

Activity Revenue

We modelled two annual events, a 'Fun Run' and a wetlands fair in celebration of International Wetlands Day. Our estimates are based on the number of people entering NEP for these events. The revenue could increase based on concessions during the events, such as food carts. The charge for participation in the events is slightly higher than the regular conservation fee because of the added activities. The users are still differentiated between citizens, residents, and tourists. No events are included in year one as we anticipate the new management entity will focus on core visitor and management activity in the first year.

Year	2	3	4	5	6	7	8	9	10
Fun Run									
Citizen #	100	100	100	100	100	100	100	100	100
Resident #	50	50	50	50	50	50	50	50	50
Tour #	10	10	10	10	10	10	10	10	10
Citizen RWF	150,000	150,000	250,000	250,000	250,000	350,000	300,000	300,000	400,000
Resident RWF	300,000	300,000	450,000	450,000	450,000	55,000	55,000	55,000	700,000
Tourist RWF	100,000	100,000	150,000	150,000	150,000	200,000	200,000	200,000	250,000
Total RWF	550,000	550,000	850,000	850,000	850,000	605,000	450,000	555,000	1,350,000
Total USD	550	550	850	850	850	605	450	555	1,350
Wetland Celebration									
Citizen #	150	150	150	150	150	150	150	150	150
Resident #	60	60	60	60	60	60	60	60	60
Tour #	15	15	15	15	15	15	15	15	15
Citizen RWF	225,000	225,000	375,000	375,000	375,000	525,000	450,000	450,000	600,000
Resident RWF	360,000	360,000	540,000	540,000	540,000	66,000	66,000	66,000	840,000
Tourist RWF	150,000	150,000	225,000	225,000	225,000	300,000	300,000	300,000	375,000
Total RWF	735,000	735,000	1,140,000	1,140,000	1,140,000	891,000	450,000	816,000	1,815,000
Total USD	735	735	1,140	1,140	1,140	891	450	816	1,815
Total RWF	1,285,000	1,285,000	1,990,000	1,990,000	1,990,000	1,496,000	900,000	1,371,000	3,165,000
Total USD	1,285	1,285	1,990	1,990	1,990	1,496	900	1,371	3,165

Table 24: Annual projection of two events held in NEP, starting in year 2.

The total revenue including conservation fees, events and the concession fee from the restaurant is consolidated in Table 25.

	Years									
Revenue (USD)	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Conservation Fees	189,696	189,696	189,696	300,269	300,269	300,269	413,400	413,400	413,400	563,098
Restaurant	26,832	31,581	40,550	52,067	66,854	85,840	110,219	141,521	181,712	233,319
Events	0	1,285	1285	1990	1990	1990	1496	900	1371	3165
Total Revenue	216,528	222,562	231,531	354,325	369,112	388,099	525,115	555,821	596,483	799,581

Table 25: Total revenue including conservation fees, events and the restaurant concession over a ten-year period.

SECTION 5: OTHER REVENUE MODELS

In addition to generating revenue through visitors, as described in Section 4, there are several other revenue models that could be developed for NEP. While REMA and other GoR relevant authorities have made it clear they do not require and/or envision NEP to generate profit for use in other conservation areas, but to cover its own cost, it would make NEP even more relevant if it did generate revenue for other wetlands and conservation areas in Rwanda. Table 26 outlines potential revenue models.

	Potential Revenue	Advantages	Disadvantages	Immediate Opportunity	CAPEX Required	ROI
5.1	Corporate Sponsorship	Revenue generated can be used for management and conservation activities Might result in volunteer days, litter clean up, exotic species removal	Requires clarity around potential expectations of use	Immediate	Medium	High
5.2	Biodiversity Offsets	Revenue generated can be used for management and conservation activities	Expensive to set up to determine offset Requires very clear monitoring and evaluation Requires 'additionality'	Medium Term	Medium	Medium
5.3	Payment for Ecosystem Services	Revenue generated can be used for management and conservation activities	Need to calculate the ecosystem value, such as waterflow, and structure the payment Requires robust monitoring and evaluation Requires 'additionality'	Medium Term	Medium	Medium
5.4	Carbon Offset	Revenue generated can be used for management and conservation activities	Expensive to calculate the carbon sequestration Requires robust monitoring and evaluation Requires 'additionality'	Medium Term	Medium	Medium

Table 26: Potential revenue generating options for NEP.

5.1 Corporate Sponsorship

The industrial zone is directly adjacent to NEP, which lends itself well to potential corporate partners. Many of the companies operating in the industrial zone have corporate social responsibility (CSR) programmes and may be interested in supporting NEP (Table 27). A sponsorship programme could be developed to engage corporate partners and certain benefits may be provided such as their logo on a sponsorship sign or map; a reduced price for employee access; or space provided for corporate events. The sponsorship programme should not be limited to those located in the industrial zone, as there will be other corporate partners keen to engage in NEP. Different sponsorship packages could be designed (gold, silver, bronze,) with different pricing and benefits provided.

Business name	Business purpose
Strawtech	Produce affordable and environmentally sustainable building materials
Ameki Color	Rwanda's largest paint producer (also producing furniture)
Aquasan	Production of water pipes and tanks
Bhavesh	Non-ferrous metal recycling
Plascon	Paint manufacturer
Papyrus	Manufacture commercial furniture
Volkswagen	Car assembly
Alpha Media	Outdoor advertising, hoardings[13], digital printing
Serena	Promoting brand and place for guests to visit
Radisson Hotel	Promoting brand and place for guests to visit
Sheraton Hotel	Promoting brand and place for guests to visit
Bank of Kigali	Brand recognition and corporate engagement
Equity Bank Rwanda	Brand recognition and corporate engagement
Alpha Media	Outdoor advertising, hoardings, digital printing

Table 27: Examples of potential corporate partners for NEP, some (shaded light grey) are based in the Industrial Zone.

Implementation Steps

A. **Determine the sponsorship levels, rates and what they receive in return for their sponsorship**

Given the companies operating in Rwanda we suggest the following annual sponsorship rates:

Platinum: $75,000
Gold: $50,000
Silver: $25,000
Bronze: $10,000

In return for sponsorship, the companies could be provided certain benefits (Table 27).

Sponsorship Benefits	Platinum	Gold	Silver	Bronze
A signed thank you / certificate from the Minister of Environment indicating the level of sponsorship	√	√	√	
Number of free entrances passes per annum for staff	25	20	15	10
Logo on the entrance sign indicated level of sponsorship	√	√	√	√
Bi-annual updates from the management team	√	√	√	√
Recognition on the website and in any communication materials of level of sponsorship	√	√	√	√
Membership to a high-profile park advisory board for the year of sponsorship	√	√		

Table 27: NEP Corporate sponsorship benefits.

[13] Hoarding is defined as a temporary boarded fence in a public place, usually erected around a building site. These are used to protect the public from site works whilst also being used to display advertisements.

RDB has created a successful corporate sponsorship program for the annual Kwita Izina event (Figure 19). Their materials and approach be reviewed and adapted for NEP. While the best sponsorship is unrestricted, given the desire to support educational visits, sponsorship could support a specific activity, such as school visits and the education program. For example, the sponsorship appeal might indicate: 'For $X,000, you can support 400 primary school children visiting NEP'. In return NEP management could send pictures of the visit to the company, which they could use in their materials, and/or they could have volunteers help with the visit. Naming rights may be considered, but this should not deter others from getting involved in NEP.

Figure 19: Sample sponsors from the 2019 Kwita Izina.

2018 EVENT SPONSORS

GOLD SPONSORS:

Supported in part by a grant from:
National Geographic Society
Disney Conservation Fund
Born Free Foundation
Chester Zoo, UK
Anonymous Corporation
Charles & Judy Tate
Dan & Pam Baty

SILVER SPONSORS:

Zoo Basel, CH
Great Plains Conservation

BRONZE SPONSORS:

Great Plains Foundation

CONTRIBUTING SPONSORS:

Marleen Groen
Safari Professionals of the Americas
Safarilink

MAASAI OLYMPICS SPONSORSHIP

BECOME A MAASAI OLYMPICS SPONSOR

PLATINUM SPONSOR: $50,000+
GOLD SPONSOR: $25,000+
SILVER SPONSOR: $10,000+
BRONZE SPONSOR: $5,000+
CONTRIBUTING SPONSOR: below $5,000

To learn more about sponsoring the Maasai Olympics, please email donations@biglife.org

YOUR CONTRIBUTION WILL HELP US FUND

$50 - 1st place cash prize for one of the six events on Olympics Day
$100 - Olympics Day uniform and shoes for two warriors
$500 - Conservation education meeting for 50+ warriors on one group ranch in lead-up to Olympics Day
$600 - One girls school (20 girls) to participate in Olympics Day
$1,000 - Team selection competition day for 1 of the 4 manyattas
$1,500 - Practice equipment and gear for days leading up to Olympics
$2,500 - Breeding bull for winning manyatta on Olympics Day
$3,000 - Regional competition between 2 manyattas

Figure 20: Sample sponsorship program from the Kenya Maasai Olympics.

B. Develop sponsorship materials

Sponsorship materials will be used to invite corporate participation. Materials should be professionally designed, high quality and clearly articulate why companies should consider annual sponsorship and what they get in return. Sponsorship materials may include how engaging in NEP will help achieve some of the Sustainable Development goals, which many corporate partners track and report on. In addition, sponsorship materials may include various ecosystem service targets and measurements, which are routinely reported on by businesses.

C. Expand list of sponsorship partners

Kigali is a hub for international companies, and many have CSR programs. Corporates are more inclined to sponsor a programme and/or a place if they have a connection. For example, companies in the industrial zone or corporates that work in the water sector would be more inclined to sponsor NEP and therefore, they should be prioritised. Businesses also want to associate themselves with other leading companies. For example, if a company such as Apple is an anchor sponsor, this will help attract other high-quality sponsors. Therefore, finding that anchor sponsor is a critical first step.

A list should be developed of potential sponsors along with the individual in the company who has the ability to make sponsorship decisions (Managing Director or CEO; CSR Director; Partnership Manager) as well as potential anchor sponsors.

D. Meet with potential sponsors

Personal meetings should be held with potential platinum, gold and silver sponsors. It is very easy to throw a solicitation letter away and / or delete an email. One on one meetings should be held with a very clear and compelling pitch as to why sponsoring NEP is a positive investment for the company. Once platinum, gold and silver sponsors are lined up, requests should be made to the bronze sponsors. These might be done via email or letter, with a follow up call. Companies that are already sponsors should be included in the materials, which will help incentivise engagement.

Conservation Capital assessed three scenarios for the sponsorship programme: low, medium and high (Table 28 and Figure 21.) Each scenario uses the same annual sponsorship rate, but the number of sponsors increases from low to medium to high. We increased the sponsorship level for all three scenarios every three years. Given year one is a start-up, the first increase in sponsorship level is in year four.

Kigali is a hub for international companies, and many have CSR programs.

Rates (USD)	***Year***									
Year	1	2	3	4	5	6	7	8	9	10
Platinum	75,000	75,000	75,000	75,000	97,500	97,500	97,500	126,750	126,750	126,750
Gold	50,000	50,000	50,000	50,000	65,000	65,000	65,000	84,500	84,500	84,500
Silver	25,000	25,000	25,000	25,000	32,500	32,500	32,500	42,250	42,250	42,250
Bronze	10,000	10,000	10,000	10,000	13,000	13,000	13,000	16,900	16,900	16,900
# Sponsors	**Low Scenario**									
Platinum		0	1	1	1	2	2	3	4	4
Gold		1	1	1	2	2	2	3	4	4
Silver		1	1	2	3	4	5	6	7	8
Bronze		1	1	2	4	5	6	7	8	9
Total										
Platinum		0	75,000	75,000	97,500	195,000	195,000	380,250	507,000	507,000
Gold		50,000	50,000	50,000	130,000	130,000	130,000	253,500	338,000	338,000
Silver		25,000	25,000	50,000	97,500	130,000	162,500	253,500	295,750	338,000
Bronze		10,000	10,000	20,000	52,000	65,000	78,000	118,300	135,200	152,100
Total Low		**85,000**	**160,000**	**195,000**	**377,000**	**520,000**	**565,500**	**1,005,550**	**1,275,950**	**1,335,100**
# Sponsors	**Medium Scenario**									
Platinum		1	1	1	2	2	3	3	4	4
Gold		1	1	2	2	2	3	3	4	4
Silver		1	3	4	5	5	7	7	9	10
Bronze		2	4	6	7	7	9	9	9	12
Total	1	2	3	4	5	6	7	8	9	10
Platinum		75,000	75,000	75,000	195,000	195,000	292,500	380,250	507,000	507,000
Gold		50,000	50,000	100,000	130,000	130,000	195,000	253,500	338,000	338,000
Silver		25,000	75,000	100,000	162,500	162,500	227,500	295,750	380,250	422,500
Bronze		20,000	40,000	60,000	91,000	91,000	117,000	152,100	152,100	202,800
Total Med		**170,000**	**240,000**	**335,000**	**578,500**	**578,500**	**832,000**	**1,081,600**	**1,377,350**	**1,470,300**
# Sponsors	**High Scenario**									
Platinum		1	1	1	2	2	3	3	4	5
Gold	1	1	1	3	3	3	3	3	4	5
Silver	1	2	3	4	5	6	7	8	10	10
Bronze	1	2	3	4	5	6	7	8	9	10
Total										
Platinum	0	75,000	75,000	75,000	195,000	195,000	292,500	380,250	507,000	633,750
Gold	50,000	50,000	50,000	150,000	195,000	195,000	195,000	253,500	338,000	422,500
Silver	25,000	50,000	75,000	100,000	162,500	195,000	227,500	338,000	422,500	422,500
Bronze	10,000	20,000	30,000	40,000	65,000	78,000	91,000	135,200	152,100	169,000
Total High	**85,000**	**195,000**	**230,000**	**365,000**	**617,500**	**663,000**	**806,000**	**1,106,950**	**1,419,600**	**1,647,750**

Table 28: NEP corporate sponsorship, three scenarios based on level of interest from corporate partners. All using the same sponsorship rate, the number of sponsors change in each scenario. Corporate sponsorship presents a significant opportunity for NEP.

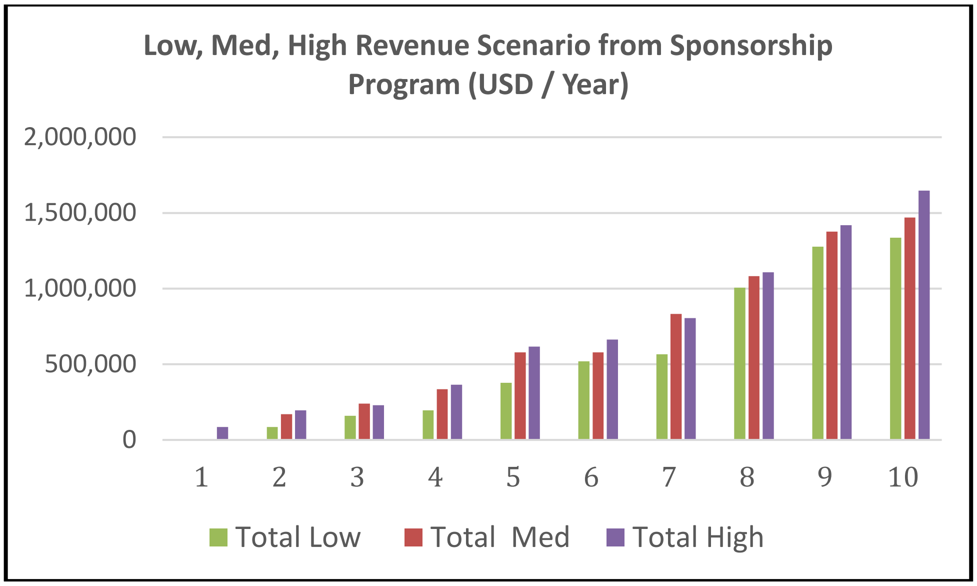


Figure 21: NEP corporate sponsorship, three scenarios based on level of interest from corporate partners.

E. Revenue Including Corporate Partnership

The total revenue for NEP adding in corporate sponsorship is in Table 29.

Revenue (USD) Medium	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Conservation Fees	189,696	189,696	189,696	300,269	300,269	300,269	413,400	413,400	413,400	563,098
Sponsorship	0	170,000	240,000	335,000	578,500	578,500	832,000	1,081,600	1,377,350	1,470,300
Restaurant	26,832	31,581	40,550	52,067	66,854	85,840	110,219	141,521	181,712	233,319
Events	0	1,285	1,285	1,990	1,990	1,990	1,496	900	1,371	3,165
Total Revenue	**433,056**	**615,125**	**703,063**	**1,043,651**	**1,316,725**	**1,354,698**	**1,882,229**	**2,193,241**	**2,570,317**	**3,069,463**

Table 29: Total revenue including conservation fees, corporate partnership programme (medium scenario), concession fee from the restaurant and events.

5.2 Biodiversity Offsets

Biodiversity offsets are measurable conservation outcomes designed to compensate for adverse and unavoidable impacts of projects, in addition to prevention and mitigation measures already implemented. Biodiversity offsets are only appropriate for projects that have rigorously applied the mitigation hierarchy (Figure 22) framework**,** a widely accepted approach for biodiversity conservation.

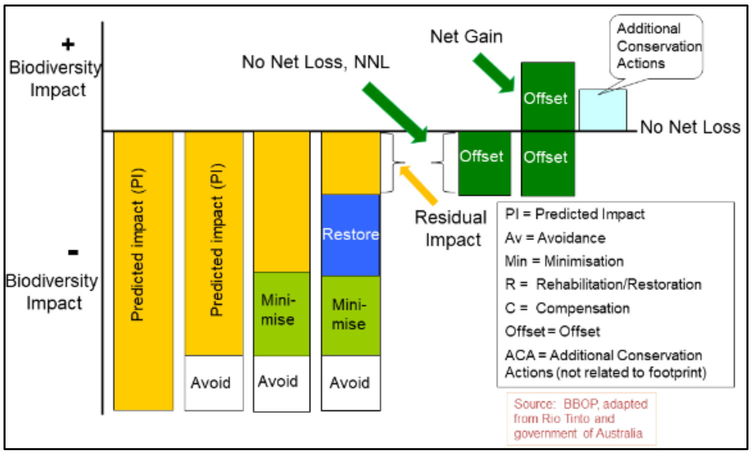


Figure 22: Mitigation hierarchy framework. (Forest Trends).

The aim of offsets is to achieve No Net Loss (NNL) and preferably a Net Gain (NG) of biodiversity when projects take place. Measures that are not designed to result in NNL and preferably NG are not biodiversity offsets. The achievement of NNL/NG is dependent on measurable, appropriately implemented, monitored, evaluated, and enforced offset schemes**.** Biodiversity offsets should be a measure of last resort; and in certain cases, offsets are not appropriate and should not be used.

Public and private sector investments in projects such as infrastructure development, mining and oil exploration are among the current drivers of economic growth. However, biodiversity is not well accounted for under the present economic system and such projects can have important impacts on species and ecosystems more generally.

Measures to compensate for negative impacts, for example by protecting threatened forests or restoring wetlands, collectively known as biodiversity offsets, are increasingly being used by governments and the private sector. Biodiversity offsets are conservation actions intended to compensate for the residual, unavoidable impact on biodiversity caused by projects, to ensure at least a NNL of biodiversity and, where possible, a NG. There is growing interest by governments and the private sector to look for ways of compensating for biodiversity impacts to achieve a NNL and preferably a NG of biodiversity when projects take place.[14] As a result, there is an increase in policies that support and/or require offsets (Figure 23). However, the site where the offset takes place, such as NEP, needs to demonstrate additionality. Meaning, the conservation activity that takes place in NEP needs to achieve a net conservation gain, which may be difficult given the wetland is well managed and already protected.

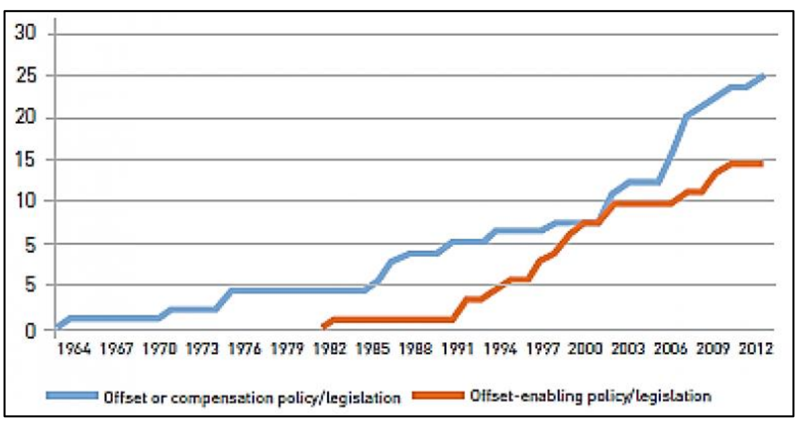


Figure 23: Policy trends biodiversity impact offset.

[14] https://www.iucn.org/resources/issues-briefs/biodiversity-offsets

While Rwanda does not require biodiversity offsets, the 2018 Rwanda Environmental Law outlines a polluter pay principle; "the polluter pays principle aims to deter activities impairing environment and punish any person who violates regulations.[15]"

In addition to regulatory requirements, there are a diversity of drivers that result in offsets:

- **Shareholder pressure**. Corporate shareholders demanding offsets.

- **Lender Pressure:** Bankers and other lenders requiring biodiversity offsets, such as the International Finance Corporation Performance Standard 6 governing Biodiversity Conservation & Sustainable Management of Living Natural Resources.

- **Consumer pressure**. Consumers demanding corporate responsibility

- **Employee activism,** especially from younger generations, who are demanding compliance with environmental and social standards, resulting in recruitment and retention challenges.

- **Reputational consideration**. Private companies concerned with their reputation as it pertains to environmental outcomes, and recognition of positive brand recognition with regards to NNL and NG.

Section 5.1 highlights some of the businesses operating directly adjacent to NEP in the industrial zone. Some of these companies may need to offset their operations, in terms of ecological footprint, carbon emissions, water use or wastewater flow into the wetland. This could be done by supporting NEP. The other likely offset partner would be an offset involving a wetland. In many cases, 'like for like' is required, meaning if a wetland is being destroyed, a wetland should be protected as an offset (Figure 24). Any offset needs to demonstrate additionality, meaning, if offset financing is going into NEP, an improvement of conservation outcomes needs to be demonstrated. As noted, given NEP is relatively well managed, this may be problematic for an offset in NEP. The GoR and the city of Kigali are planning for the conservation and management of all the wetlands in the city; therefore, assessing the overall value of the city wetlands and engaging in an offset for the whole wetland system may make more sense, given scale and the cost of developing offsets. The revenue generated from an offset depends on a variety of contexts (location, impact, and scale); thus, this is not included in the revenue model but is something the future manager should consider.

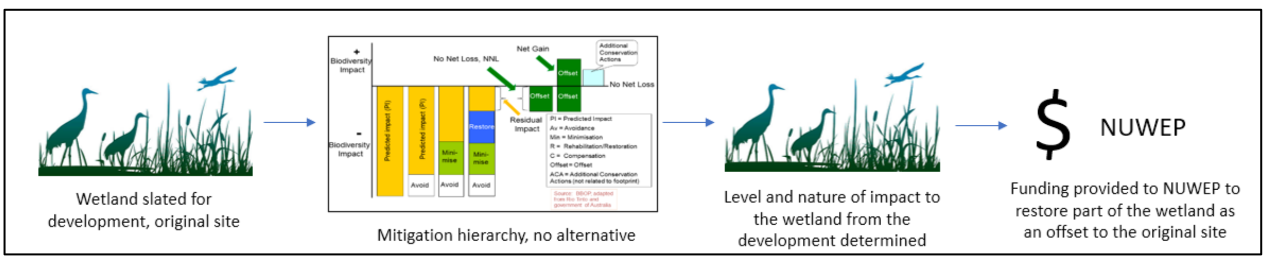


Figure 24: Hypothetical biodiversity offset that could finance the restoration of additional wetland area in NEP.

[15] https://waterportal.rwb.rw/sites/default/files/2018-10/Water%20law%20gazetted%2C2018.pdf

Implementation Steps

Steps for implementing a biodiversity offset are outlined in *Biodiversity Offsets: A User Guide*[16]. Below are some of the steps required.

1. An assessment should be done on potential biodiversity offsets suitable for NEP and/or the wetland system in Kigali, depending on the scale. This would include identifying the project proponents, assessing the scale of impact, determining the timeline of the development, and understanding the potential drivers for an offset.
2. Once potential offsets are identified, these should be ranked in terms of likelihood to engage in an offset, ROI, and ease of working with them.
3. If there is agreement between the project proponent and NEP, then the initial feasibility and development of an offset can be initiated, broadly following these high-level steps:
    a. Estimate residual biodiversity losses from the original project.
    b. Select the offset activities and conservation site(s).
    c. Prepare the biodiversity offset project component.
    d. Monitor implementation of the biodiversity offset activities and results.

5.3 Payment for Ecosystem Services

The term 'ecosystem services' refers to the diverse benefits that are derived from the natural environment. Examples include the supply of food, water and timber (provisioning services); the regulation of air quality, climate and flood risk (regulating services); opportunities for recreation, tourism and education (cultural services); and essential underlying functions such as soil formation and nutrient cycling (supporting services.)[17]

Payment for ecosystem services (PES) occurs when the beneficiaries or users of an ecosystem service make payments to the providers of that service. In practice, this may take the form of a series of payments in return for receiving a flow of benefits or ecosystem services. The basic idea is that whoever provides a service should be paid for doing so (Figure 25).

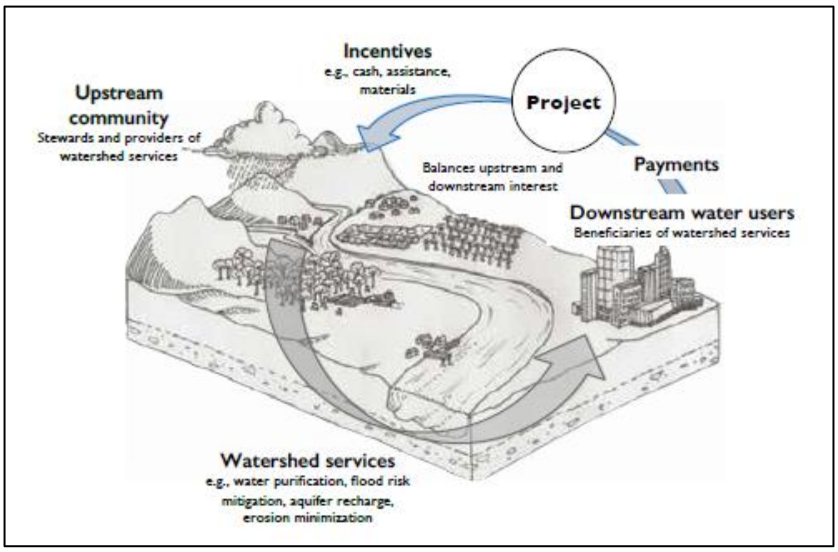


Figure 25: An example of a watershed PES (CIFOR 2014).

[16] https://www.cbd.int/financial/doc/wb-offsetguide2016.pdf.
[17] https://www.cifor.org/publications/pdf_files/Books/BFripp1401.pdf

A PES is a voluntary transaction where a well-defined ecosystem service (or a land use likely to secure that service) is 'bought' by a (minimum of one) ecosystem service buyer from a (minimum of one) ecosystem service provider, if and only if the service provider secures ecosystem service provision (conditionality).

PES constitutes one of the pillars of Rwanda vision 2050, which sets the vision for the country's economic transformation and development agenda. Rwanda and Costa Rica signed a Memorandum of Understanding (MoU) on environment cooperation that will specifically focus on exchanging experiences on PES. In March 2019, the GoR in partnership with the Netherlands launched a PES pilot program in Upper Nyabarongo catchment, which is one of the water towers of Rwanda[18]. The pilot is intended to assess pathways towards the implementation of a proposed PES scheme that is expected to be implemented countrywide.[19] Given the water source of NEP, a PES centred on water might be considered. This would also dovetail into Rwanda's Natural Capital Accounting aims.

Implementation Steps

There are ten key steps in making a PES work (Figure 26.) REMA could engage an ES expert to help identify and implement a PES if deemed feasible. Because of the level of investment required for a PES, REMA might consider assessing a group of wetlands in Kigali as opposed to just Nyandungu. The revenue that could be generated from a PES scheme is dependent upon the feasibility study and the nature of the scheme, therefore it is not included in the revenue model.

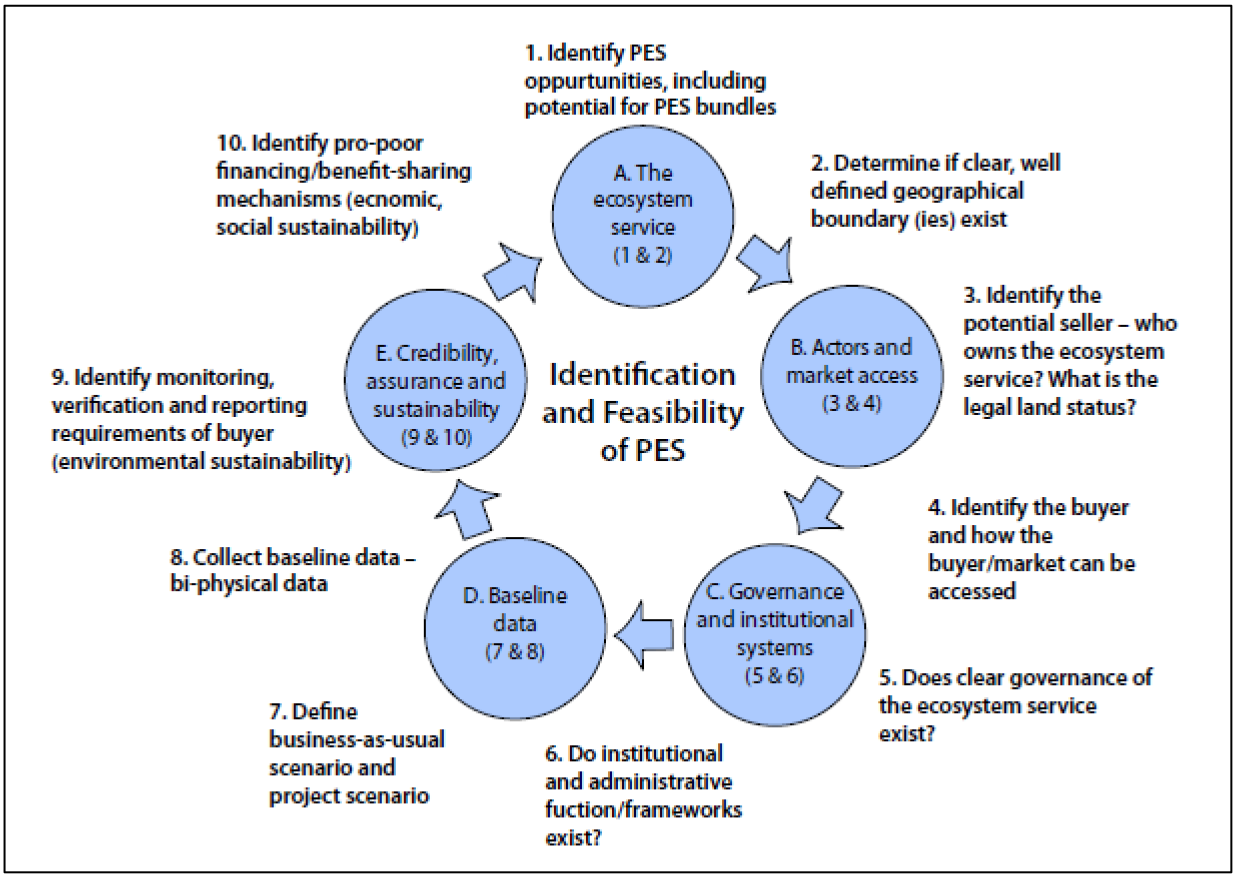


Figure 26: Ten key steps in designing and implementing a PES.

[18] https://www.greenclimate.fund/sites/default/files/document/23450-rwanda-integrated-landscape-climate-resilient-management-program-upper-nyabarongo-catchment.pdf
[19] https://drive.google.com/file/d/1DHMghAlDEOziYcfXWXF7t4IOHgrx1pZQ/view

5.4 Carbon Offsets

Carbon offsets are a type of PES (Section 5.3.) Given the level of interest in carbon, this is profiled separately.

The 1997 Kyoto Protocol and the 2015 Paris Agreement laid out international CO_2 emissions goals. With the latter ratified by all but six countries, they have given rise to national emissions targets and the regulations to back them. With these new regulations, there is growing pressure on businesses to find ways to reduce their carbon footprint, which in some cases takes time. As a result, interim solutions involve the use of the carbon markets. What the carbon markets do is turn CO_2 emissions into a commodity by giving it a price. These emissions fall into one of two categories: carbon credits or carbon offsets, and they can both be bought and sold on a carbon market. It is a simple idea that provides a market-based solution to a monumental problem[20].

In addition, nature has been recognised as contributing significantly to reducing carbon emissions by sequestering carbon. For example, The Nature Conservancy and 15 other institutions found that nature-based solutions provide up to 37% of the emission reductions needed by 2030 to keep global temperature increases under 2 degrees Celsius, which is 30% more than previously estimated. Additionally, using only cost-effective solutions, nature's mitigation potential is estimated at 11.3 billion tons in 2030—the equivalent of stopping the burning of oil globally. In addition, many climate solutions (such as wetlands) offer additional benefits, such as water filtration, flood buffering, improved soil health, protection of habitat and enhanced climate resilience. Therefore, if a company needs to offset their carbon footprint, investing in nature is one way to do so.

Rwanda has committed to a climate-resilient, low-carbon economy by 2050. Healthy wetlands play a significant role in climate change mitigation by storing carbon that would otherwise contribute to global warming, leading to the reduction of water and food resources as well as more extreme weather phenomena. NEP is approximately 121.8-hectares, and the wetland is approximately 54 hectares. Determining the annual carbon sequestration of the wetland presents an interesting revenue opportunity for the park. The industrial zone of Kigali is directly adjacent to NEP. An agreement could be entered into between NEP and the companies in the industrial zone providing them an opportunity to offset their carbon footprint.

There are two types of carbon markets:

1. Regulatory compliance market e.g., EU Trading System. This is utilised by companies and governments who, by law, are required to account for their greenhouse gas (GHG) emissions.
2. Voluntary carbon offset markets – Verified Emission Reduction credits (VERs.) The voluntary carbon market provides companies which have not yet established the ability to reduce their GHG footprint operationally or through their supply chains the opportunity to purchase carbon offsets, known as 'credits,' as a cost-effective solution to countering their greenhouse gas impact.

Wetlands are recognised as sequestering significant CO2 (Figure 27), which may present an opportunity for NEP.

[20] https://carboncredits.com/the-ultimate-guide-to-understanding-carbon-credits/#1

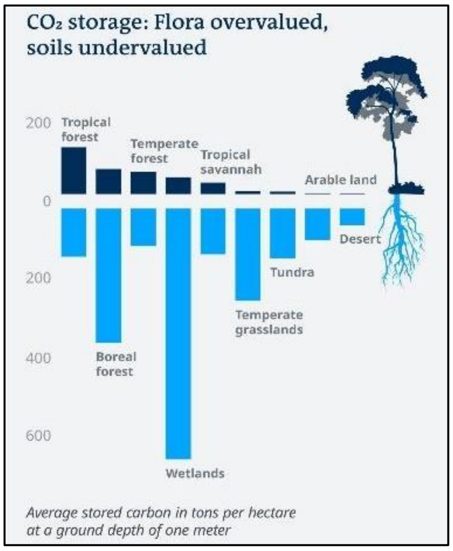
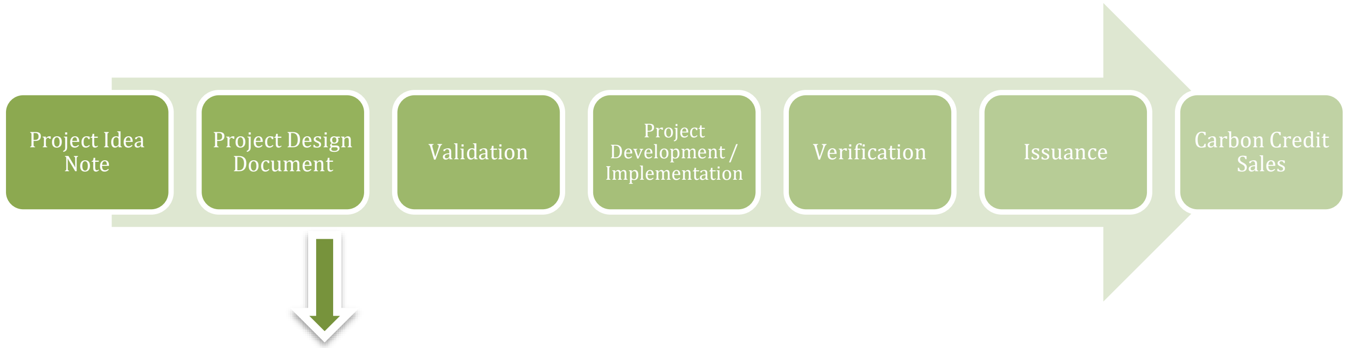


Figure 27: Average stored carbon in tons per hectare (World Climate Council IPCC).

Given the heightened awareness around climate change and the carbon market, carbon offsets may present an opportunity for Rwanda's wetlands. The challenge with both the regulatory and voluntary carbon market is the cost in initiating a carbon programme—it is expensive. This means that scale is important for a return on investment. Therefore, it is recommended that REMA consider carbon for the wetland system in Kigali as a whole, as opposed to NEP individually. The carbon market as it is currently designed rewards threat mitigation. Meaning, if a wetland is already protected and well managed, it will not qualify for the carbon market because there is no additionality. Therefore, if the GoR is going to engage in the carbon market for its wetlands now is the time as it starts to restore and protect the entire system.

Implementation Steps

To initiate and develop a verified project, REMA would need to go through the steps in Figure 28. A pre-feasibility study is a relatively inexpensive way to determine if a carbon project is even viable before investing in the PIN (Project Idea Note,), which we strongly recommend. The project development process is expensive, and scale is an important factor, which would be assessed in the pre-feasibility assessment. There are numerous carbon experts, such as Wildlife Works (www.wildlifeworks.com), with experience in Africa who could complete a pre-feasibility assessment for REMA. The barrier in the past for successful carbon programs was the lack of buyers. Given the recognition of climate change, finding a buyer for verified and compelling carbon projects is not problematic. In addition, the corporates operating in Rwanda are potential carbon credit buyers. Given the cost in developing a carbon project in Kigali's wetlands and the uncertainty of 'net-gain,' projected revenue is not included in the revenue model.



The **project development** process includes:

1. Identifying and documenting the threat to the wetland and cause of degradation.
2. Quantifying the rate of degradation over the past 10 years.
3. Designing and implementing mechanisms that will mitigate the threats.
4. Quantifying the decrease in degradation that will be achieved with interventions.
5. Demonstrating how leakage will be avoided (preventing the problem from shifting to another area.)

Figure 28: Steps to follow by REMA to the initiation and development of verified projects

The validation/verification process takes approximately 12 months, depending on several factors, such as baseline information.

Natural Capital Accounting

There is increased attention to the need to value natural capital. To operationalise certain PES and biodiversity offset schemes, as described prior, the first step is valuing natural capital. Rwanda is one of the World Bank's pilot countries for natural capital accounting (NCA)—determining the value of the non-market services provided by nature[21] and has been actively working with Wealth Accounting and the Valuation of Ecosystem Services (WAVES)[22]. Once valued, the Government of Rwanda will consider various finance mechanisms to support the management and sustainability of its natural capital. The NEP managers should track this progress to determine how NEP can be part of any payment mechanism.

5.5 Total Cost/Revenue

The total cost / revenue analysis indicates that in year one and two with revenue from conservation fees, the restaurant, events and CSR (medium scenario,) there will be a loss. This is largely due to a slow start up with revenue development in the early years, and high capex costs in project set up. However, this will be recovered by year 5. From year 5 onwards, additional revenue can be used to sponsor school trips and educational opportunities in NEP and might be considered for conservation of other wetlands. We have not included donor funding in the model, and this may be used in years 1 and 2 to support specific activities or infrastructure, such as a perimeter fence.

[21] https://snappartnership.net/teams/rwanda-natural-capital-accounting/#:~:text=Rwanda%20is%20one%20of%20the%20World%20Bank%E2%80%99s%20pilot,value%20of%20the%20non-market%20services%20provided%20by%20nature.

[22] https://www.wavespartnership.org/sites/waves/files/images/5th%20Partnership%20Meeting_WAVES%20Rwanda.pdf

	Year									
	1	2	3	4	5	6	7	8	9	10
Total Opex Capex	741,725	660,997	372,697	412,527	513,044	371,789	411,102	408,509	393,655	597,600
Total Revenue	433,056	615,125	703,063	1,043,651	1,316,725	1,354,698	1,882,229	2,193,241	2,570,317	3,069,463
Profit - Loss Medium	-308,669	-45,873	330,365	631,124	803,680	982,909	1,471,127	1,784,732	2,176,662	2,471,862

Table 30: Cost / revenue analysis of NEP operations and CAPEX.

Education Programme and Kigali Wetland Funding Support

While the budget in this CABP includes an Education Officer on the staff, it does not include other education programme costs, such as materials, transport for children unable to afford transport, and additional staffing. Adequate funding should be allocated to a formal education program once feasible.

Once the education programme is covered, a percentage of revenue might be used to support other wetlands in Kigali or the development of sustainable revenue models for the wetland system (see 5.2-5.4). Some of Kigali's wetlands will be developed for recreation and therefore generate revenue from user fees, whereas other wetlands might be used purely for conservation purposes and require funding support.

Given the GoR has developed much of the park infrastructure, it is reasonable to require the management partner, once operational costs are covered (year 3), to allocate 12% of top-line revenue to some kind of wetland fund. If the fence is financed by the GoR, the percent contribution could increase to 13%. The governance structure and operational procedures of any kind of fund would need to be created. Board representation would include for example REMA, RDB, and FONERWA. At 12%, this would generate USD 1.7 million in eight years for Kigali wetland conservation.

Table 31 includes an operations reserve fund, an education programme budget and a percentage that goes into a fund for Kigali wetlands.

Year	1	2	3	4	5	6	7	8	9	10
Total Opex / Capex	741,725	660,997	372,697	412,527	513,044	371,789	411,102	408,509	393,655	597,600
Education Program		50,000	51,500	53,045	54,636	56,275	57,964	59,703	61,494	63,339
12% revenue to Kigali Wetland Fund			84,368	125,238	158,007	162,564	225,867	263,189	308,438	368,336
Total Cost	**741,725**	**710,997**	**508,565**	**590,810**	**725,688**	**590,628**	**694,933**	**731,401**	**763,587**	**1,029,274**
Total Revenue	433,056	615,125	703,063	1,043,651	1,316,725	1,354,698	1,882,229	2,193,241	2,570,317	3,069,463
Profit - Loss Medium	-308,669	-95,873	194,498	452,841	591,037	764,070	1,187,296	1,461,840	1,806,730	2,040,188

Table 31: Cost / revenue (USD) analysis including operational cost and CAPEX, education program costs and 12% top-line revenue to a 'Kigali Wetland Fund.'

Identification of Management Partner

NEP requires professional management. As one of the first public parks in Kigali, there is no precedent. While REMA currently oversees the management of NEP, their role long-term is not in managing NEP. Operation of NEP is envisaged by GoR to be conducted by an independent operator in a Public Private Partnership (PPP) model. The GoR has several successful PPPs. For example, Akagera National Park is managed through a PPP between RDB and African Parks. CC considered several management options (Table 32,) their roles, and advantages and disadvantages of each model.

Potential Park Manager	Potential Role	Advantages	Disadvantages
The City of Kigali	The city/municipality oversees the day-to-day management of NEP	NEP is within the city of Kigali's jurisdiction / and the local municipality	The city does not have the experience, staffing or budget in place to manage NEP
REMA	REMA oversees the management of NEP	• REMA is the authority that oversees environmental projects including wetlands	• REMA does not manage national parks
RDB	RDB oversees the management of national parks in Rwanda and tourism development	• RDB is the authority that oversees national parks in Rwanda	• RDB manages national parks and in some cases works with partners on management through a PPP. Given the urban nature of NEP, they recommend a PPP
Local Community Based Organisation (sector, cell, village)	A local CBO manages the park in a community-based management model	• Local engagement and ownership	• They lack the skills and experience to manage NEP • They do not have adequate resources • They do not have connections to donors • They lack the business acumen to manage an urban park
Private Sector Partner (PPP)	Advertise for a private sector partner to take on management of NEP	• The right partner may be able to effectively manage the park and generate adequate revenue to support the park and make a profit • The right partner may be able to attract other businesses (bike hire, restaurant, events, exhibitions) • The right partner may have access to capital for further development/ improvement of NEP	• To make a profit the private sector partner may engage in activities that are not compatible with the park.

NGO (PPP)	Advertise for an NGO to take on management of NEP	• The right NGO partner may be able to effectively manage the park and generate adequate revenue to support the park and make a profit • The right NGO partner can attract donor funding • The right NGO partner may be able to attract other businesses (bike hire, zip line, restaurant, events, exhibitions)	• While there are PPPs for Park management in Rwanda, there are no PPPs for urban park management • Most NGOs lack the business acumen to manage an urban park
Corporate Consortium	Advertise for a corporate consortium to manage and operate NEP	• Managing and operating NEP requires different skill sets. A consortium of corporate partners (restaurant operator, conservation manager, outdoor activity company) may work together to operate NEP.	• Identifying and aligning a suite of operators to take on NEP collectively may be a challenge

Table 31: Scenarios for NEP management (orange indicates less feasible and green indicates feasible).

Implementation Steps

To attract the best management partner, REMA should follow international standards in public tendering. It is understood that RDB, given its role in facilitating business partnerships, would facilitate the process for REMA to attract a management partner. RDB recently undertook a public tendering process in advertising and attracting a management partner for NEP. The World Bank's Global Wildlife Program published in 2021 a Toolkit for Collaborative Management Partnerships of protected areas.[23] This next section draws from the Toolkit and outlines the steps required to engage an effective partner for NEP and the process for a competitive tendering process. A tendering process is consistent with the Rwanda legal framework, which should be reviewed. Figure 29 outlines the seven key phases involved in the tendering process.

To attract the best management partner, REMA should follow international standards in public tendering.

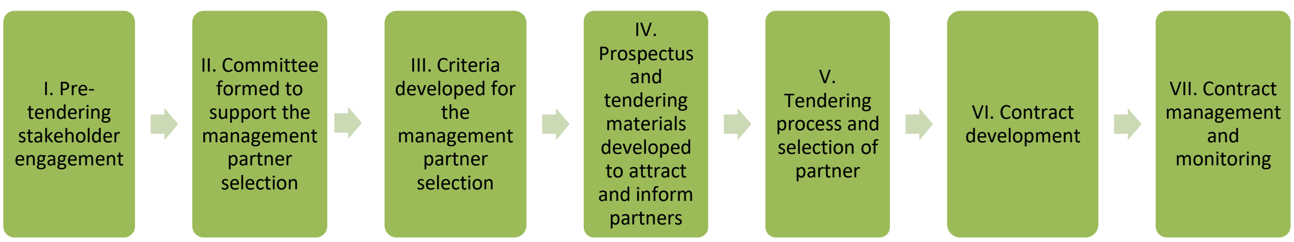


Figure 29: Seven key phases of the tendering process.

[23] (GWP 2021) https://www.worldbank.org/en/programs/global-wildlife-program/publication/collaborative-management-partnership-toolkit

Step I. Stakeholder Engagement

National protected areas have a diversity of stakeholders, ranging from citizens who view parks as public resources, local communities who rely on natural resources, private sector partners operating in and/or around the PA, local government authorities, and conservation and development organisations. Since PAs are valuable to different groups and individuals for a variety of reasons, engaging in a consultative process is important so that stakeholders understand and feel part of the process. Proactive discussion and dialogue with these stakeholders can help avoid problems and delays and ensure the effective engagement of local stakeholders, including communities, in the process. REMA has already initiated stakeholder consultation with the various feasibility studies that have proactively engaged stakeholders. REMA has a clear understanding of the key stakeholders and going forward should continue to effectively engage them in the development and management process of NEP.

Step II. Formation of the Proposal Evaluation Committee

REMA and RDB will form a proposal evaluation committee (PEC) to support the selection of the management partner. The PEC will:

- Determine criteria for the management partner selection (Step III)
- Review tendering documents prior to publication (Step IV)
- Review expressions of interest and full invited bids from potential partners (Step IV)

In some countries, the gazettement of the PEC is required. We recommend that the PEC is made public for transparency purposes and to avoid challenges in the future, regardless of the legal requirement. The PEC should contain adequate representation from individuals with the requisite technical skills and experience to properly appraise bids. Ideally, there should also be independent experts with urban PA and wetland management, as well as revenue development experience. The committee should be relatively small to ensure effective operations and management. All members of the PEC should sign a conflict-of-interest declaration. (GWP 2021)

Step III. Criteria for Partner Selection

The PEC will develop criteria for the partner selection process (Step III). This will help the PEC weigh expressions of interest and full proposals against clear criteria. Some things REMA will want to consider are outlined in Box I. (GWP 2021)

Box I. Potential criteria for management partner selection (GWP 2021)

Organisational Structure

- Ability to operate in Rwanda or a plan to ensure ability to operate.
- Structure enables the partner to accept donations.

Technical Experience

- Successful urban PA development and management experience, or experience with other relevant projects
- Familiarity with wetland management.
- Experience engaging in effective partnerships with government, private sector, communities and NGOs
- Proven professional project management skills, including budget development, staff management and oversight, environmental management, and human resource development.
- Progressive nature-based revenue development in PAs
- Proven knowledge and application of contemporary, innovative conservation financing mechanisms.
- Successful donor fundraising, management, and networks
- Successful, effective public relations and communications systems
- Demonstrable professional and rigorous fiscal management and control

Financial Capacity

- Proof of funding required to support the initial period of NEP management, and the skills required to attract additional resources and develop sustainable revenue models
- Provision of a detailed, realistic, and professional NEP business plan
- Proof of the partner's ability to commit long-term (20 years)

Social Impact

- Proven local community engagement, integration, and related economic development
- Proven history of optimizing local employment opportunities
- Plan for capacity development of staff

Shared Vision and Philosophy

- Demonstrated commitment to REMA's vision and philosophy
- Demonstrated commitment to equitable access and education

Step IV. Development of Prospectus Material

To promote the management opportunity a suite of tendering materials will be developed. The information will include the scope of the partnership and a detailed description of the project and desired responsibilities of the partner. REMA will indicate that it is seeking a partner to manage NEP and include its vision for the PA. REMA can review the tendering materials RDB used for the tendering process for Nyungwe National Park. In addition to the actual tendering notice, REMA should develop information on the partnership opportunity that can be included in the advertisement and made available on its website to inform potential partners about NEP and attract top candidates. At a minimum, the following information should be included:

- Park location and map
- Park size
- Access
- Vision
- Ecosystem types
- Wildlife
- Threats
- Commercial operations
- Infrastructure
- Number of visitors
- Management opportunity description

Step V. Tendering Process and Partner Selection

The tendering process will follow the Rwandan regulations around PPPs and tendering procedures. The steps include a request for an expression of interest, vetting by the PEC, solicitation of full proposals from successful candidates, vetting by the PEC and due diligence on the selected candidate. The 2021 GWP CMP Toolkit has sample tendering documents, evaluation frameworks, and due diligence tools that REMA can adapt and use. Figure 30 outlines the three main steps along with the estimated timeframe. It is important to provide adequate time for bidders to consider the PPP and submit a suitable bid.

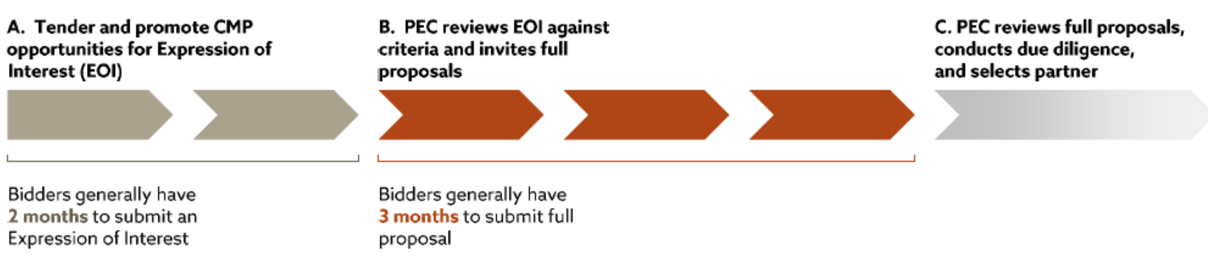


Figure 30: Public tendering process to attract and select a suitable PA management partner (GWP 2021)

Step VI. Contracting

Once the PEC selects the partner and REMA has completed due diligence on the potential partner, REMA will negotiate and develop the management contract. The content of the contract will be guided by Rwanda's legal framework, and we recommend looking at other PPP contracts between RDB and management partners. Contracts should be concise and explicit to avoid any confusion and conflicts of interpretation. Figure 31 includes some of the headings that might be included in a management contract. We recommend the duration of the management agreement be at a minimum 10 years with a renewal clause. Like all contracts, clear indicators will be included and if the management partner is not fulfilling its obligation, REMA will have the ability to terminate the agreement (Step VII).

• Parties	• Law Enforcement
• Background	• Community Relations
• Definitions / Interpretations	• Establishing Park Fees
• Objectives	• Existing Commercial Relationships and New Concessions and Enterprise Development
• Governance Structure	• Assets
• Geographical Area	• Liability and Indemnity
• Delegation of Management	• Conflict Resolution
• Duration, Start Date, and Renewal	• Perfomance Review
• Integration of Staff	• Termination
• Staff Recruitment	• Data Ownership
• Reserved Matters	• Communication
• Donor Funding and Revenue Management	• Other Sections
• PA Management Roles and Responsibilities	
• Non-Management Responsabilities	

Figure 31: Sections that might be included in the NEP management agreement (GWP 2021)

Step VI. Contract Management and Monitoring

Managing a contract helps ensure that a mutually beneficial relationship exists between the PA authority and the partner (GWP 2021.) Contract management also enables the partners to swiftly mitigate challenges, adapt as needed, and in the worst-case scenario, terminate a CMP agreement if needed. Contract management and performance monitoring includes two key components: monitoring and evaluation (M&E) and audits (GWP 2021.) The PA team generally completes the M&E, and an external expert completes the audit. Using both approaches in NEP is recommended.

Monitoring and Evaluation (Section 7)

M&E should be done regularly by the management team covering management performance, including conservation, social, and ecological targets, and compliance with the obligations of the contract. Monitoring reports should be provided routinely to REMA and/or a NEP board, if this is formed (see next section.) The reporting timeline and frequency of monitoring reports will be outlined in the contract, and at a minimum, the board should receive monitoring reports twice a year. Monitoring is done against clear targets outlined in the annual work plan and against the annual budget. (GWP 2021)

At a minimum, an external auditor should review the management every three years. The auditor should be an internationally recognised urban PA management expert. The audit should assess and appraise the implementation and compliance by the management partner regarding their obligations in the contract and evaluate the general performance and achievement of the contracts intended goals and objectives. The auditor will make recommendations for areas of improvement. The report should be submitted to the board or REMA for approval. The cost of the evaluation should be built into NEP's operational budget.

Governance Board

A governance board would be put in place to oversee the management of NEP. This board would review annual budgets, monitoring, audits and overall compliance with the management contract. They would not be involved directly in management; their role is governance. The management board should be kept small to ensure efficiency and might include the following representatives:

1. REMA representative
2. RDB representative
3. Representative from City of Kigali
4. Representative from local government
5. Corporate partner
6. Wetland expert
7. Local community representative
8. Non-voting seat to the management partner

SECTION 7: MONITORING AND KEY PERFORMANCE INDICATORS

REMA has outlined very clear ecological, economic and social targets for NEP. These will be incorporated into the management contract. NEP's manager will develop joint annual work plans, a management plan, and a five to ten year business plan. To enable assessment of progress and performance, these plans will include targets that are specific, measurable, assignable, realistic, and time-related (SMART). (GWP 2021)

Key performance indicators should include targets within the following categories:

1. ecological
2. financial
3. social
4. operational

SMART targets will enable the partners to adequately assess performance against the contract and help to resolve any claims of non-performance. The establishment of baselines in the beginning of the CMP, if they do not already exist, is essential to enable project monitoring. (GWP 2021)

Given the short-term needs in establishing a management partner and getting the park open, we have included short term indicators (Table 33) and longer-term indicators (Table 34).

Short Term Indicators			
Indicator	**Description**	**Milestones**	**Metrics**
Selection of management partner for NEP operation	The award of a management contract with suitable operator must be put into place before any further improvements or subleases can be initiated at NEP	• Formation of a PEC • Call for EOIs advertised • Selection of short list candidates • Formal / binding proposals from short list submitted • Selection of winning bidder • Contracting	• Signed contract for management
Selection of sub-concessionaires (if the manager is not going to manage all operations) by management partner	Depending on the capabilities of the management partner, sub-concessionaires will/may be needed for a variety of services and activities within NEP, such as the restaurant	• Call for EOIs by the management partner • Selection of short list • Formal / binding proposals from short list • Selection of winning bidders • Contracting	• Signed contracts for concessions
Completion of infrastructure by relevant parties	The visitor centre and restaurant need full shop fitting and furnishing. Under some business models Sectors 3 and 4 will need to be enclosed by a secure fence. Trash bins are needed throughout the park.	• Completed /ready to open infrastructure such as restaurant and visitor centre	• Operational infrastructure

Communication platform and digital presence	A professional and compelling website and social medica platform that informs the public about the park and on-going activities	• Website • Social media platform	• Website • Social media platform
Soft launch/es	Provide organised access to NEP for carefully selected trial audience/s to test functionality and solicit feedback	• Advertisement of soft launch • Soft launch/es	• Soft launch/es • Feedback from visitors • Implementation of feedback
Official launch	Hold official opening event for invited guests from key constituencies including government, press, media, influencers etc.	• Official launch	• Official launch • PR/media reach (number of articles, social media engagement etc)
Actual opening	Opening the park to access for the public.	• Actual opening	• Actual opening • # of visitors • Reaction of visitors

Table 32: NEP Short Term Management Indicators.

For each indicator in Table 33, five and ten year targets should be established. For example, if the baseline number of alien species is 10, the five-year target might be to have zero alien species. These will need to be developed following baseline assessments.

Performance Indicator	Indicator Definition	Base-line Year	Method Of Data Collection	Data Acquisition		Data Analysis & Reporting	
				Frequency	By Whom	Frequency	By Whom
Ecological Indicators							
Comprehensive conservation plan	The existence of a comprehensive conservation plan covering conservation status, threats and targets and mechanisms to achieve these	2022	N/A Plan updated every 5 years	N/A	N/A	Annual	Management Partner
Water volumes & consistency of flow	Does the wetland function as intended in terms of flood mitigation? How long does the wetland absorb and retain storm water for before releasing it?	2022	Measurement of rate of flow at multiple points along the wetland system	Monthly, and specifically after heavy rainfall	Management Partner, ideally engaging students / conserva-tion partners	Annual	Management Partner

Water quality	Does the wetland function and filter water?	2022	Measurement of water quality at inflow points and outflow points of the wetland	Quarterly	Management Partner, ideally engaging students / conservation partners	Annual	Management Partner
Ecological restoration	As the overall park matures in its ecological restoration a wider array of indigenous species will recolonise	2022	Bird, amphibian, mammal and butterfly surveys	Monthly	Management Partner	Annual	Management Partner
Absence of alien species	Alien species can significantly compromise the integrity of indigenous ecosystems and should be limited wherever possible	2022	Botanical surveys	Bi-Annual	Management Partner	Annual	Management Partner
Presence of rare, threatened, and endangered (RTE) species	Tracking RTE species is important for ecological and visitor management, and for GoR reporting	2022	Bird, amphibian, botanical and butterfly surveys	Monthly and bi-annual	Management Partner	Annual	Management Partner
Commercial Indicators							
Tourism revenue generated	Total sum of revenues generated by activities in NEP	2022	Financial Book records	Monthly	Management Partner / Accountant	Yearly	Management Partner / Accountant
Revenue covering operating costs	Percentages of operation & maintenance costs covered by revenue generated on site	2022	Financial Book records	Monthly	Management Partner / Accountant	Yearly	Management Partner / Accountant

NEP contributing to regional economic growth	Amount paid in taxes and salaries	2022	Financial Book records	Monthly	Managem ent Partner / Accoun-tant	Yearly	Management Partner / Accountant
NEP contributing to employment	# of people work on the site, total salary expenditure, employer profile	2022	Financial Book records	Monthly	Managem ent Partner / Accoun-tant	Yearly	Management Partner / Accountant
Corporate sponsorship program successfully supporting NEP	# of corporate sponsors at the different levels	2022	Financial Book records	Quarterly	Managem ent Partner / Accoun-tant	Yearly	Management Partner / Accountant
Education							
NEP providing quality educational opportunities	# of academic institutions partnered with and # of graduate and postgraduate programmes and projects conducted on NEP	2022	Mainten ance of register	Yearly	Managem ent Partner	Yearly	Management Partner
NEP hosting quality educational opportunities	# of education programs hosted by NEP	2022	Mainten ance of register	Yearly	Managem ent Partner	Yearly	Management Partner
Increased public awareness of the importance of wetlands	# of students visiting NEP and participating in educational programs	2022	Mainten ance of register	Yearly	Managem ent Partner	Yearly	Management Partner
Visitor Numbers and Satisfaction							
NEP well visited and used	# of visitors, number of repeat visitors, broken down by citizen, resident, tourists	2022	Mainten ance of registry	Daily	Managem ent Partner	Bi-annual	Management Partner
NEP visitors satisfied with visits	Satisfaction rate, broken down by citizen, resident, tourists	2022	Survey	Daily	Managem ent Partner	Bi-annual	Management Partner
Management has a clear understanding of visitor use	# of visitors, time of day, and activity	2022	Survey	Daily	Managem ent Partner	Bi-annual	Management Partner

Stakeholder Support							
Key stakeholders well informed and supportive	# of stakeholder meetings, and communication mechanisms	2022	Survey	Annual	Management Partner	Yearly	Management Partner
Local communities supportive of and engaged in NEP	# of local jobs, managed access for resource collection, trained local guides and number of complaints	2022	Survey	Annual	Management Partner	Yearly	Management Partner
High awareness about NEP, its activities, programs and goals	# of visitors to the website and social media engagement, as well as media mentions in print and digital	2022	Maintenance of register	Monthly	Management Partner	Yearly	Management Partner
Operations							
NEP safe and free of encroachment	# of incidents, crimes and encroachment	2022	Management / incident reports	Monthly	Management Partner	Yearly	Management Partner
NEP effectively managed by managing partner	Contract complied with, suitable staff trained and in place	2022	Management reports	Monthly	Management Partner	Yearly	Management Partner
NEP powered by green energy	Amount of green energy used at the site, and analysis of green energy vs not	2022	Management reports	Monthly	Management Partner	Yearly	Management Partner
NEP governed by an independent board	NEP governance board in place, meeting bi-annually and providing useful oversight	2022	Board minutes / reports	Bi-annual	Management Partner	Yearly	Management Partner
NEP effectively managed by managing partner and in compliance with management contract	NEP assessed by an independent auditor	2025	Audit	Every three years	Auditor	Every three years	Auditor

Table 33: NEP Long Term Indicators

SECTION 8: MARKETING

Marketing is the process through which the unique features of a product are communicated to both existing and potential future markets (businesses or consumers) in a manner that promotes the desirability and, ultimately, sales of that product in an honest and sustainable way. In the case of NEP, marketing is the process whereby the eventual operator of the park promotes visitation. Depending on the target market, several mechanisms and platforms can be used to generate interest and demand.

There are several well-known frameworks for establishing a marketing plan, ranging from 'the 4 Ps,' to 'the 5 Ps,' 'the 7Ps' and even 'the 8 Ps'. In the context of NEP, we believe the 5 Ps of Product, Price, Positioning, Place and Promotion are sufficient and these are outlined in relation to NEP in this section (Figure 32).

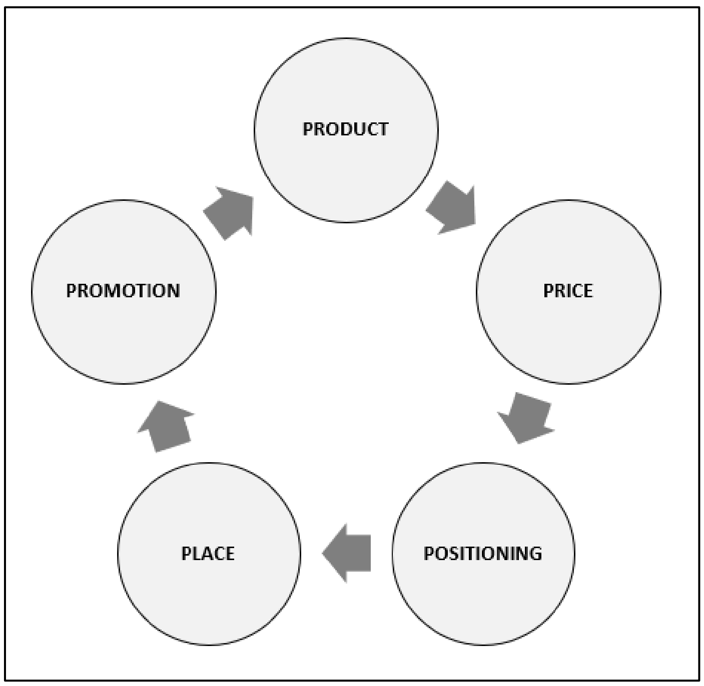


Figure 32: The NEP marketing mix and cycle with the '5 Ps.'

Product

Together with the rest of the management structure, the NEP marketing department needs to begin by defining exactly what 'product' or experience is being offered to the user, and to define the user or target market (this is often done through a segmentation process). This is a multi-directional process that needs to take the circumstance of NEP into account ('what can it offer?'), the market demand in the different identified user segments ('who wants it?') and how to marry the two. As outlined in this business plan we believe that for the target market the NEP 'product' is an urban park that offers the user the opportunity to enjoy natural space in the middle of a developed urban environment and to do this in a variety of ways ranging from exercise to nature appreciation to socialising. Adapting the product and keeping it relevant to the user is important and can take the form of ongoing user engagement through surveys and focus groups to be able to adapt to current market trends and demands.

Price

Pricing of the product is a critical part of the marketing plan. What is the target market prepared to pay for the identified product? Usually, the first process in determining this is to conduct a competitor analysis, ensuring that all similar product available to the target market and its pricing is understood and thereafter determining where your own product fits. In the Kigali context there are no directly similar products but analysing those that are similar, such as Sunday Park, MEZ or Fazenda Sengha, is an important part of the process. What do these products offer, who uses them and what do they pay? In the NEP case consideration needs to be given to the fact that pricing is not necessarily driven by what the market can bear, but rather by the notion that charging a fee is an important part of denoting value for a natural resource, and that access cannot be restricted by pricing. We suggest pricing based on our analysis of similar products, consultation with stakeholders and assessment of NEP's goals.

Positioning

Having conducted the above two processes, it is important to be able to position the product in a way where its unique virtues can easily be discerned by the target market in relation to the competition. To do this the Unique Selling Points (USPs) need to be identified, defined, and expressed in as succinct and catchy a way as possible. Just presenting the product and indicating its price is insufficient. The USPs need to be easily apparent and emphasised in marketing materials such as website, social media, advertising and so on. In a sense one needs to be able to convey to the potential user what their benefits of usage will be, how they will feel and why this is the product for them. In an ideal scenario all of this would be conveyed in a logo for NEP (Figure 33), this is another element that the marketing team needs to develop.

Place

Place usually applies to the location in which the product is sold, such as a retail store for example. In the NEP context, place cannot be modified; NEP is located where it is and cannot be moved. Users simply must access and utilise it in situ. As such the marketing plan needs to consider the element of access. If one cannot take the product to where the user is, one needs to enable to user to access the product in the most time and cost effective/convenient way. Thus, the marketing plan for NEP needs to consider the different modes and means of access (car, bike, bus, on foot) and where necessary establish joint marketing campaigns to promote them.

Promotion

No matter how good the product is, if people do not know about it, they cannot use it. Promotion describes the different mechanisms that are applied in raising the profile of the product and ensuring that the target market is reached and is aware of the product. In the NEP context these mechanisms are most likely to include aspects such as a website, a social media presence, a database of users and their contact details (email or WhatsApp,) as well as advertising (e.g., paid spots on radio, or billboards in Kigali, or newspapers) and PR and media (e.g., unpaid exposure achieved through hosting media, organising launch functions, engaging influencers, issuing relevant press releases and so on.) Additional mechanisms in the social media age are things like 'selfie frames' where users are encouraged to take their photograph in a way that makes their use of NEP clear, and then to tag or hashtag appropriately (Photo 3). More direct and overt marketing or promotion can be conducted through meetings with potential user groups, organisations like Kigali DMCs and tour operators.

Component	NEP Definitions and Mechanisms
Product	Product / offering: An urban park centred around a wetland (that is an important contributor to flood mitigation) and perfect for Kigali residents and visitors to use in connecting with nature, exercising or socialising. Target market / user segmentation (further detail will be garnered by ongoing engagement with users): • Kigali residents: exercise, socialising, contemplation • Expatriate residents: exercise, socialising, birding • Tourists: birding, leisure, down time during business visits, exercise • School children: education
Price	• Rwanda citizens: RWF 1,000 conservation fee o Nominal pricing that does not significantly limit accessibility, but does help manage numbers and denotes a value to conservation and ecosystem services • Expatriate residents: RWF 5,000 conservation fee o Significantly higher than citizen fee, but still an 'affordable' level that will be accepted by most expatriates as a means to contribute to their 'adopted' city • Foreign tourists: RWF 10,000 conservation fee o Double the expatriate fee but still reasonable and acceptable to foreign tourists
Positioning	Unique Selling Points - Kigali: • One of Africa's cleanest, best organised and greenest cities • Remarkably crime free and safe • Friendly and welcoming • Visionary approach to integrating natural ecosystem services – like wetlands – into the function and form of the city • High quality Unique Selling Points - NEP • The only significantly sized natural open space within Kigali where you can feel as if you are not in the city at all • The only space with infrastructure development specifically for cyclists and runners. • Far more affordable than other existing outdoor venues in Kigali and thus accessible to everyone • The only outdoor venue in Kigali that actively contributes to the city's wellbeing. • It is very accessible within the city, particularly from the Kicukiro and Gasabo Districts but also from Nyarugenge • 'Not only is NEP good for you but it is good for your city' Identity & logo • The identity and logo for NEP must emphasise the sense of nature, space, healthy activity and healthy social interaction all for the good of Kigali (flood mitigation; green space; beautification), as well as the broader Rwandan environment (biodiversity; water management) (Figure 33) • A pay offline or slogan is also something that can be considered by the marketing department as a means of conveying the identity and USPs to the target market

Place	As referenced prior in this document, the component of *place* in the NEP context speaks to the need to ensure that the park is accessible and that means of access are well known and advertised to the target market Aspects that should be considered by NEP marketing department are: • road signage on routes leading to NEP from key locations in all of Kicukiro, Gasabo and Nyarugenge, using the logo and associated marketing themes; and • working with bus companies to ensure that NEP is included on key routes and bus stops are integrated into the NEP access points
Promotion	Materials/Press Kit: • Ensure a good collection of promotional material that is also made accessible to partners who will promote Nyandungu, e.g., the media. Such materials would include professional photographs of landscape and infrastructure with and without people (representing the target market) in them, and also including other attractions such as charismatic bird species. This collection of promotional materials is often called a 'press kit' and needs to include easily readable and quotable facts about the product (e.g., how many trees have been planted, how many species have returned, how big it is, what the USPs are) • Imagery must be professionally generated, high quality/resolution and must feature the key USPs of NEP Launches: • Official opening – this must include relevant dignitaries from Ministry of the Environment, REMA, RDB, City of Kigali, Kicukiro and Gasabo Districts, Nyarugunga, Ndera and Kimironko Sectors, and the relevant cells. In addition, all relevant print, digital, radio and TV media should be invited and provided with the press kit described above making it easy for Nyandungu itself to help guide the narrative about the park. Also, necessary to consider is the inclusion of high-profile influencers whose presence can amplify the promotional reach. The website and social media platform should be ready prior to the launch • Ongoing public events – 'launches' do not end after the first official opening but must be continued at regular intervals to launch new aspects of the park, or simply to host a new influencer. Essentially any opportunity to inform the public narrative must be taken • An example of an event might be media open days where members of the media are permitted free entry and are hosted by a member of the NEP team to take a tour across the park and learn about its activities and purpose. The same open day concept could be applied to hosting Kigali-based tour operators and DMCs and other key stakeholders, as well as producers of 'The Map: Kigali' and other guides to the city (including country guides published by the likes of Bradt, Petit Fute, and Lonely Planet.) Digital: • NEP website. NEP needs an independent website—e.g., www.nyandungu.com—that provides all necessary information on the product (including opening times, prices, activities, access) to the target market; updated as necessary, as well as relevant updates, events and on-going activities • NEP social media channels (Twitter, Instagram, Facebook, e.g., @NyandunguPark) —that develop followings of users, reinforce the USPs, and

	keep users updated of events, changes, new activities etc.; updated twice a week. • Selfie Frame—a mechanism aligned to social media where a physical picture frame, branded with NEP logo and hashtags is installed in a key place within NEP allowing people to have their photo taken with an identifiable NEP attraction (such as the ponds) in the background (Photo 3) • NEP e-newsletter (WhatsApp & email)—an additional means of communication to the user database that is generated through a sign-up form on the website, on social media and at park entrance gates and which allows NEP to communicate directly with repeat users (all relevant privacy laws will need to be complied with.) Also, could include a membership club of people who buy annual passes and are sent privileged information prior to public release; sent monthly • Press releases—a means of communicating with the media and other non-user stakeholders who can amplify the message (e.g., newspapers, news websites, radio stations, television stations, tour operator associations etc); sent as required Print/Signage: • Printed promotional collateral like brochures and pamphlets are very useful in raising awareness of NEP and can be simply and affordably produced (e.g., as a 'z fold' pamphlet) to be distributed across key points in Kigali where an intersection with the target market is possible/likely. An example is in hotel lobbies where there is often a space where attractions within the city are promoted • The billboard campaign for Akagera National Park was very successful in raising awareness for the park's renewal in Rwanda. The same should be attempted for Nyandungu, with a handful of billboards rented around the city (including one on the main road outside NEP itself) to ensure that a 'buzz' and awareness about Nyandungu is created, especially around the time of opening • Similarly, signage at key points on the road journey to NEP should be erected for directions and as a kind of billboard campaign • Corporate partners will be interested in sponsoring various marketing materials such as billboards given their logo can be part of the signage. PR & Media: • Aside from the mechanisms covered above, an additional way to generate 'earned media/exposure' is to enter NEP into awards that recognise innovative urban developments. Being recognised by such awards would give NEP independent endorsement. Competing in these awards and winning them will depend on the continued rehabilitation of the wetland and its biodiversity and having the data available to document this through ongoing engagement with academic or NGO institutions

Table 34: NEP marketing components and mechanisms.

In addition to the above it should be understood that literally every touch point of the target market experience of the product can be considered a marketing opportunity or risk. To this end every single employee working for NEP needs to be well versed in the USPs of the product and its benefit to the user.

Additionally, the promotional aspect of marketing should be seen as a cyclical one. Marketing doesn't end with reaching the target market and selling them the product. Rather the promotional process should consider the message and engagement with the target market in the initial 'awareness' phase, then during the 'experience' phase while they are actually at NEP, and lastly the 'loyalty' phase where they have experienced and enjoyed the product and are receptive to greater degrees of engagement and can themselves even act as advocates or brand ambassadors if the messaging and tools are correct.

We are aware that there is a logo for NEP with an acacia tree. The samples provided below depict the Nyandungu wetland, the heart of NEP, and in some cases the recreational use of the park. The term Ecotourism is also not used in the logo below as the logo is cleaner without the term.

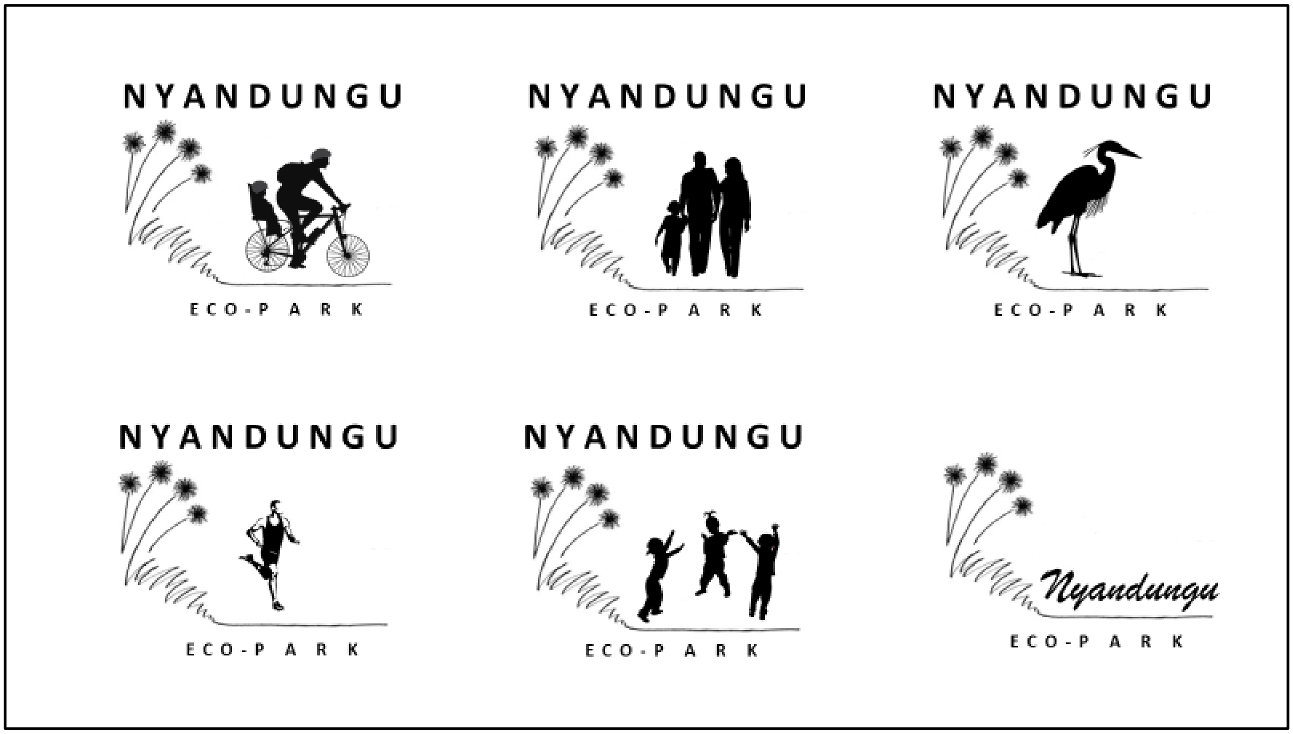


Figure 33: Hypothetical logo design direction for NEP, which will be used in all promotional materials and at the site.

Not only is Nyandungu Eco-Park good for you – it is good for your city.

SECTION 9: NEXT STEPS

Management Partner

The first step for REMA is to work with RDB to embark on a tendering process to attract a management partner. This will enable RDB and REMA to enter into a management contract for NEP. The steps to engage in a tender process are outlined in Section 6.

Soft Opening

We understand there is pressure to open NEP to the public. If REMA needs to open prior to engaging a management partner, we strongly recommend that REMA has a 'soft' opening and charges the conservation fees as recommended in this CABP. The launch can be promoted as an opportunity for citizen of Kigali to provide feedback. Users should have access to a survey when they leave the park, on their phone or at the gate, so that information can be gathered and used to inform the park design and management in the future (Annex I).

Onboarding of Management Partner

Once the management partner is on board, they will developing a management and operations plan, set up the governance board, hire and train relevant staff, develop the marketing materials needed prior to the official launch, review the data compiled by users following the soft opening, recruit concessionaries (if needed) to manage the restaurant and other facilities and activities (shops, bikes, yoga classes) and then, when ready, work with REMA on the official opening.

ANNEX I: NEP User Questionnaire Sample

Nyandungu Eco-Park User Questionnaire

Email: ____________________

Date: (DD/MM/YYYY_________

Age
- ☐ <10
- ☐ 11-19
- ☐ 20-30
- ☐ 31-40
- ☐ 41-50
- ☐ 51-60
- ☐ 61-70
- ☐ 71-79
- ☐ >80

Gender
- ☐ Male
- ☐ Female
- ☐ Other

Have you visited Nyandungu before?
- ☐ No
- ☐ Yes

  If Yes, how many time?
  - o 1-4 times
  - o > 5 times
  - o > 10 times
  - o > 15 times

Who are you?
- ☐ Citizen
- ☐ Resident
- ☐ Tourist

How did you get here?
- ☐ Car and parked at the site
- ☐ Public transport
- ☐ Walked
- ☐ Biked
- ☐ Other

Why did you visit?
- ☐ To enjoy open space
- ☐ To connect with nature
- ☐ To socialize with friends and family
- ☐ To exercise
- ☐ To learn about wetlands
- ☐ Other

What time did you arrive?
- ☐ 5 am – 8 am
- ☐ 8 am – 11.30 am
- ☐ 11.30 am – 2. 30 pm
- ☐ 2.30 pm – 5 pm
- ☐ 5 pm – 8 pm

How long did you stay?
- ☐ 1 hour
- ☐ 2 hours
- ☐ 3 hours
- ☐ Half day
- ☐ Full day

The conservation fee is:
- ☐ I expected to pay less
- ☐ I was happy to pay the amount
- ☐ I would pay more for nature

Which activity/activities did you do?

- ☐ Running
- ☐ Walking
- ☐ Biking
- ☐ Birding
- ☐ Café
- ☐ Sitting
- ☐ Reading
- ☐ Pope's Garden
- ☐ Visitor Centre
- ☐ Medicinal Garden
- ☐ Picnic
- ☐ Gift shop
- ☐ Other

Rate your overall experience

- ☐ Fantastic
- ☐ Good
- ☐ OK
- ☐ Could have been better
- ☐ Bad

Did you learn anything about the importance of urban wetlands?

- ☐ No
- ☐ Yes

Will you come back?

- ☐ No
- ☐ Yes

What other activities do you want to see in NEP?

____

How can we improve your experience in NEP?

____

ANNEX II: NEP Operational and Capital Expenditure Budget

Budget Detail	Operating Budget / Years									
	1	2	3	4	5	6	7	8	9	10
Staff Costs										
NEP Manager	18,000	18,540	19,096	19,669	20,259	20,867	21,493	22,138	22,802	23,486
Senior Garden Technician	9,000	9,270	9,548	9,835	10,130	10,433	10,746	11,069	11,401	11,743
Garden Technician	14,400	14,832	15,277	15,735	16,207	16,694	17,194	17,710	18,241	18,789
Security Officer	4,200	4,326	4,456	4,589	4,727	4,869	5,015	5,165	5,320	5,480
Communication Officer	650	670	690	710	732	754	776	799	823	848
Site Foreman	350	361	371	382	394	406	418	430	443	457
Botanist	4,200	4,326	4,456	4,589	4,727	4,869	5,015	5,165	5,320	5,480
Security officers	90,000	92,700	95,481	49,173	50,648	52,167	53,732	55,344	57,005	58,715
Maintenance Manager	7,800	8,034	8,275	8,523	8,779	9,042	9,314	9,593	9,881	10,177
Tourism Officer	7,800	8,034	8,275	8,523	8,779	9,042	9,314	9,593	9,881	10,177
Education Officer	6,000	6,180	6,365	6,556	6,753	6,956	7,164	7,379	7,601	7,829
Maintenance Staff	9,000	9,270	9,548	9,835	10,130	10,433	10,746	11,069	11,401	11,743
Medical cover	1,510	1,555	1,602	1,650	1,700	1,751	1,803	1,857	1,913	1,970
Long service award	5,000	5,000	5,150	5,305	5,464	5,628	5,796	5,970	6,149	6,334
Funeral assistance	500	515	530	546	563	580	597	615	633	652
Subtotal staff	***178,410***	***183,612***	***189,121***	***145,622***	***149,990***	***154,490***	***159,125***	***163,898***	***168,815***	***173,880***
Management										
Office supplies	4,800	4,944	5,092	5,245	5,402	5,565	5,731	5,903	6,080	6,263
Maps and materials production	500	515	530	546	563	580	597	615	633	652
IT maintenance and software (internet)	1,000	1,030	1,061	1,093	1,126	1,159	1,194	1,230	1,267	1,305
Communications (mobile)	6,000	6,180	6,365	6,556	6,753	6,956	7,164	7,379	7,601	7,829
Communications (website, social media)	1,000	1,030	1,061	1,093	1,126	1,159	1,194	1,230	1,267	1,305
Meetings	1,000	1,030	1,061	1,093	1,126	1,159	1,194	1,230	1,267	1,305
Annual audit	5,000	5,150	5,305	5,464	5,628	5,796	5,970	6,149	6,334	6,524
Legal fees	5,000	5,150	5,305	5,464	5,628	5,796	5,970	6,149	6,334	6,524
Liability insurance	11,000	11,330	11,670	12,020	12,381	12,752	13,135	13,529	13,934	14,353
Health & Safety	2,500	2,575	2,652	2,732	2,814	2,898	2,985	3,075	3,167	3,262
Medical supplies	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Media engagement	1,000	1,030	1,061	1,093	1,126	1,159	1,194	1,230	1,267	1,305

Management training	2,500	2,575	2,652	2,732	2,814	2,898	2,985	3,075	3,167	3,262
Management systems	1,500	1,545	1,591	1,639	1,688	1,739	1,791	1,845	1,900	1,957
Consultant input	2,500	2,575	2,652	2,732	2,814	2,898	2,985	3,075	3,167	3,262
Bank charges	1,000	1,030	1,061	1,093	1,126	1,159	1,194	1,230	1,267	1,305
Subtotal central/HQ management	***47,500***	***48,925***	***50,393***	***51,905***	***53,462***	***55,066***	***56,717***	***58,419***	***60,172***	***61,977***
Fleet Management										
Vehicle fuel and maintenance	7,200	7,416	7,638	7,868	8,104	8,347	8,597	8,855	9,121	9,394
Vehicle insurance	800	824	849	874	900	927	955	984	1,013	1,044
Vehicle tire replacement	1,500	1,545	1,591	1,639	1,688	1,739	1,791	1,845	1,900	1,957
Rotary mower fuel and maintenance	6,000	6,180	6,365	6,556	6,753	6,956	7,164	7,379	7,601	7,829
Rotary mower insurance	65	67	69	71	73	75	78	80	82	85
Brush cutter fuel / maintenance	3,000	3,090	3,183	3,278	3,377	3,478	3,582	3,690	3,800	3,914
Brush cutter insurance	9,000	9,270	9,548	9,835	10,130	10,433	10,746	11,069	11,401	11,743
Edge Trimmer maintenance	1,000	1,030	1,061	1,093	1,126	1,159	1,194	1,230	1,267	1,305
Chainsaw maintenance	3,000	3,090	3,183	3,278	3,377	3,478	3,582	3,690	3,800	3,914
Industrial mower fuel and maintenance	2,100	2,163	2,228	2,295	2,364	2,434	2,508	2,583	2,660	2,740
Subtotal transport	***33,665***	***34,675***	***35,715***	***36,787***	***37,890***	***39,027***	***40,198***	***41,404***	***42,646***	***43,925***
Field Support										
Workshop consumables	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Welding supplies	600	618	637	656	675	696	716	738	760	783
Solar/electricity maintenance	1,250	1,288	1,326	1,366	1,407	1,449	1,493	1,537	1,583	1,631
Water supply maintenance	500	515	530	546	563	580	597	615	633	652
Generator (fixed) fuel and maintenance	3,600	3,708	3,819	3,934	4,052	4,173	4,299	4,428	4,560	4,697
Generator (portable) fuel / maintenance	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Fence maintenance	4,500	4,635	4,774	4,917	5,065	5,217	5,373	5,534	5,700	5,871

Training - logistics staff	2,500	2,575	2,652	2,732	2,814	2,898	2,985	3,075	3,167	3,262
Uniform replacement - Security	3,250		3,348		1,724		1,776		1,829	
Performance Audit	5,000		5,150			5,305			5,464	
Subtotal field support	***23,600***	***15,811***	***24,782***	***16,773***	***19,001***	***23,099***	***20,104***	***18,879***	***26,738***	***20,028***
Conservation & Management										
Uniform replacement Staff	3,250		3,348		3,448		3,551		3,551	
Compost	1,000	1,030	1,061	1,093	1,126	1,159	1,194	1,230	1,267	1,305
Trees	0	0	0	0	0	0	0	0	0	0
Plants		0	0	0	0	0	0	0	0	0
PPE for brush cutters	0	0	0	0	0	0	0	0	0	0
Garden refuse collection bag	0	0	0	0	0	0	0	0	0	0
Brush cutter nylon	0	0	0	0	0	0	0	0	0	0
Monitoring training	1,250	1,288	1,326	1,366	1,407	1,449	1,493	1,537	1,583	1,631
Ecological monitoring costs	1,250	1,288	1,326	1,366	1,407	1,449	1,493	1,537	1,583	1,631
Habitat regeneration projects	2,500	2,575	2,652	2,732	2,814	2,898	2,985	3,075	3,167	3,262
Invasive and alien species control	1,250	1,288	1,326	1,366	1,407	1,449	1,493	1,537	1,583	1,631
Subtotal conservation & Management	***10,500***	***7,468***	***11,039***	***7,922***	***11,608***	***8,405***	***12,208***	***8,917***	***12,735***	***9,460***
Civic Engagement										
Community meetings	1,500	1,545	1,591	1,639	1,688	1,739	1,791	1,845	1,900	1,957
District meetings	2,400	2,472	2,546	2,623	2,701	2,782	2,866	2,952	3,040	3,131
Education & awareness programs	5,000	5,150	5,305	5,464	5,628	5,796	5,970	6,149	6,334	6,524
Community resource projects	15,000	15,450	15,914	16,391	16,883	17,389	17,911	18,448	19,002	19,572
Subtotal civic engagement	***23,900***	***24,617***	***25,356***	***26,116***	***26,900***	***27,707***	***28,538***	***29,394***	***30,276***	***31,184***
Business Management										

Website maintenance	3,000	3,090	3,183	3,278	3,377	3,478	3,582	3,690	3,800	3,914
Promotional literature	2,500	2,575	2,652	2,732	2,814	2,898	2,985	3,075	3,167	3,262
Advertising	2,500	2,575	2,652	2,732	2,814	2,898	2,985	3,075	3,167	3,262
Corporate sponsorship management	5,000	5,150	5,305	5,464	5,628	5,796	5,970	6,149	6,334	6,524
Subtotal business management	***13,000***	***13,390***	***13,792***	***14,205***	***14,632***	***15,071***	***15,523***	***15,988***	***16,468***	***16,962***
TOTAL	**330,575**	**328,497**	**350,197**	**299,330**	**313,482**	**322,864**	**332,413**	**336,899**	**357,849**	**357,416**

Table 35: NEP 10-year operating budget (USD).

Budget Detail	CAPEX Budget / Years									
	1	2	3	4	5	6	7	8	9	10
Management Infrastructure										
Office Develop	150,000									
Office Renovation				25,750				26,523		
Generator	25,000				25,750					26,523
Subtotal management infrastructure	***175,000***	***0***	***0***	***25,750***	***25,750***	***0***	***0***	***26,523***	***0***	***26,523***
Conservation Infrastructure										
Fence Installation		300,000								
Fence renovation			20,000			20,600			21,218	
Solar Installations	20,000			20,600			21,218			21,855
Water Provisions	10,000			10,300			10,609			10,927
Subtotal conservation infrastructure	***20,000***	***300,000***	***20,000***	***20,600***	***0***	***20,600***	***21,218***	***0***	***21,218***	***21,855***
Transport										
Rotary mower industrial	50,000				51,500					53,045
Brush cutter Stihl, industrial	30,000				30,900					31,827
Industrial mower	30,000				30,900					31,827
Truck	40,000				41,200					42,436
Subtotal transport	***150,000***	***0***	***0***	***0***	***154,500***	***0***	***0***	***0***	***0***	***159,135***
Communications										
Phone replacement	1,250			1,288			1,326			1,366
Website Development	15,000			15,450			15,914			16,391
Subtotal communications	***16,250***	***0***	***0***	***16,738***	***0***	***0***	***17,240***	***0***	***0***	***17,757***
Office and IT Equipment										
Laptops	3,000			3,090			3,183			3,278
Printers	400			412			424			437
Furniture	2,000			2,060			2,122			2,185
Subtotal office and IT equipment	***5,400***	***0***	***0***	***5,562***	***0***	***0***	***5,729***	***0***	***0***	***5,901***
Field Equipment										
Spades	1,000			1,030			1,061			1,093
Rakes, metal	700			721			743			765
Watering cans	600			618			637			656
Forks	90			93			95			98

Lasher/ machete	120			124			127			131
Secateurs	640			659			679			699
Loppers	80			82			85			87
Shears	240			247			255			262
Hoe and handle	600			618			637			656
Broom, industrial	400			412			424			437
Pick and handles	400			412			424			437
Pipes and fittings for water pumps	80			82			85			87
Wheelbarrow	500			515			530			546
Knapsack sprayer	625			644			663			683
Blower backpack	625			644			663			683
Edge trimmer	400			412			424			437
Water pump	250			258			265			273
Chainsaw - small	900			927			955			983
Subtotal field equipment	***8,250***	***0***	***0***	***8,498***	***0***	***0***	***8,752***	***0***	***0***	***9,015***
Roads, bridges, trails and water culverts										
Gravelling of new roads		1,000			1,030			1,061		
Riprap		750			773			796		
Small bridges		750			773			796		
Water Culverts		2,500			2,575			2,652		
New trails		2,500			2,575			2,652		
***Subtotal* bridges, trails and water culverts**	***0***	***7,500***	***0***	***0***	***7,725***	***0***	***0***	***7,957***	***0***	***0***
Tourism and Enterprise										
Restaurant renovation (private sector partner)										
Information Centre Development (maps, signs)				25,750				26,523		
Gate Renovation				10,300				10,609		
Sign Renovation			2,500			2,575			2,652	
Interpretation signs	25,000					25,750				
Interpretive signs		25,000					25,750			
Garbage cans	11,250				11,588				11,935	
Subtotal tourism and enterprise	***36,250***	***25,000***	***2,500***	***36,050***	***11,588***	***28,325***	***25,750***	***37,132***	***14,587***	***0***
TOTAL	**411,150**	**332,500**	**22,500**	**113,197**	**199,563**	**48,925**	**78,689**	**71,611**	**35,805**	**240,185**

Table 36: NEP 10-year CAPEX Budget (USD).

CAPEX Budget Detail	Item	# Items	Item/Unit	Cost Unit	# Units	Total
Management Infrastructure						
Office Development	Unit	1	Lumpsum	150,000	1	150,000
Office Renovation	Unit	1	Lumpsum	25,000	1	25,000
Generator	Unit	1	Lumpsum	25,000	1	25,000
Subtotal management infrastructure						***200,000***
Conservation Infrastructure						
Fence Installation	Unit	1	Sectors III, IV, V	300,000	1	300,000
Fence renovation	Unit	1	Lumpsum	20,000	1	20,000
Solar Installations	Unit	1	Lumpsum	20,000	1	20,000
Water Provisions	Unit	1	Lumpsum	10,000	1	10,000
Subtotal conservation infrastructure						***340,000***
Transport						
Rotary mower industrial	Vehicle	1	Lumpsum	50,000	1	50,000
Brush cutter Stihl, industrial	Vehicle	1	Lumpsum	30,000	1	30,000
Industrial mower	Tractor	1	Lumpsum	30,000	1	30,000
Truck	Tractor	1	Lumpsum	40,000	1	40,000
Subtotal transport						***150,000***
Communications						
Phone replacement	Unit	1	Lumpsum	250	5	1,250
Website Development	Lumpsum	1	Lumpsum	15,000	1	15,000
Subtotal communications						***16,250***
Office and IT Equipment						
Laptops	Laptop	1	Lumpsum	1,000	3	3,000
Printers	Printer	1	Lumpsum	200	2	400
Furniture	Assorted	1	Lumpsum	2,000	1	2,000
Subtotal office and IT equipment						***5,400***
Field Equipment						
Spades, flat straight edge	Unit	1	Lumpsum	50	20	1,000
Rakes, metal	Unit	1	Lumpsum	35	20	700
Watering cans	Unit	1	Lumpsum	30	20	600
Forks	Unit	1	Lumpsum	5	18	90
Lasher/ machete	Unit	1	Lumpsum	10	12	120
Secateurs	Unit	1	Lumpsum	40	16	640
Loppers	Unit	1	Lumpsum	40	2	80
Shears	Unit	1	Lumpsum	40	6	240
Hoe and handle	Unit	1	Lumpsum	40	15	600
Broom, industrial	Unit	1	Lumpsum	40	10	400

Pick and handles	Unit	1	Lumpsum	40	10	400
Pipes and fittings for water pumps	Unit	1	Lumpsum	40	2	80
Wheelbarrow	Unit	1	Lumpsum	50	10	500
Knapsack sprayer	Unit	1	Lumpsum	125	5	625
Blower backpack	Unit	1	Lumpsum	125	5	625
Edge trimmer	Unit	1	Lumpsum	200	2	400
Water pump	Unit	1	Lumpsum	250	1	250
Chainsaw - small	Unit	1	Lumpsum	300	3	900
Subtotal field equipment						***8,250***
Roads, Bridges, trails and water Culverts						
Gravelling of new roads	Lumpsum	1	Year	1,000	1	1,000
Riprap	Drift	1	Lumpsum	750	1	750
Small bridges	Bridge	1	Lumpsum	750	1	750
Water Culverts	Drift	1	Lumpsum	2,500	1	2,500
New trails	Unit	1	Lumpsum	250	10	2,500
Subtotal roads, bridges and airstrips						***7,500***
Tourism and Enterprise						
Restaurant renovation (private sector partner)	Restaurant	1	Lumpsum			
Information Centre Development (maps, interpretation)	Info Centre	1	Lumpsum	25,000	1	25,000
Gate Renovation	Gates	1	Lumpsum	5,000	2	10,000
Sign Renovation	Signs	1	Lumpsum	2,500	1	2,500
Interpretation sign design	Consultant	1	Lumpsum	25,000	1	25,000
Interpretive signs	Signs	1	Lumpsum	2,500	10	25,000
Garbage cans	Uni	1	Lumpsum	750	15	11,250
Subtotal tourism and enterprise						***98,750***
TOTAL						**826,150**

Table 37: NEP CAPEX unit costs (USD).

REFERENCES

Afrilandscapes Interview – November 2021, Kigali, Rwanda

Bodin, M., & Terry, H. (2003). Does the outdoor environment matter for psychological restoration gained through running? *Psychology of Sport and Exercise, 4*(2), 141-153.

Carr, S., Francis, M., Rivlin, L. G., & Stone, A. M. (1992). Public space. *Cambridge: Cambridge University Press.*

CBD Rwanda Wetlands Governance Profile: https://chm.cbd.int/api/v2013/documents/6EC46755-C2FF-9FFF-AF37-7FA66D88638E/attachments/213149/Rwanda-Wetlands-Governance-and-Management-Profile%202019.pdf

CBD Wetland & Ecosystem Service Report (2015). https://www.cbd.int/waters/doc/wwd2015/wwd-2015-press-briefs-en.pdf

Conservation Capital (2022). Updated NUWEP Feasibility Study. Kigali.

Gabuka, A. (2012). NUWEP Feasibility Study - Final Report. Kigali.

Gasabo 3D (2017). NUWEP Feasibility Study. Final Report Volume 1: Developing an Improved Design Plan of the Park and Construction Supervision of Nyandungu Urban Wetland Eco-Tourism Park. Kigali.

Gasabo 3D (2017). NUWEP Feasibility Study. Volume 2: LANDSCAPINGS & ECOLOGICAL ENHENCEMENT AND ARCHTECTURAL.pdf.

Gasabo 3D (2017). NUWEP Feasibility Study. Volume 3: MECHANICAL AND SANITARY DRAWINGS. Kigali: Gasabo 3D.

Government of Rwanda (2020). National Land Use & Development Masterplan 2020-2050. Kigali.

Kawawa, R., Obiri, J., Muyekho, F., Omayio, D., Agevi, H., & Mwaura, A. et al. (2016). *Allellopathic potential of invasive Psidium guajava L., against selected native tree species in Kakamega Tropical Forest, Western Kenya*. IOSR Journal of Pharmacy And Biological Sciences, 11(05), 80-86. doi: 10.9790/3008-1105028086

Krishnasamy, S., Chhetri, L., Mchugh, A. &Thosani, D. (2019). Kigali Master Plan 2050 – edition 2020. 1st ed. Kigali: City of Kigali.

McInnes R.J. (2016) Climate Regulation and Wetlands: Overview. In: Finlayson C. et al. (eds) The Wetland Book. Springer, Dordrecht. https://doi.org/10.1007/978-94-007-6172-8_231-1

Ministry of Lands (2004). National Land Policy for Rwanda. Kigali: Republic of Rwanda.

Ministry of Trade and Industry (2009). Rwanda Tourism Policy Kigali: Republic of Rwanda.

Mvukiyumwami, J. (2019). Inventory of plants existing in Nyandungu Urban Wetland Ecotourism Park in January 2019. Afrilandscapes, unpublished internal report.

National Institute of Statistics of Rwanda (2018). Rwanda Natural Capital Account. NCA – Land. Kigali: NISR, WAVES, World Bank, Ministry of Environment.

National Institute of Statistics of Rwanda (2019). Rwanda Natural Capital Accounts. NCA - Water. Kigali: NISR, WAVES, World Bank, Ministry of Environment.

Niyokwiringirwa, P. (2021). Extreme Flood Events in Rwanda: The Case of Kigali City. https://storymaps.arcgis.com/stories/1f814d6184a141e2a03a7bd7707c14c5

Rugege, D. (2004). Environmental Impact Assessment of the proposed establishment of free zones in the Nyandungu valley and adjacent areas in the City of Kigali, Rwanda.

Rwanda Environment Management Authority (2010). Rwanda State of Environment and Outlook. Kigali: Republic of Rwanda.

Rwanda Environment Management Authority (2010). Operationalization of National Fund for Environment (FONERWA) in Rwanda: Final Report. Supported by Poverty Environment Initiative (PEI). Kigali: Republic of Rwanda.

Rwanda Environment Management Authority (2011). Environmental and Social Management Framework for Rwanda under LVEMP II (Final report). Kigali: Republic of Rwanda.

Sullivan, W., C. (2001). Environment and crime in the inner city. *Environment and Behaviour, 33*(3), 343-367.

Tsinda, A & Gakuba, A. (2010). Sustainable Hazards Mitigation in Kigali city (Rwanda), 46th ISOCARP Congress, 2010. Retrieved on 2012/12/10, http://www.isocarp.net/Data/case_studies/1829.pdf

Tsinda, A. & Ilunga, L (2006). *Facteurs physiques du ruissellement à Kigali (Rwanda):* Geo-Eco-Trop, 2004, 28, 1 2: pp.53-60.

UNFCC Updated Rwanda NDC (2020): https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Rwanda%20First/Rwanda_Updated_NDC_May_2020.pdf

US Department of State (2003). Overview of sustainable ecotourism in the United States of America. World Ecotourism Summit: UNEP.

USAID Rwanda Policy Research brief n. 5 (2016): Balancing Wetland Sustainable Use and Protection Through Policy In Rwanda: https://land-links.org/wp-content/uploads/2016/09/USAID_Land_Tenure_Rwanda_Policy_Research_Brief_No-5.pdf

Wall, G. (1997). Is Ecotourism Sustainable? Environmental Management, 21(4), 483-491.

Wells, M.P. (1997). Economic Perspectives on Nature Tourism, Conservation and Development. Environmental Economics.

World Bank Global Wildlife Program (2021). Collaborative Management Partnership Toolkit. Washington DC.

NYANDUNGU ECO-PARK
KIGALI, RWANDA